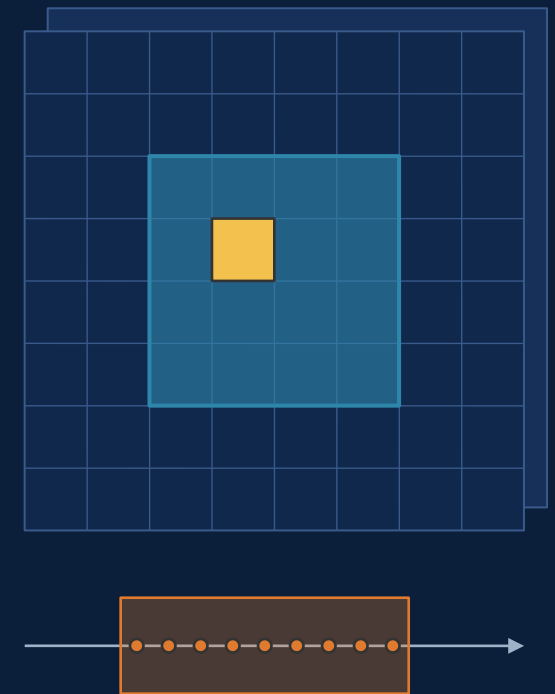


Geospatial representation models

A stencil-level overview of AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 and IBM's Prithvi-based ocean model (Granite-Geospatial-Ocean)

What area in space and what span of time goes into each embedding — and what each model actually hands you

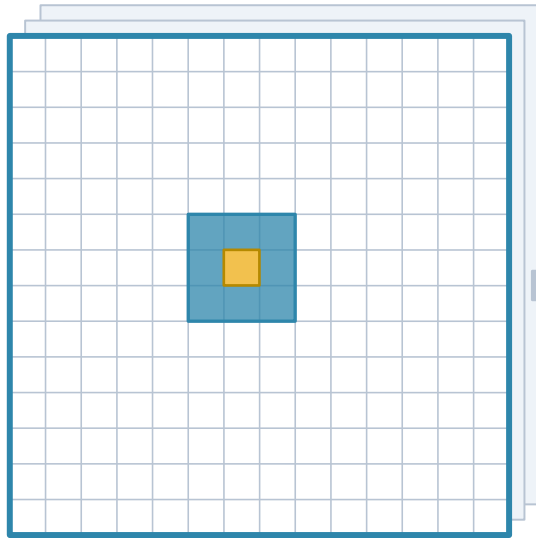


How to read the stencil diagrams

Every model slide uses the same three-panel convention

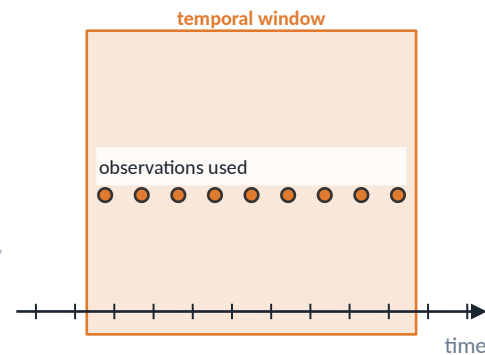
Stencil — what feeds one embedding

SPACE



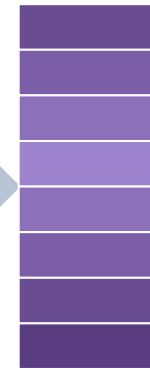
Grid = input tile the encoder sees at once (blue outline).
Blue square = one token / receptive field. Yellow = the output pixel.
Stacked sheets = co-registered modalities / channels.

TIME



Shaded band = window summarised by one embedding.
Dots = observations sampled inside it. A single tall bar = one snapshot acquisition.

OUTPUT



D dims

one vector per pixel, per token, or per tile

Four questions we ask of each model

- Per-pixel or patch? Does the embedding of a pixel see its neighbours (patch/ViT) or only its own time series (pixel-wise)?
- Snapshot or period summary? One acquisition, a handful of dates, or a whole year folded into one vector?
- Product or model? Do we get precomputed global embeddings, open weights we can run on our own grid, or both?
- Does it touch the ocean? Land-only masking is the norm; only one of the six is trained on sea pixels.

AlphaEarth Foundations (Google DeepMind)

An 'embedding field': every 10 m land pixel $\rightarrow$ 64-D unit vector per year, 2017–2024

Who / when DeepMind + Google Earth Engine; arXiv 2507.22291 (Jul 2025, v2 Sep 2025).

Inputs Sentinel-2, Landsat 8/9, Sentinel-1, PALSAR-2, GEDI lidar, ERA5-Land, GRACE, GLO-30 DEM, NLCD, plus text (Wikipedia, GBIF) — >3 B observations.

Architecture Space-Time-Precision (STP) encoder — 15 blocks, each with three operators at three resolutions (spatial attention @ L/16, time-axial attention @ L/8, 3×3 convs @ L/2; widths 1024 / 512 / 128) exchanging state through a learned Laplacian pyramid (next slide). Teacher/student video embedders + text alignment; ~480 M deployed (1 B trained). Losses: reconstruction, batch-uniformity on S^{63} , consistency, text-contrastive.

Stencil Output 10 m pixel. Input frame 1.28 km × 1.28 km (128 × 128 px; supplement S2.1/S8.2): inference runs on 960 m tiles buffered by 160 m, so a pixel's context is at most the 1.28 km frame, with spatial attention global inside it. Continuous 'valid period' $[t_0, t_1]$ — arbitrary windows in principle; released product = calendar-year summaries.

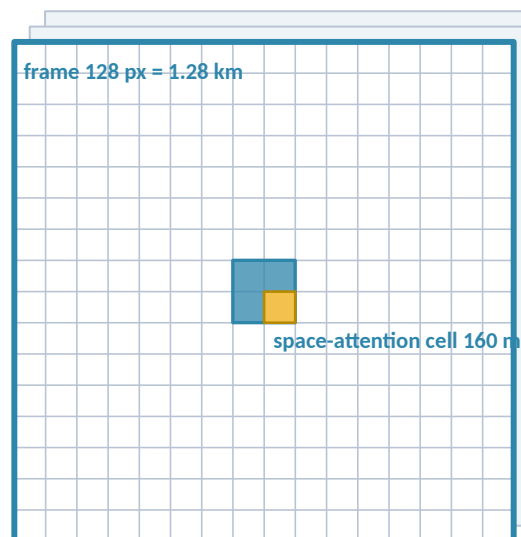
Output 64-D, unit-norm, int8 in the public dataset; ~1.4 T footprints / yr. Land + coastal/inland water; no open ocean.

Availability Dataset only (CC-BY 4.0) in Earth Engine / GCS. Weights and code not released.

Results ~23.9 % average error reduction vs. prior featurizations over 15 evaluations (10.4 % at 10-shot, 4.2 % at 1-shot); evapotranspiration regression R^2 0.58. Per-dataset scores are charts only.

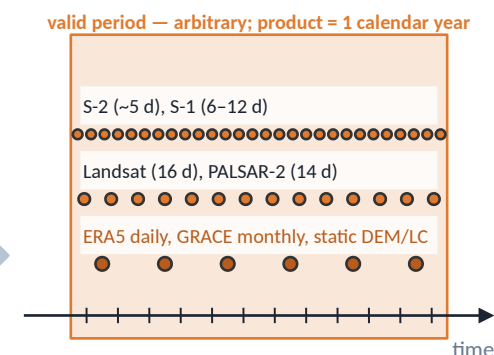
Stencil — what feeds one embedding

SPACE



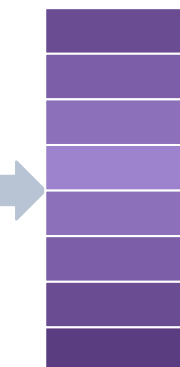
Output pixel: 10 m × 10 m (drawn 8× too large). Context: the 1.28 km frame. Spatial attention is global inside it on an 8 × 8 grid of 160 m cells; the outer 80 m is trimmed at inference, so no pixel sees further than ~640 m. ~10 co-registered sources at 10 m ... 300 km native resolution.

TIME



All observations in the window are summarised — a period embedding, not a snapshot. Model supports interpolation/extrapolation in time; only annual layers are public.

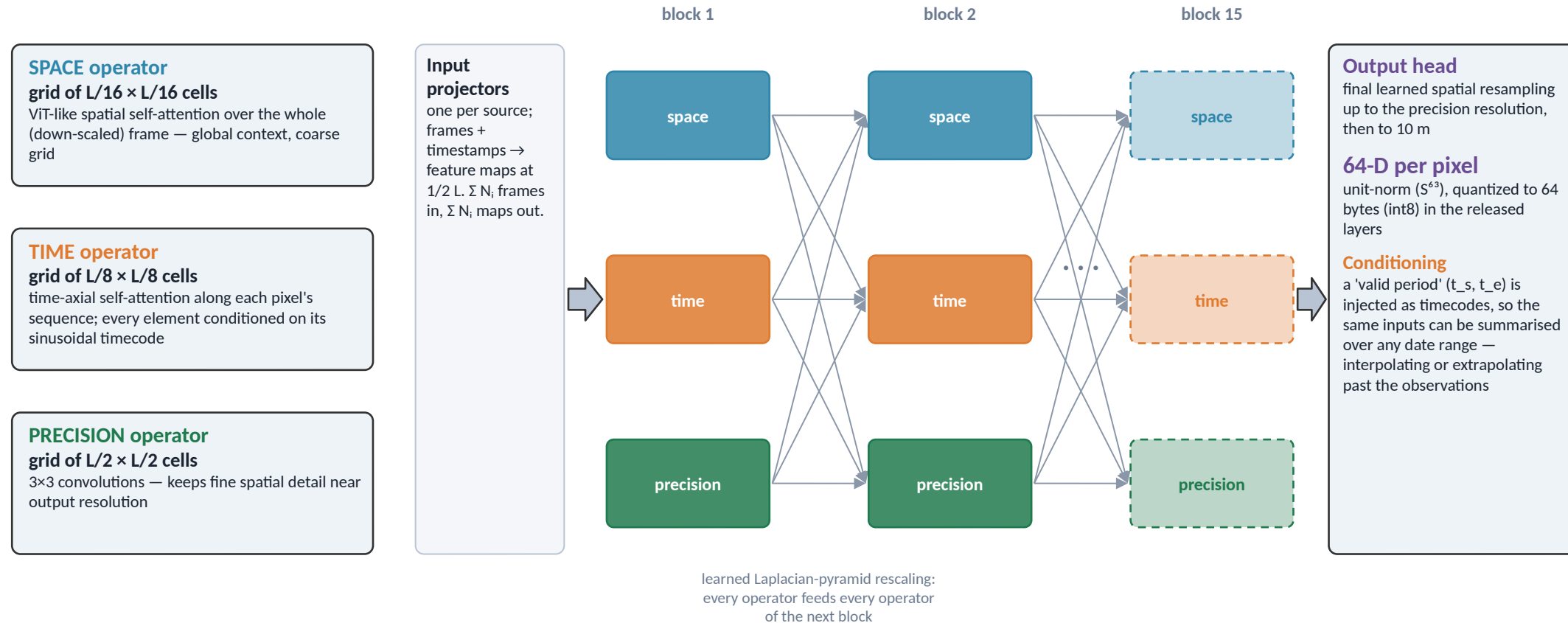
OUTPUT



64-D
per 10 m pixel per year
int8, unit sphere

AlphaEarth — inside the STP encoder

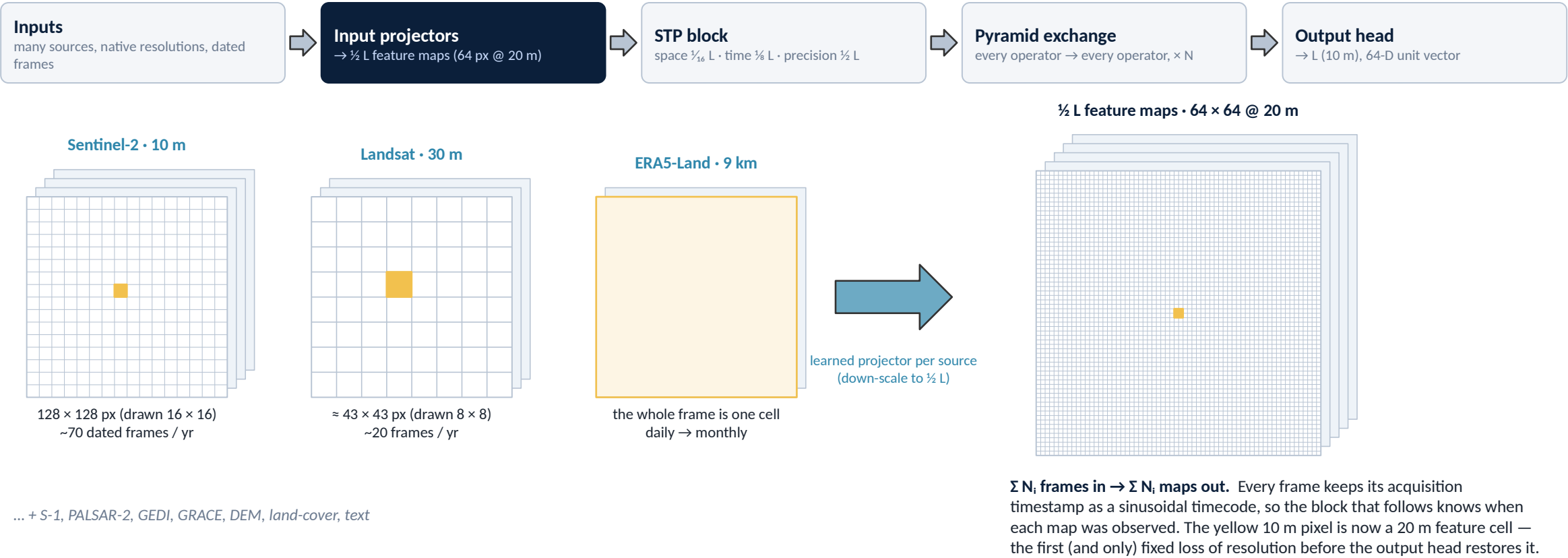
L = side length of the square input frame, in pixels — $L = 128$ (1.28 km at 10 m) for training and inference (supplement S2.1, S8.2). Three operators at $L/16$, $L/8$ and $L/2$ exchange state through a learned Laplacian pyramid after every one of 15 blocks.



Training trio teacher video embedder with implicit decoders · student with identical architecture (consistency + batch-uniformity on the sphere) · text-alignment model with a frozen language model (CLIP-style). ~1 B and ~480 M variants trained; the 480 M one is deployed. **In the supplement (S2.4, S8.2):** 15 STP blocks; widths $D_S = 1024$ (space), $D_T = 512$ (time), $D_P = 128$ (precision); implicit decoders = 2-hidden-layer MLPs of width 512; vMF bottleneck $\kappa = 8 \cdot 10^3$; inference on 960 m tiles buffered to 1.28 km, outer 80 m trimmed. So the spatial receptive field is at most 1.28 km — the whole frame. **Still not stated:** the exact pyramid up/down-sampling kernels and how the three states are merged.

STP, step 1 of 4 — input projectors bring every source to $\frac{1}{2} L$

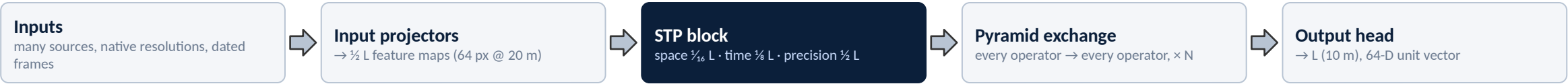
One frame of the same 1.28 km footprint from each sensor, at its own native resolution, becomes a 64×64 feature map at 20 m



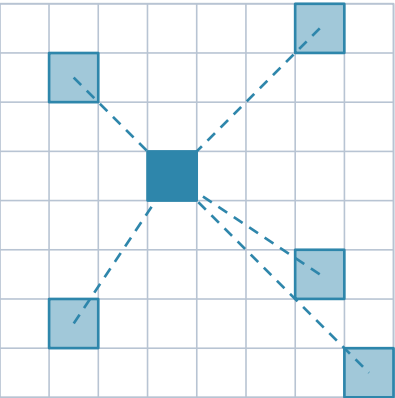
L = side length of the square input frame in pixels. $L = 128 \text{ px} @ 10 \text{ m}$ (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is $N = 15$ blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 2 of 4 — three operators, three resolutions

Same 1.28 km footprint, three grids: what one cell can 'see' in each operator

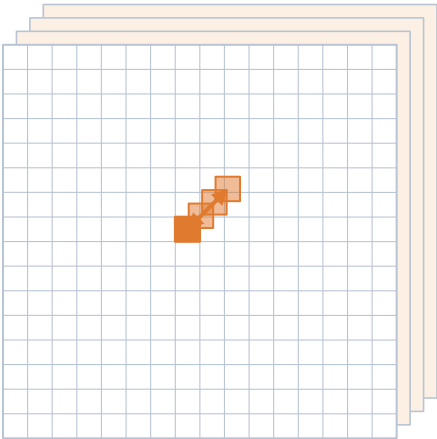


SPACE operator · $\frac{1}{16} L = 8 \times 8$ cells of 160 m



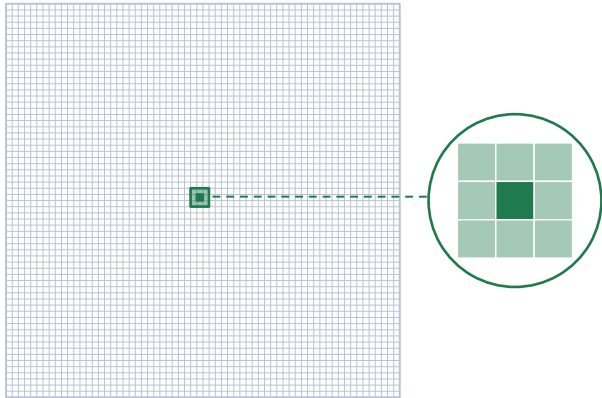
ViT-like self-attention: the cell attends to every other cell of the frame. Receptive field = whole frame, at coarse grain.

TIME operator · $\frac{1}{8} L = 16 \times 16$ cells of 80 m



Time-axial self-attention: the cell attends only to itself across all dated frames (stacked sheets), each tagged with its timecode.

PRECISION operator · $\frac{1}{2} L = 64 \times 64$ cells of 20 m



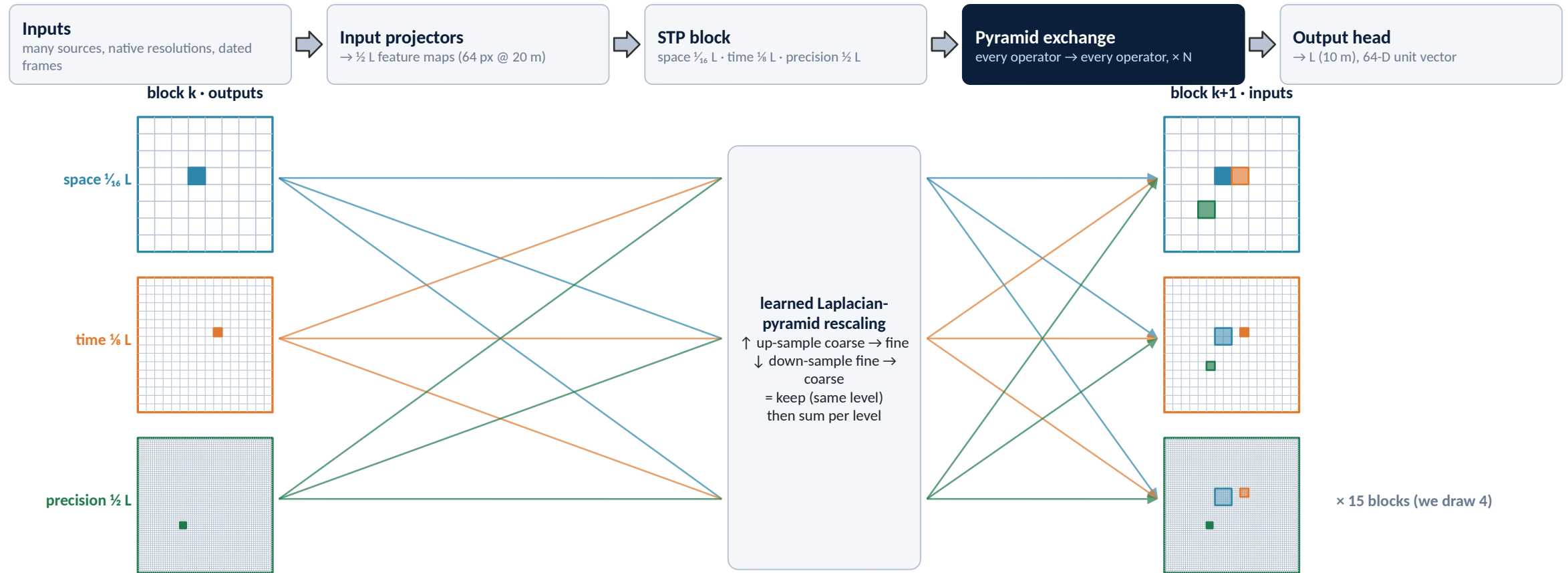
3 × 3 convolutions: the cell sees only its immediate neighbours (60 m), but at the finest grain the block carries.

Three views of one place, computed simultaneously. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture — none of them alone is the stencil; the exchange in step 3 is.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 3 of 4 — the pyramid exchange between blocks

End of block k : each of the three maps is resampled to all three resolutions and summed into block $k+1$'s inputs — repeat for N blocks

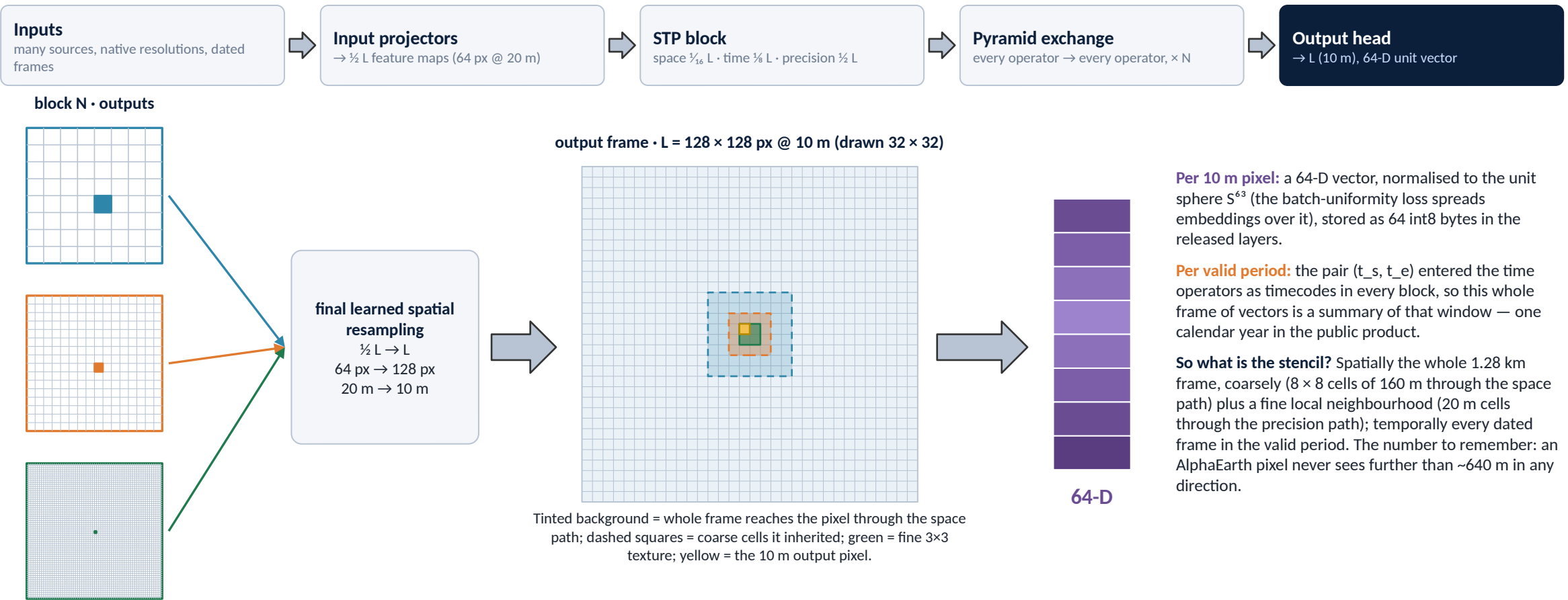


Follow the colours. The blue 160 m space cell arrives in the fine map as an 8×8 block of 20 m cells; the green 3×3 precision patch (60 m) arrives in the coarse map as a fraction of one 160 m cell; the orange time cell lands as $2 \times 2 / 4 \times 4$. After the sum, every operator in block $k+1$ starts from all three views. Repeat N times — the receptive field of the fine path grows to the whole frame while it keeps 20 m grain.

L = side length of the square input frame in pixels. $L = 128 \text{ px @ } 10 \text{ m}$ (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is $N = 15$ blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 4 of 4 — back to 10 m, 64-D per pixel

After block N: the $\frac{1}{2} L$ maps are resampled to the precision resolution and then to L; each output pixel is a unit vector on S^{63}



Per 10 m pixel: a 64-D vector, normalised to the unit sphere S^{63} (the batch-uniformity loss spreads embeddings over it), stored as 64 int8 bytes in the released layers.

Per valid period: the pair (t_s, t_e) entered the time operators as timecodes in every block, so this whole frame of vectors is a summary of that window — one calendar year in the public product.

So what is the stencil? Spatially the whole 1.28 km frame, coarsely (8 × 8 cells of 160 m through the space path) plus a fine local neighbourhood (20 m cells through the precision path); temporally every dated frame in the valid period. The number to remember: an AlphaEarth pixel never sees further than ~640 m in any direction.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

TESSERA (University of Cambridge)

Pixel-wise 10 m annual embeddings from S-1 + S-2 time series — fully open (CC0), CVPR 2026

Who / when Cambridge Centre for Earth Observation; arXiv 2506.20380 (Jun 2025), CVPR 2026. Code github.com/ucam-eo/tessera, access via geotessera.

Inputs Sentinel-2 L2A (10 bands, 60 m bands dropped, resampled to 10 m) + Sentinel-1 RTC (VV, VH). Two modality-specific branches, fused late.

Architecture Per-pixel temporal transformer (4 layers) + GRU pooling per modality; encoder ~46 M params (1.34 B projector used only in training). Barlow Twins self-supervision on ~800 M pixel-series from 3,012 MGRS tiles; ~6,200 GPU-h on MI300X.

Stencil Strictly one 10 m pixel — no spatial context at all. One calendar year; 40 valid dates sampled per modality with learnable day-of-year embeddings (irregular cadence tolerated).

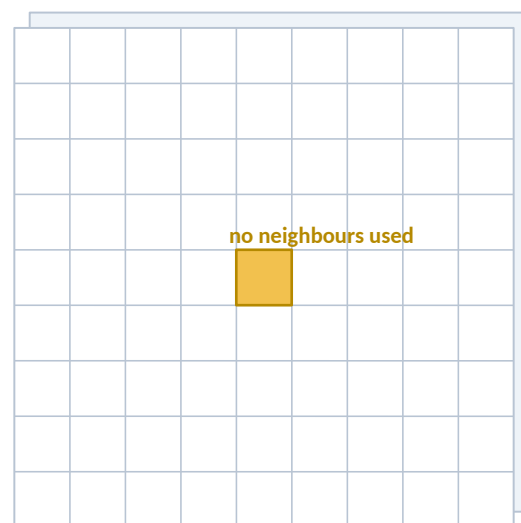
Output 128-D per pixel-year, int8 (QAT). Years 2017–2025, global land (water masked); tile-sampled coverage, gaps on request.

Availability Weights (CC0), code (MIT), precomputed embeddings via GeoTessera CDN. Self-run: ~10 h per 110 km tile-year (128-core CPU + A30).

Results Frozen + MLP head: PASTIS-R mIoU 50.7, TreeSatAI-TS F1 78.0, canopy height RMSE 12.2 m; holds up until <~20 valid S-2 obs/yr.

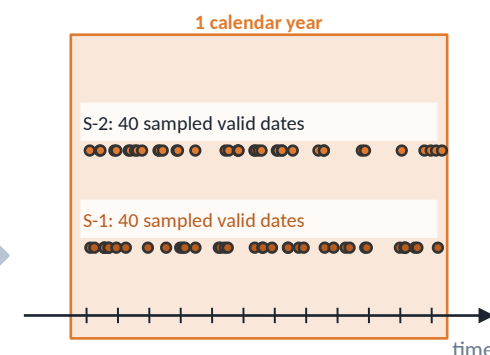
Stencil — what feeds one embedding

SPACE



Output pixel: 10 m × 10 m.
Receptive field = the same single pixel. Neighbouring pixels never enter.
Two modality sheets: S-2 (10 bands) and S-1 (2 bands).

TIME



Fixed window; dates irregular, encoded by day-of-year.
Sampling with replacement pads pixel-years with <40 clear observations.

OUTPUT



128-D
per 10 m pixel per year
int8

OlmoEarth (Ai2)

Open multimodal space-time ViT family (Nano→Large); embeddings computed on demand, not precomputed

Who / when Allen Institute for AI (+ UW, ASU, UBC). v1 Nov 2025 (arXiv 2511.13655), v1.1 May 2026, v1.2 later 2026. Weights on Hugging Face; custom 'OlmoEarth Artifact License' (no military/extractive use).

Inputs Sentinel-1, Sentinel-2, Landsat-8 monthly time series, all resampled to 10 m; static layers (OSM, WorldCover, CDL, SRTM, canopy height, WorldCereal) as inputs/targets.

Architecture ViT over space × time × band-group tokens (FlexiViT patches 1–8 px in v1; single-bandset token merging in v1.1, RoPE in v1.2). 'Latent MIM Lite': masked modelling with patch-discrimination + instance-contrastive losses. Sizes: Nano 1.4 M · Tiny 6.2 M · Base 89 M · Large 308 M (encoder).

Stencil Tile 2.56 km × 2.56 km = 256 × 256 px @ 10 m; token = 1–8 px patch (≤ 80 m). Up to 12 monthly steps over one year, missing months masked.

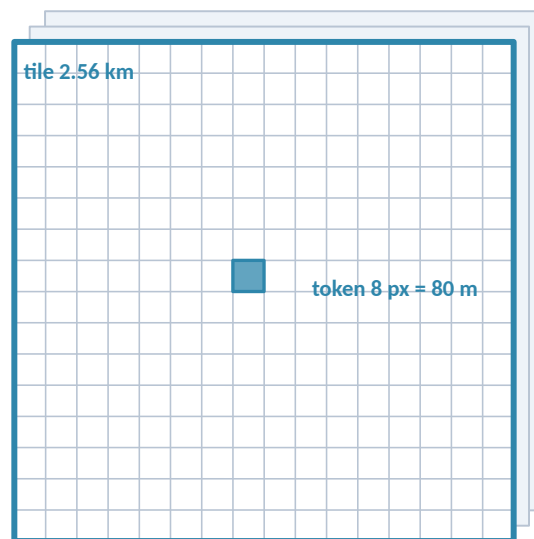
Output Per-patch, per-timestep tokens (Base 768-D, Large 1024-D) + pooled vector. No standing global product; Studio exports embeddings for any AOI at 10–80 m, 1–12 monthly steps (int8 COGs).

Results Best on 15/24 tasks (kNN/linear) and 19/29 (fine-tuned) vs. Prithvi v2, TerraMind, Galileo, DINOv3; matches/exceeds AlphaEarth after fine-tuning on 5 tasks.

Pretraining data 285 k samples × 2.56 km × 1 yr, stratified over 120 OSM classes (~10 TB) — land/infrastructure-biased.

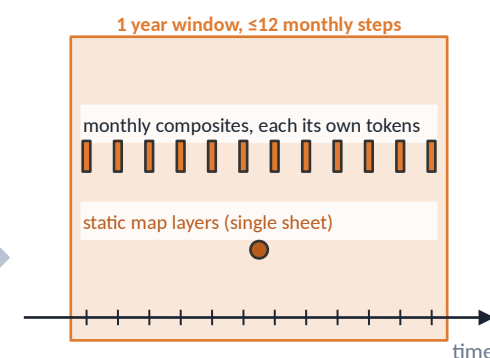
Stencil — what feeds one embedding

SPACE



Tile: 256 × 256 px @ 10 m = 2.56 km (drawn 16 × 16 for legibility).
Token: flexible 1–8 px patch → 10–80 m per token, full attention across the tile.
Sheets: S-1, S-2, Landsat + static maps.

TIME



Each month is tokenised separately; attention runs across space, time and modality.
Temporal masking (v1.2) trains the model to fill missing months.

OUTPUT



768-D

per patch per month (Base)
+ pooled tile vector

TerraMind (IBM Research + ESA Φ-lab)

Any-to-any generative multimodal EO model; 'Thinking in Modalities' — ICCV 2025

Who / when IBM Research Europe, ESA Φ-lab, ETH Zürich, FZ Jülich, NASA IMPACT, Univ. of Iceland. arXiv 2504.11171 (Apr 2025); TerraMesh dataset 2504.11172. Apache 2.0, weights on Hugging Face (ibm-esa-geospatial), TerraTorch integration. Tiny/Small edge variants Jun–Jul 2025.

Inputs Pixel-level: S-2 L2A & L1C, S-1 GRD & RTC, DEM, RGB, NDVI. Token-level: LULC classes, captions, geolocation. All co-registered at 10 m in 224 px tiles.

Architecture Dual-scale encoder–decoder: token path (cross-modal correlation) + pixel path (fine detail). Per-modality FSQ-VAE tokenizers (16 k codebook). Objective: masked cross-modal token reconstruction over 500 B tokens from ~9 M TerraMesh samples. Base: 6 d, Large: 10 d on 32 A100s.

Stencil Tile 224×224 px @ 10 m = 2.24 km; token 16×16 px = 160 m → $14 \times 14 = 196$ tokens per modality. Single co-registered timestamp per sample — no multi-temporal fusion.

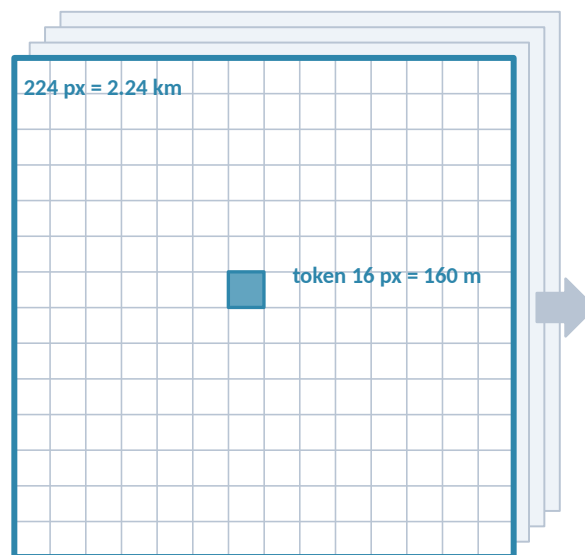
Output 196 tokens per modality of 768-D (Base) / 1024-D (Large), mergeable to one vector; plus generated rasters of missing modalities (TiM). No precomputed embedding archive.

Results PANGAEA avg mIoU: Large 59.6, Base 58.4 (≥ 3 pp over CROMA, DOFA and the other GeoFMs in that table); only GeoFM beating task-specific U-Nets across the benchmark.

Ocean TerraMesh is land/ecoregion-stratified; no marine modalities. Open-ocean behaviour untested.

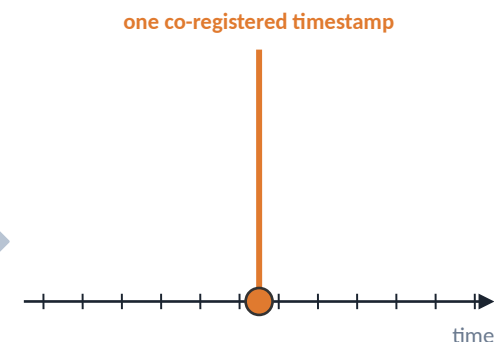
Stencil — what feeds one embedding

SPACE



Tile: 224×224 px @ 10 m = 2.24 km × 2.24 km.
Token: 16×16 px = 160 m; 14×14 grid per modality.
Up to ~9 modality sheets, each with its own tokenizer.

TIME



All modalities come from the same date and place.
A sample is a snapshot: every modality is aligned to one acquisition.
Temporal change is not part of the pretraining stencil.

OUTPUT



768-D

per 160 m token, per modality (Base; Large 1024-D) or merged per tile

Prithvi-EO 2.0 (IBM + NASA)

Multi-temporal HLS masked autoencoder at 30 m; 300 M and 600 M, with temporal-location (TL) variants

Who / when IBM Research + NASA IMPACT, trained at JSC; arXiv 2412.02732 (Dec 2024, v3 Mar 2026). Weights on Hugging Face (ibm-nasa-geospatial); fine-tuning via TerraTorch. Succeeds Prithvi-EO 1.0 (100 M).

Inputs HLS v2 (Harmonized Landsat-Sentinel-2) at 30 m, 6 bands (B02–B07: blue, green, red, NIR-narrow, SWIR1, SWIR2). 4.2 M training samples, 2014–2023; sea-only tiles removed with Fmask, Greenland excluded.

Architecture ViT-L (300 M: 1024-D, 24 layers) / ViT-H (600 M: 1280-D, 32 layers) masked autoencoder, mask ratio 0.75, 3D sin-cos positions. TL variants add lat/lon + date encodings with metadata dropout.

Stencil Tile 224×224 px @ 30 m = 6.72 km. Token 16×16 px = 480 m (300 M) or 14×14 px = 420 m (600 M); tubelet depth 1, so each of 4 timestamps gets its own token grid. Frames 1–6 months apart (seasonal), not a fixed window; single-frame inference supported.

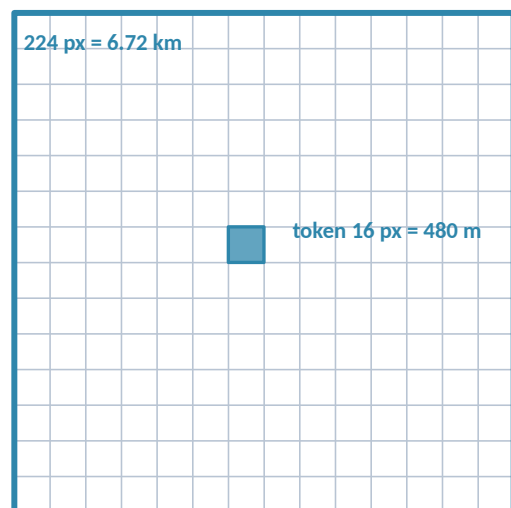
Output 4×196 (or 4×256) tokens of 1024/1280-D. No precomputed embedding product; typical use is full fine-tuning of encoder + light decoder.

Results GEO-Bench: 600 M-TL best on average, ~8 % over v1.0 and ahead of six other GFMs incl. DOFA and Scale-MAE (per-set numbers are charts); tasks: floods, burn scars, crops, landslides, PASTIS, Sen4Map.

In orbit May 2026: deployed on Kanyini satellite and ISS IMAGIN-e for onboard flood/cloud detection (still the 2.0 architecture).

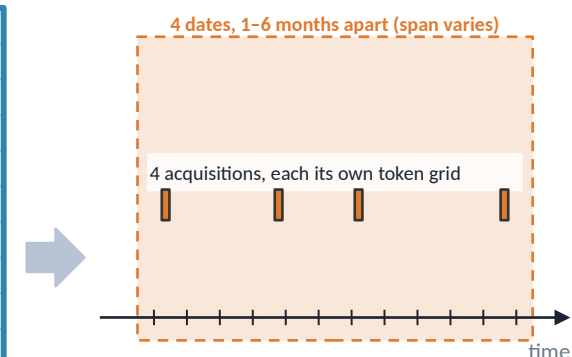
Stencil — what feeds one embedding

SPACE



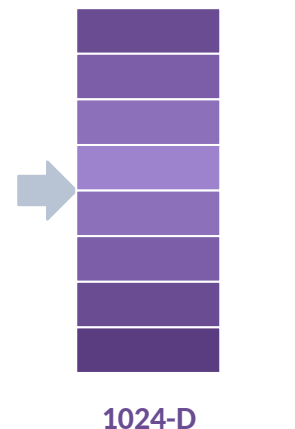
Tile: 224×224 px @ 30 m = 6.72 km $\times$ 6.72 km.
Token: 16×16 px = 480 m (300 M) $\cdot$ 14×14 px = 420 m (600 M).
One sheet: 6 HLS bands.

TIME



Time is a stack of 4 snapshots with 3D position codes — not a period summary. TL variants inject acquisition date + lat/lon as extra encodings.

OUTPUT



per 480 m token, per date
300 M: 1024-D $\cdot$ 600 M: 1280-D

Granite-Geospatial-Ocean (IBM · STFC · PML · Exeter)

The Prithvi-EO recipe moved onto Sentinel-3 ocean colour + SST — github.com/ibm-granite/geospatial

Who / when IBM Research Europe (UK), STFC Hartree Centre, Plymouth Marine Lab, Univ. of Exeter (UK HNCID). Released 6 Oct 2025; arXiv 2509.21273 'A Sentinel-3 foundation model for ocean colour'. Apache 2.0; weights on Hugging Face, code + TerraTorch notebook at github.com/ibm-granite/geospatial.

Inputs Sentinel-3 OLCI Level-2 reflectance, 16 bands (Oa01–Oa12, Oa16–18, Oa21) + SLSTR sea-surface temperature = 17 channels at 300 m. ~512 k tiles (the paper also gives 470 k train + 50 k val), 2017–2021, balanced over 83 Longhurst provinces and months.

Architecture Prithvi-EO ViT-MAE with the tile shrunk for the coarser sensor: embed 512-D, 12 layers, 8 heads, decoder 256-D × 4; ~50 M params (larger gave no gain). RMSE reconstruction loss on masked patches.

Stencil Tile 42 × 42 px @ 300 m = 12.6 km × 12.6 km; token 2 × 2 px = 600 m → 21 × 21 = 441 tokens. num_frames = 1: a single acquisition, no temporal stacking (fine-tuning pairs each in-situ sample with cloud-free imagery from a 6-day window centred on it).

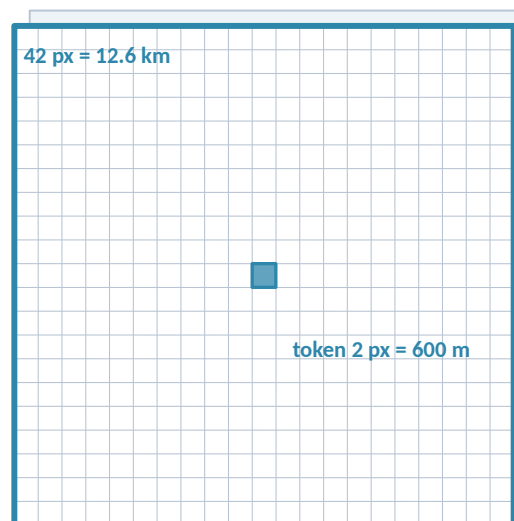
Output 441 × 512-D tokens per tile; downstream pixel-wise regression heads (TerraTorch) for chlorophyll-a and primary production.

Results Chl-a (188 in-situ points): RMSE 0.14 ± 0.05 vs 0.16 ± 0.10 log₁₀ mg m⁻³ from scratch; primary production (103 points): 0.39 ± 0.04 vs 0.42 ± 0.07 log₁₀ mgC m⁻² d⁻¹. Gains grow as labels shrink (useful at 25 % of labels).

Caveats Surface-only, cloud-limited optical; single snapshot; authors say not yet competitive with operational algorithms; under-predicts bloom-level chl-a.

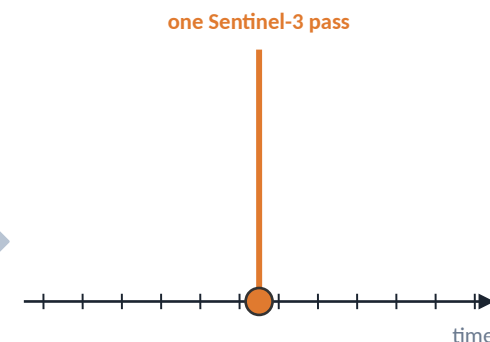
Stencil — what feeds one embedding

SPACE



Tile: 42 × 42 px @ 300 m = 12.6 km × 12.6 km (21 × 21 tokens drawn to scale).
Token: 2 × 2 px = 600 m.
Two sheets: 16 OLCI bands + 1 SST band.

TIME



num_frames = 1 — no temporal stacking. Single-timestep stencil: the 4-frame Prithvi option is switched off. Fine-tuning pairs each in-situ measurement with imagery from a 6-day window centred on it.

OUTPUT



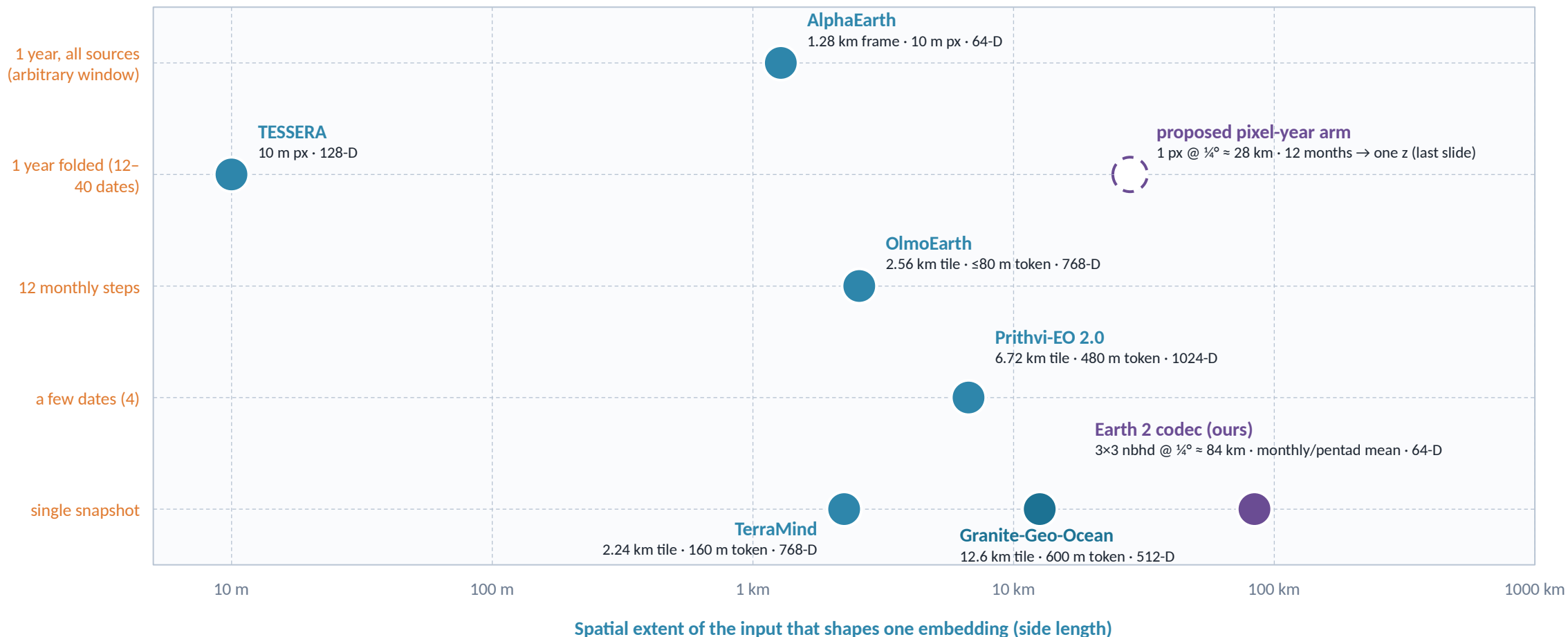
512-D

per 600 m token
per single acquisition

Stencils side by side

Spatial footprint that feeds one embedding (log scale) vs. how much time it summarises

Temporal window per embedding



Earth 2 shown for reference: the codec's channel token sees a 3×3 neighbourhood of $\frac{1}{4}^\circ$ (or 1°) cells at one monthly or 5-day mean — a snapshot stencil $\sim 7\times$ wider than the widest EO tile here (Granite, 12.6 km).

Summary I — what goes into each embedding

Inputs, spatial stencil, temporal stencil, output vector

Model	Inputs (sensors / modalities)	Spatial stencil	Temporal stencil	Output per embedding
AlphaEarth (DeepMind)	S-2, Landsat 8/9, S-1, PALSAR-2, GEDI, ERA5-Land, GRACE, DEM, NLCD, text	10 m output pixel inside a 1.28 km (128 px) frame; attention global within the frame	Period summary; arbitrary valid window in the model, calendar year in the product	64-D unit vector, int8; per pixel per year
TESSERA (Cambridge)	S-2 L2A (10 bands) + S-1 RTC (VV, VH)	Single 10 m pixel — no spatial context	1 calendar year; 40 sampled dates per modality, day-of-year encoded	128-D int8; per pixel per year
OlmoEarth (Ai2)	S-1, S-2, Landsat-8 monthly + static maps (OSM, WorldCover, CDL, SRTM, canopy, WorldCereal)	2.56 km tile (256 px @ 10 m); 1–8 px tokens (10–80 m)	≤12 monthly steps over 1 year, each step tokenised; temporal masking	768-D (Base) / 1024-D (Large) per patch per step + pooled vector
TerraMind (IBM/ESA)	S-2 L2A/L1C, S-1 GRD/RTC, DEM, RGB, NDVI, LULC, captions, geolocation	2.24 km tile (224 px @ 10 m); 16 px = 160 m tokens, 14×14 grid	Single co-registered timestamp	196 × 768-D tokens per modality; merged 768-D; generated modalities
Prithvi-EO 2.0 (IBM/NASA)	HLS v2 at 30 m, 6 bands (blue, green, red, NIR, SWIR1, SWIR2)	6.72 km tile (224 px @ 30 m); 480 m (300 M) or 420 m (600 M) tokens	4 dated frames, 1–6 months apart, tubelet depth 1; single frame allowed	1024-D (300 M) / 1280-D (600 M) per token per frame
Granite-Geo-Ocean (IBM/STFC/PML)	Sentinel-3 OLCI L2, 16 bands + SLSTR SST (17 ch) at 300 m	12.6 km tile (42 px @ 300 m); 2 px = 600 m tokens, 21×21 grid	Single acquisition (num_frames = 1)	512-D per token; regression heads for chl-a, primary production

Summary II — what each project hands you

Weights, precomputed embeddings, licence, ocean coverage, and the intended way to use it

Model	Open weights / code	Precomputed embeddings	Licence	Ocean?	Intended use / evaluation style
AlphaEarth	No — model and code unreleased	Yes — global 10 m annual 2017–2024 in Earth Engine + GCS	CC-BY 4.0 (data)	Coastal/inland water only	Frozen embeddings + linear/shallow heads on sparse labels; 'featurization' replacement
TESSERA	Yes — weights CC0, code MIT, full pipeline	Yes — 10 m annual 2017–2025 via GeoTessera (tile-sampled)	CC0 / MIT	Land only (water masked)	Frozen per-pixel embeddings + MLP/conv heads; label-efficient mapping
OlmoEarth	Yes — 4 sizes on Hugging Face	On demand — Studio exports for any AOI; no global archive	OlmoEarth Artifact License (use restrictions)	Not in pretraining sample	Both: kNN/linear probes and full fine-tuning; platform for NGOs/agencies
TerraMind	Yes — tiny/small/base/large + tokenizers	No	Apache 2.0	Land-stratified TerraMesh	Fine-tuning via TerraTorch; any-to-any generation; edge/onboard variants
Prithvi-EO 2.0	Yes — 300 M / 600 M, $\pm$ TL	No	Apache 2.0 (HF) / MIT (GitHub) — verify	Sea-only tiles removed	Full fine-tuning of encoder + decoder (TerraTorch); GEO-Bench
Granite-Geo-Ocean	Yes — HF weights + GitHub notebook	No	Apache 2.0	Yes — open ocean, surface optical + SST	Fine-tune regression heads on ~100–200 in-situ points; ocean-colour biogeochemistry

Who embeds the ocean — and how deep?

Representation models whose pretraining contains sea pixels: the published ones stop at the surface; Earth 2 is the one with the interior (checked Aug 2026)

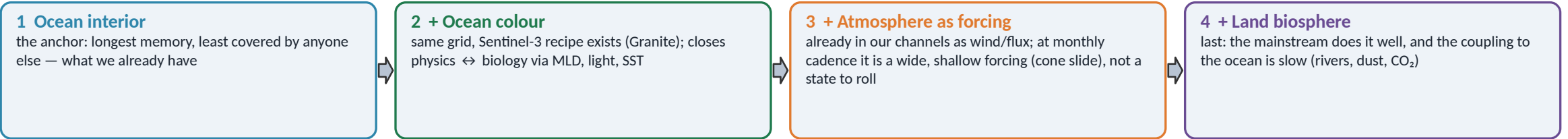
Model (org, year)	Ocean data in pretraining	Open ocean? depth?	Stencil	Embedding / availability	What it is for
Earth 2 codec (ours, 2026)	Gridded ocean state at $\frac{1}{4}^\circ$ / 1° : T and S at depth from Argo (plus pre-Argo gaps marked 'never measured'), SSH, currents, wind and surface forcing; 14–25 channels, monthly or pentad means	Yes — open ocean, surface to ~2000 dbar interior	3×3 patch per channel token; one pixel-month snapshot; 24-month history in stage 2 with a 222–4444 km stencil	64-D per pixel-month (128-D variant training); weights + embeddings ours to release	Ocean-state field prediction; frozen-embedding probes to AMOC transport (RAPID, MOVE)
Granite-Geospatial-Ocean (IBM • STFC • PML • Exeter, 2025)	Sentinel-3 OLCI 16 bands + SLSTR SST, 300 m; 512 k tiles 2017–2021, balanced over 83 Longhurst provinces	Yes — surface optical only	42 px = 12.6 km tile; 600 m tokens; single acquisition	512-D per token; Apache 2.0 weights on HF + TerraTorch notebook	Ocean-colour biogeochemistry: chl-a, primary production (surface optical)
OceanSAR-1 / -2 (Galeio, Apr 2025 / Jan 2026)	Sentinel-1 Wave-Mode SAR vignettes over open sea, ~12 M images 2015–2024, VV, downsampled to 50 m	Yes — surface roughness only	256×256 px @ 50 m = 12.8 km; single acquisition; DINO / DINOv2 with dynamic dataset curation	ResNet50 2048-D • ViT-S/B; v2 ViT 384-D; v1 Apache 2.0 on HF	Sea-state: TenGeoP geophysical classes, wave height (RMSE 0.63–0.72 m), wind speed (RMSE $\sim 1.4 \text{ m s}^{-1}$), icebergs
Copernicus-FM / Copernicus-Pretrain (TUM, 2025)	S-1, S-2, S-3 OLCI (300 m), S-5P (1 km), DEM on a 0.25° ERA5 grid; 18.7 M images, ~312 k grid cells	Partial — 'whole land surface and near-land ocean'	per-modality patches inside 0.25° cells; dynamic hypernetwork accepts any spectral/non-spectral input + metadata	ViT encoders; code, data and weights released (github zhu-xlab); Copernicus-Bench, 15 tasks	Unified multi-sensor EO backbone; coastal seas incidental, not a target
Weather/climate FMs (Prithvi WxC, Aurora, ClimaX, AtmoRep)	Reanalysis grids (MERRA-2, ERA5) whose surface fields exist over ocean cells; Aurora has an ocean-wave fine-tune (HRES-WAM)	Ocean cells present, but as reanalysis background — SST/ocean state not the learning target	0.25° – 0.5° global grids; 1–2 lead-time steps	Open weights (varies); no ocean embedding product	Atmospheric forecasting; wave forecasting (Aurora)
Location encoders (SatCLIP, GeoCLIP)	Coordinates $\leftrightarrow$ imagery/photos; SatCLIP samples S-2 globally, GeoCLIP uses geotagged photos	Effectively no — sea coordinates unverified or absent	point location $\rightarrow$ vector	Open	Geographic priors, not ocean state

Verdict. Two purpose-built ocean encoders exist besides ours — both surface, both single-snapshot, one optical (colour + SST) and one radar (sea state). Everything else masks the sea out or carries it as reanalysis background; forecast-only systems (GLONET, Xihe, WenHai, AI-GOMS, Samudra) learn ocean dynamics but publish forecasts, not reusable embeddings. Earth 2 is, to our knowledge, the only embedding model whose inputs include the ocean interior through time.

Four interacting spheres — who embeds which

Atmospheric weather · ocean weather (incl. the interior) · ocean biosphere · land biosphere. Nobody represents two of them jointly today.

Sphere	Who embeds it today	Native cadence · memory	Observing geometry	Couples to the others through
a) Atmospheric weather	Forecast FMs: Prithvi WxC, Aurora, ClimaX, GraphCast/AIFS/Pangu (states, not reusable embeddings). AlphaEarth ingests ERA5-Land only as an input.	hours–days · ~10 d predictability; slow modes (ENSO, NAO) months	dense gridded reanalysis; radiances and stations upstream	wind stress, heat & freshwater flux → ocean; precipitation, radiation → both biospheres
b) Ocean weather	Surface only: OceanSAR (sea state), Granite (SST as one band), AIFS-ocean 2026 (SST/SSS/SSH/currents, forecast). Interior: Earth 2 — the only embedding with T, S at depth.	days (surface) to decades (interior) · mixed layer months, interior years; top 2.5 m hold the heat of the whole atmospheric column	gridded SST/SSH from satellites; sparse Argo profiles below; nothing under ice	SST → atmosphere; mixed-layer depth, nutrients, light → ocean biosphere; carbon uptake → land via CO ₂ (slow)
c) Ocean biosphere	Granite-Geospatial-Ocean (colour → chl-a, primary production); carbon-sink mappers SOM-FFN, CSIR-ML6; CANYON-B for BGC-Argo — all task models, one FM	days–seasons · blooms weeks, seasonal cycle, decadal sink trends	cloud-gapped optical swaths (Sentinel-3, MODIS); BGC-Argo sparse; pCO ₂ ship tracks	phytoplankton ← MLD, SST, wind (physics); CO ₂ flux → atmosphere; export → interior carbon
d) Land biosphere / surface	The mainstream: AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 — all fold a year, none touch the sea	seasonal–annual · phenology, multi-year land-use memory	dense 10–30 m tiles, frequent revisits; LiDAR, SAR	runoff, dust, nutrients → coastal ocean; evapotranspiration, albedo → atmosphere



Three design consequences. (i) Memories differ by orders of magnitude, so each sphere needs its own (v, τ) cone and the coupling variables (SST, fluxes, chlorophyll–light–MLD) are the L-shaped unions. (ii) Biospheres run on the calendar, physics on the clock: a joint model needs period tokens (pixel-year, slide on the TESSERA arm) for c) and d) and dated-state tokens for a) and b). (iii) The observing systems differ in geometry — grids, sparse profiles, cloud-gapped swaths, 10 m tiles — and our masked-vs-never-measured token distinction is the primitive that lets one encoder take all of them.

Head-to-head I — frozen embeddings, one protocol

OlmoEarth paper, Table 2: kNN (k = 20, cosine) for single-image classification, linear probe for segmentation and multi-temporal tasks. Five of our six appear.

Model (frozen encoder)	BigEarthNet μ F1	So2Sat acc	EuroSAT acc	CropHarvest-Togo S1+S2 acc	m-cashew mIoU	SA-crop mIoU	PASTIS (S2) mIoU	MADOS mIoU	Sen1Floods11 mIoU
OlmoEarth Base (89 M)	62.4	67.7	94.7	82.0	32.3	28.9	50.6	67.2	79.2
OlmoEarth Large (308 M)	62.0	68.2	96.3	79.7	30.9	28.5	51.8	66.4	79.8
TerraMind Base	63.9	46.7	85.6	77.5	46.0	30.4	40.9	66.0	78.7
TerraMind Large	63.9	47.4	90.0	75.2	50.4	31.2	41.3	67.5	78.4
Prithvi-EO 2.0 300M	51.6	34.7	82.2	–	46.8	24.7	37.2	50.9	–
Prithvi-EO 2.0 600M	51.0	34.0	81.2	–	45.8	26.6	37.5	52.2	–
TESSERA (precomputed)	–	–	–	81.0	–	–	–	–	–
Galileo Base	58.3	55.7	92.8	77.5	28.9	25.3	39.6	68.4	79.4
DINOv3-Sat 7B	61.6	50.1	91.3	–	54.1	31.7	26.3	59.7	–
Panopticon Base	64.9	60.5	95.2	76.5	32.7	27.3	30.2	66.1	78.0
Copernicus-FM Base	64.6	50.3	84.7	70.6	32.2	28.4	32.1	63.9	77.6
CROMA Large	59.2	48.2	85.8	81.0	27.0	30.4	42.7	66.4	78.8

Reading the table. Green = best in column (all rows). Shaded rows are the models in this deck; grey rows are context. On frozen features OlmoEarth leads the classification columns (So2Sat, EuroSAT, Sen1Floods11, PASTIS), TerraMind leads BigEarthNet and the cashew/SA-crop segmentation, and Prithvi-EO 2.0 trails on every classification column but is competitive on m-cashew. TESSERA appears once (CropHarvest-Togo, 81.0 — within 1 pp of OlmoEarth Base). AlphaEarth is not in this table (its embeddings are annual and 10 m, so the authors compared it separately — next slides). Caveat: authored by the OlmoEarth team.

Head-to-head II — fine-tuning and PANGAEA

Top: OlmoEarth Table 3 (full fine-tuning, encoder unfrozen after 20 % of epochs). Bottom: TerraMind Table 6 (PANGAEA, frozen encoder + UPerNet, mIoU).

Fine-tuned — accuracy / mIoU (OlmoEarth paper, Table 3)

Model (fine-tuned)	BigEarthNet	So2Sat	EuroSAT	m-cashew	SA-crop	PASTIS	MADOS	Sen1Floods11
OlmoEarth Base	72.0	68.6	98.7	79.8	39.6	64.3	77.8	79.8
OlmoEarth Large	72.4	68.1	98.5	80.6	40.8	66.3	81.8	79.8
TerraMind Base	72.6	66.1	97.6	80.9	39.2	59.9	73.2	79.5
TerraMind Large	74.0	65.4	97.8	81.3	41.1	60.9	71.5	79.5
Prithvi-EO 2.0 600M	70.6	64.7	96.8	81.1	38.8	58.6	69.3	-
Galileo Base	69.2	64.7	97.8	78.8	35.7	61.2	71.9	79.7
Copernicus-FM	71.3	66.8	98.5	78.7	33.6	54.6	66.0	78.6
OlmoEarth Base, random init (no pretraining)	61.0	48.9	80.3	43.0	27.5	43.9	45.6	77.0

PANGAEA — mIoU, frozen encoder + UPerNet (TerraMind paper, Table 6; Prithvi appears only as v1.0)

Model	HLS Burn Scars	MADOS	PASTIS-R	Sen1Floods11	FiveBillionPx	DynamicEarthNet	CropType S.Sudan	SpaceNet 7	AI4SmallFarms	Average	Avg rank
TerraMind Large	82.93	75.57	43.13	90.78	63.38	37.89	55.04	59.98	27.47	59.57	3.44
TerraMind Base	82.42	69.52	40.51	90.62	59.72	37.87	55.80	60.61	28.12	58.35	3.94
Prithvi-EO 1.0 100M	83.62	49.98	33.93	90.37	46.81	27.86	43.07	56.54	26.86	51.00	11.00
CROMA	82.42	67.55	32.32	90.89	51.83	38.29	49.38	59.28	25.65	55.29	6.61
DOFA	80.63	59.58	30.02	89.37	43.18	39.29	51.33	61.84	27.07	53.59	8.22
U-Net trained from scratch	84.51	54.79	31.60	91.42	60.47	39.46	47.57	62.09	46.34	57.58	4.89

Reading. Once fine-tuned, the gap closes: TerraMind Large and OlmoEarth Large split the columns (TerraMind: BigEarthNet, cashew, SA-crop; OlmoEarth: So2Sat, EuroSAT, PASTIS, MADOS) and Prithvi-EO 2.0 600M sits 1–3 pp behind on the classification columns but 8–13 pp behind on PASTIS and MADOS. The random-init row is the size of the pretraining effect itself (+10 to +37 pp). On PANGAEA, TerraMind is the only foundation model that beats a U-Net trained from scratch on average (59.6 vs 57.6); Prithvi-EO 1.0 loses to it. Prithvi-EO 2.0’s own GEO-Bench table (600M-TL best on 6 of 12 sets vs DOFA / Scale-MAE / DeCUR / Satlas) contains none of the other models here.

Head-to-head III — the two embedding products

TESSERA paper, Table 1 (frozen embeddings + light head, 100 % labels unless noted) · OlmoEarth paper, Table 7 · an independent Local-Climate-Zone study (Bern)

TESSERA vs AlphaEarth vs frozen RSFMs (TESSERA paper) — F1 / mIoU ↑ , RMSE ↓

Model	TreeSatAI-TS F1	Austrian crop F1 (30 % labels)	PASTIS-R mIoU	Austrian crop seg. mIoU	BioMassters RMSE (t/ha)	Borneo canopy RMSE (m)
TESSERA (128-D, frozen)	77.96	82.09	50.68	53.12	27.43	12.21
AlphaEarth (64-D, frozen)	76.90	56.36	51.08	25.70	29.59	16.11
Prithvi-EO (frozen)	65.86	–	34.09	16.25	41.12	19.71
Presto (frozen)	67.81	57.89	–	34.04	–	17.88
Galileo (frozen)	69.44	–	27.92	22.80	38.52	20.48
SkySense (frozen)	69.35	–	32.87	26.82	32.52	16.56
U-Net baseline (frozen features)	71.39	–	30.16	27.00	36.41	17.79

Reading. TESSERA beats AlphaEarth on 5 of these 6 tasks in TESSERA's own table (AlphaEarth edges PASTIS-R by 0.4 mIoU), with the largest margins on pixel-wise crop tasks where per-pixel time series matter most; both precomputed products beat every frozen ViT-style RSFM including Prithvi by wide margins. Against OlmoEarth, AlphaEarth wins only where OlmoEarth is also frozen (Nandi frozen+decoder); once OlmoEarth is fine-tuned it wins all five — and AlphaEarth cannot be fine-tuned. The independent Bern study puts the two products within 1 pp of each other and 2–5 pp above raw composites. AlphaEarth's own paper reports a 23.9 % average error reduction vs the next-best method over 15 evaluations (per-dataset values are charts only), losing on just one (LCMAP land-use change).

AlphaEarth vs OlmoEarth Base (OlmoEarth paper, Table 7)

Model · protocol	Nandi acc	AWF acc	Ecosystem acc	LFMC L1 ↓	Solar farm mIoU
AlphaEarth · kNN	55.6	81.0	60.6	–	–
AlphaEarth · frozen + decoder	66.0	75.9	61.2	23.1	77.5
AlphaEarth · full fine-tune	n/a	n/a	n/a	n/a	n/a
OlmoEarth · kNN	66.2	82.0	59.3	–	–
OlmoEarth · frozen + decoder	62.9	84.0	61.1	19.9	84.8
OlmoEarth · full fine-tune	82.2	86.0	62.4	17.9	86.7

Independent: LCZ mapping, Bern (arXiv 2606.20034)

Input to Attention U-Net	Accuracy	IoU
Sentinel-1 + 2 composites	0.87	0.77
AlphaEarth 64-D	0.89	0.81
TESSERA 128-D	0.90	0.82

Head-to-head IV — the ocean encoders

Granite-Geospatial-Ocean, Table 1 (5-fold CV RMSE, log₁₀ units) · OceanSAR-1, Tables 2–3 (Sentinel-1 wave-mode tasks)

Granite-Geospatial-Ocean — fine-tuned FM vs from scratch vs random forest (RMSE ↓)

Model	Chl-a RMSE log ₁₀ mg m ⁻³	Primary production RMSE log ₁₀ mgC m ⁻² d ⁻¹
OLCI + SST · foundation model, fine-tuned	0.14	0.39
OLCI only · foundation model, fine-tuned	0.16	0.39
OLCI + SST · same net from scratch	0.16	0.42
OLCI only · same net from scratch	0.16	0.43
Random forest on pixel values	0.16	0.40

± spreads: chl-a 0.03–0.10, PP 0.04–0.07 across folds — the FM gain (0.02 / 0.03–0.04) is inside the fold spread; the authors' stronger claim is label-efficiency (gain grows as labels shrink to 25 %) and spatial pattern (SSIM vs the operational L2 product: FM 0.88, scratch 0.82, decision tree 0.68). No comparison to OC4Me-type algorithms is given.

OceanSAR-1 — kNN on frozen features vs other SAR FMs (acc ↑ , RMSE ↓)

Model	TenGeoP acc %	Wave height RMSE (m)	Wind speed RMSE (m s ⁻¹)
OceanSAR-1 ViT-B/8	83.6	0.63	1.37
OceanSAR-1 ViT-S/8	82.1	0.64	1.38
OceanSAR-1 ResNet50	75.5	0.75	1.62
SoftCon ViT-B/14	74.8	1.01	2.08
CROMA ViT-B/8	65.4	0.78	1.95
DOFA ViT-L/16	63.4	0.95	2.43
MoCo ResNet50	60.9	0.83	1.98

With linear-probe / LoRA fine-tuning OceanSAR-1 ViT-B/8 reaches 88.3 % / 0.54 m / 1.29 m s⁻¹ (MoCo-FT 86.5 / 0.77 / 1.80). Ocean-only pretraining beats land-trained SAR FMs by 9–23 pp on sea-state classification and cuts wind-speed error by 30–45 %.

Why no cross-model ocean table. The two ocean encoders measure different physics (optical colour vs radar roughness) against different in-situ truths, and neither has been run on a land benchmark or on each other's tasks. The only common denominator with the land models is the protocol — frozen features + small head, and 'FM vs same net from scratch' as the pretraining control — which is also the protocol of our attribution matrix. Our numbers (RAPID r 0.617 pixel / 0.672 patch vs raw 0.613 / 0.659) are the same kind of statement as Granite's 0.14 vs 0.16: a pretraining margin measured against a matched from-scratch control, and in both cases a small one.

Who wins where — a digest of the tables

Rankings restricted to the models in this deck, per protocol; every source table was written by one of the contestants

Protocol · metric family	Ranking among our models	Margin / evidence	Source & caveat
Frozen features · classification accuracy / μF1	OlmoEarth > TerraMind > Prithvi-EO 2.0	So2Sat 68 vs 47 vs 35; EuroSAT 96 vs 90 vs 82; BigEarthNet TerraMind 63.9 edges OlmoEarth 62.4	OlmoEarth Table 2 (own paper)
Frozen features · segmentation mIoU	split: TerraMind (cashew 50.4, SA-crop 31.2) · OlmoEarth (PASTIS 51.8, MADOS 67.2, floods 79.8) · Prithvi last	Prithvi-EO 2.0 competitive only on m-cashew (46.8)	OlmoEarth Table 2
Fine-tuned · accuracy / mIoU	TerraMind-L $\approx$ OlmoEarth-L > Prithvi-EO 2.0 600M	columns split 3/4 between the two leaders; Prithvi 1–3 pp behind on classification, 8–13 pp on PASTIS/MADOS; pretraining itself worth +10 to +37 pp (random-init row)	OlmoEarth Table 3
PANGAEA · mIoU, frozen + UPerNet	TerraMind-L 59.6 > TerraMind-B 58.4 > U-Net scratch 57.6 > ... > Prithvi-EO 1.0 51.0	only GFM above the from-scratch U-Net; Prithvi 2.0 not evaluated	TerraMind Table 6
GEO-Bench-2 · capability ranks (of 14)	Prithvi 600M-TL core rank 5 · TerraMind-L 6 · Prithvi 300M-TL 9 · TerraMind-B 11; TerraMind-L rank 1 on multi-temporal and multi-spectral	ranks only, no raw scores	GEO-Bench-2, arXiv 2511.15658 (independent); no AlphaEarth / TESSERA / OlmoEarth / Granite
Precomputed products · frozen + head	TESSERA > AlphaEarth on 5/6 TESSERA tasks; $\approx$ tie on Bern LCZ (0.82 vs 0.81 IoU); both $\gg$ frozen Prithvi	Austrian crop F1 82 vs 56; canopy RMSE 12.2 vs 16.1 m; PASTIS AlphaEarth +0.4	TESSERA Table 1 (own) · LCZ study (independent)
Product vs fine-tunable model	fine-tuned OlmoEarth > AlphaEarth on 5/5; frozen-vs-frozen mixed (AlphaEarth wins Nandi 66.0 vs 62.9)	Nandi 82.2 vs 66.0; solar farm 86.7 vs 77.5	OlmoEarth Table 7 (own)
Ocean · RMSE vs in-situ, FM vs scratch	Granite: 0.14 vs 0.16 (chl-a), 0.39 vs 0.42 (PP) · OceanSAR-1: 83.6 % vs 74.8 % best land-trained SAR FM	Granite gain inside fold spread; OceanSAR gain large	own papers; no shared task with land models

Two structural caveats. No benchmark contains all six models — the widest single table (OlmoEarth's) has five and omits AlphaEarth; the only independent multi-model source (GEO-Bench-2) has two. And no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem; the transferable lesson is the protocol — frozen probe vs fine-tune, and a from-scratch control at matched receptive field, which is what our attribution matrix already does.

What this means for our ocean representation work

Six models, two design axes, one gap

1 Axis 1 — pixel vs. patch

TESSERA proves a per-pixel encoder with zero spatial context is competitive on land; every other model buys context with a ViT patch grid. Our attribution matrix: single-pixel raw 0.613 vs codec 0.617; 3×3 raw 0.659 vs codec 0.672 (two seeds). Probe numbers move by 0.04–0.25 between seeds, so the neighbourhood's $\approx +0.05$ is at the edge of the noise band and pretraining's $\approx +0.01$ inside it — the former probably real, the latter unresolved.

2 Axis 2 — snapshot vs. period

TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our stage-1 codec is a snapshot of a monthly (or 5-day) mean and the month-block codec folds only ~ 6 pentads. The year-folded design is the untested axis — hence the proposed arm on the last slide.

3 Product vs. model

AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both weights and embeddings. For the ocean nothing precomputed exists — whatever we release is the first embedding-field product for the ocean interior.

4 The gap

Only two published encoders are trained on sea pixels — Granite-Geospatial-Ocean (colour + SST) and OceanSAR (sea-state radar) — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state (T, S, SSH, currents) through time — the territory our codec occupies.

Proposed arm: a TESSERA-style pixel-year codec

Fold the year, not the neighbourhood — the one TESSERA ingredient the programme has not yet tested

Why not a 1×1 spatial ablation Already done. The attribution matrix has $p = 1$ and 3×3 at both raw and codec: 0.613 / 0.617 vs 0.659 / 0.672 — single seed, inside the probe noise band (ml/CLAUDE.md §3b): read as parity. The temporal axis is unmeasured.

What TESSERA does that we don't One embedding per pixel-YEAR: a 4-layer temporal transformer + GRU pooling over ~40 dated observations of a single pixel, day-of-year embeddings, Barlow Twins objective (no reconstruction), int8 QAT. Our stage-1 z is one pixel-MONTH; stage 2 then runs over the monthly z 's.

Arm A — pixel-year codec Encoder over the 12 (or 24) pixel-months of one $\frac{1}{4}^\circ$ pixel: channel tokens carry month-of-year; $p = 1$ to isolate the temporal fold; emit one $z \in \mathbb{R}^{64}$ per pixel-year (or rolling 12-month window ending at t). Decoder queries (channel, Δmonth) up to ± 12 . Token count $\approx 12 \times (C+2) \approx 310$ at 24 channels — same order as the 282-token month-block codec, so it fits the existing recipe.

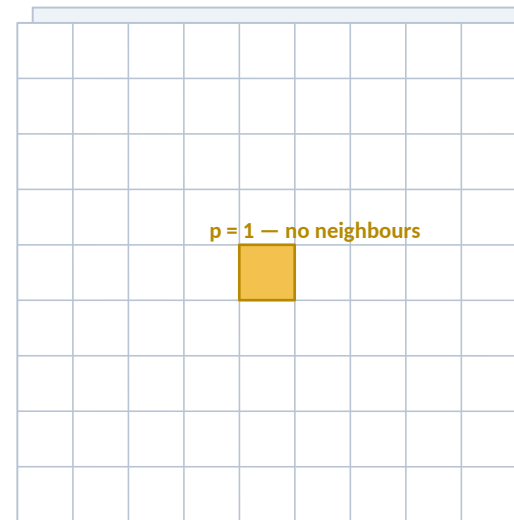
Arm B — objective swap Same encoder, Barlow Twins on two random date-subsamples of the same pixel-year (TESSERA's augmentation) instead of masked reconstruction. Tests whether a non-reconstructive objective keeps the deep-density detail the decoder had to be re-taught to read.

Controls (matched) Mean of the 12 monthly z 's (pooling control) · stage-2 over monthly z 's (current path) · raw 12-month stack into the same attention head (end-to-end control, as in the attribution matrix). Two codec seeds minimum — head numbers are never quoted from one.

Read-outs Probe ladder → RAPID r and RMSE (Sv); round-trip variance lost per channel; rolled skill under the corrected pool (MSSS per lead vs climatology and damped persistence; paper v8, 2 Sep 2026) with a year-folded z as the state. Success = beats the control beyond bootstrap intervals, $n \geq 3$ seeds.

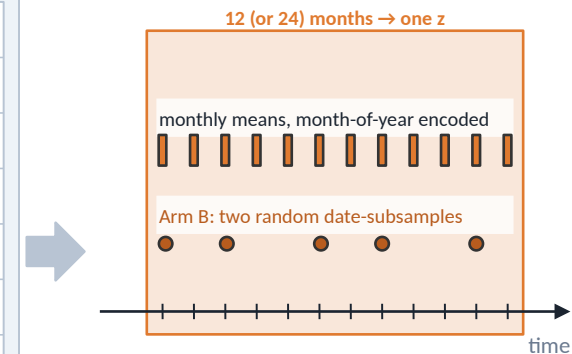
Stencil — what feeds one embedding

SPACE



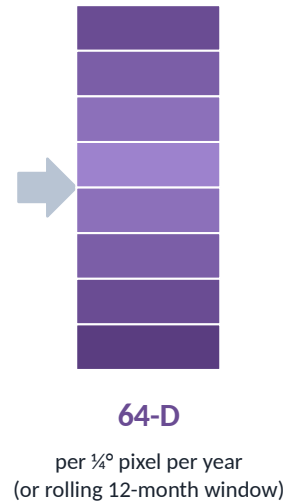
Output pixel: one $\frac{1}{4}^\circ$ cell (≈ 28 km).
Receptive field = the same cell; the 3×3 patch is deliberately switched off so only the time axis changes.
Sheets: the 24–25 ocean channels (T, S, SSH, wind, ...).

TIME



A period embedding of the ocean state, TESSERA-style, instead of a snapshot of one monthly mean.
Moves our point on the space × time map from 'single snapshot' to '1 year folded'.

OUTPUT

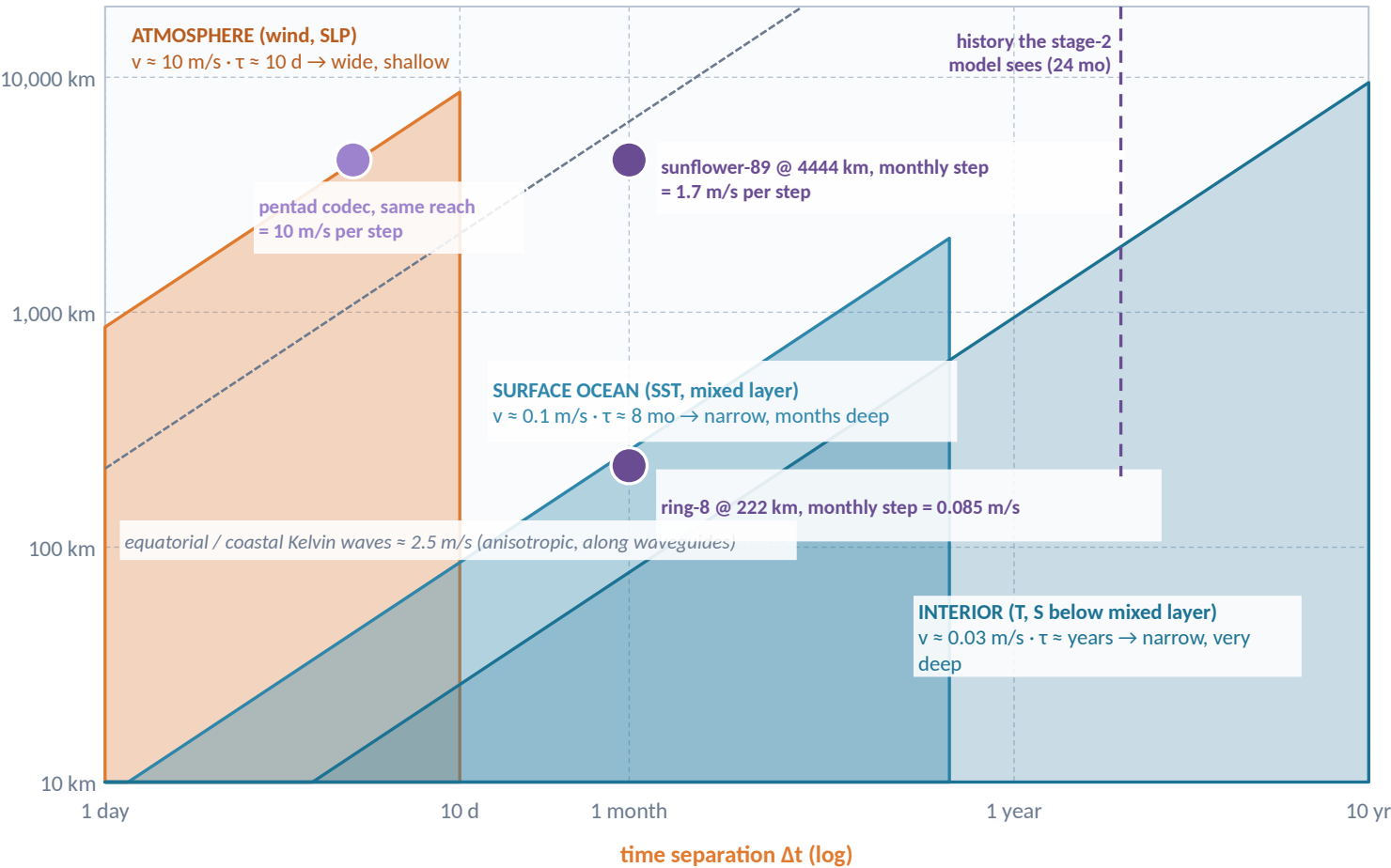


SUPERSEDED (2 Sep 2026). The pixel-year arm is the special case of the cone-native codec with zero spatial reach and a 72-pentad inner window ($r_{in} = 0$, $L_{in} = 72$). Slides 40–44 and E-069 replace it: the inner cone carries the temporal fold AND the near field, and the '28 km' here is only the $\frac{1}{4}^\circ$ cell (27.8 km at the equator, 21 km at 40° N), not a physical scale.

A velocity-specific dependency cone per channel

What can influence one pixel Δt ahead is bounded by how fast each process propagates (v) and how long it remembers (τ): $\Delta x \leq v \cdot \Delta t$, $\Delta t \leq \tau$

distance Δx (log)



The model

For a process p with signal speed v_p and memory τ_p , the state at $(x, t+\Delta t)$ can depend on the past only inside $C_p(\Delta t) = \{ (\Delta x, \ell) : \Delta x \leq v_p (\Delta t + \ell), \Delta t + \ell \leq \tau_p \}$. Δt is the forecast step (we predict $t+\Delta t$ from data up to now, t); ℓ is the lag of an input — it was observed at $t-\ell$, so it sits $\Delta t+\ell$ before the target ($\ell=0$ is the newest input). $\Delta x \leq v \cdot (\Delta t+\ell)$ is the domain of dependence (the CFL / light-cone argument); $\Delta t+\ell \leq \tau$ says that beyond the process's own predictability its history is noise, not signal.

The union rule

A channel c is forced by several processes, so its cone is the union: $C_c = \bigcup_{p \in P(c)} C_p$. Temperature is forced by the atmosphere (fluxes) and by currents (advection), so $C_{\text{SST}} = C_{\text{atm}} \cup C_{\text{ocean}}$ — wide-and-shallow plus narrow-and-deep, which is not a cone but an L-shape in $(\Delta x, \Delta t)$.

Where our stencils sit

A stencil is a cylinder — the same reach at every lag. ring-8 @ 222 km per month is an ocean-advection speed (0.085 m/s); sunflower-89 @ 4444 km is 1.7 m/s, the speed of the fast ocean waveguides and inside the atmosphere's cone at $\Delta t = 1$ month. The rolled forecast rebuilds the cone recursively (dependency-cone note, §1), so reach-per-step $\times$ horizon is what the target finally sees.

Notation t = now (last observed step) · Δt = forecast step · ℓ = lag of an input (observed at $t-\ell$; $\ell=0$ newest) · both in months here.

What the cone predicts — and how to test it on our stencil

Per channel group: dominant processes, the useful reach per monthly step, the useful history, and the falsifiable consequence

Channel group	Processes (v , τ)	Useful reach per monthly step	Useful history	Design consequence
10 m wind, SLP, fluxes	synoptic advection 10 m/s, $\tau \approx 5\text{--}10$ d; planetary modes (NAO, ENSO teleconnections) $\tau \approx$ months, forced by SST	causal reach in one month = the globe, but $\tau < \Delta t$: monthly wind is not predictable from its own history — only via slow ocean modes	1 step (its own); the ocean's history through SST	treat as wide, shallow forcing: large stencil, no depth. At pentad $\Delta t \approx \tau$ the wind becomes partly self-predictable — a pentad-only gain to look for
SST, mixed-layer T	surface currents 0.1–0.3 m/s (260–800 km/mo); air–sea forcing (atmospheric cone); mixed-layer memory $\tau \approx 3\text{--}12$ mo	advective 300–800 km, plus the atmospheric cone via forcing $\approx$ synoptic correlation length 1,000–3,000 km	several months (re-emergence up to a year)	union: needs BOTH the wide wind stencil and months of history — the one group that justifies the full cylinder
Interior T, S (below ML)	gyre advection 0.02–0.1 m/s (50–260 km/mo); midlatitude Rossby 0.02–0.05 m/s; $\tau \approx$ years–decades	50–300 km — a 3×3 patch at $\frac{1}{4}^\circ$ already covers it; 4444 km is 15–90 $\times$ too wide for one step	24 months and more; skill should keep rising with history	narrow, deep: small reach, long history. Reach can be cut with no loss — the concrete answer to the open question in the cone note (§5)
SSH	baroclinic Rossby 0.03 m/s (mid-lat) $\rightarrow \sim 0.3$ m/s (10°); Kelvin / coastal waves 1–3 m/s along waveguides; barotropic adjustment in hours (averaged out at monthly Δt)	interior: hundreds of km; along coasts and the equator: thousands of km in one step	months	anisotropic: an along-boundary / along-equator arm on the stencil, narrow elsewhere — a shape the sunflower approximates by brute force
Transport target (RAPID 26°N)	thermal wind = east–west density difference across the basin; set by interior Rossby adjustment (yrs) and by fast boundary waves (weeks)	the whole section, both boundaries	years for the interior arm; the current step for the boundary arm	the read-out must see the whole section, not its spatial mean, which averages away the east–west contrast: the target's cone is the section (mechanism; the measured pooled-vs-unpooled gap is not quotable as evidence)

Four falsifiable predictions (each an ablation we can run)

- 1 Group-specific reach: cutting reach to ~ 500 km for interior T/S slots costs no rolled skill (MSSS per lead); cutting it for wind/SST slots does.
- 2 Group-specific history: wind slots beyond lag 1 add nothing; interior slots gain from 24 $\rightarrow$ 48 months.
- 3 A cone-shaped stencil (reach $\propto \Delta t + \text{lag}$) matches the 4444 km cylinder at a fraction of the slots.
- 4 A pentad stage-2 will show one-step wind skill the monthly one cannot, because $\Delta t \approx \tau_{\text{atm}}$ — untested: the earlier pentad rolls were withdrawn with the paper reset.

Where the simple picture needs care

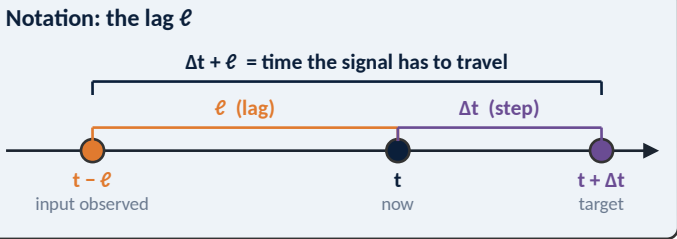
v is the fastest relevant signal speed, not the mean flow — waves (Rossby, Kelvin, barotropic) outrun advection. τ is predictability, not causality: the cone is truncated by memory, so wind's 'large area' is bounded by $v \cdot \tau$ ($\sim 4,000\text{--}8,000$ km), not by $v \cdot \text{horizon}$. Cones are tilted, not symmetric: upstream along the flow, westward for Rossby waves, along boundaries for coastal waves. Time-averaging to monthly means turns anything with $\tau < \Delta t$ into spatially-correlated forcing — the atmosphere is a stochastic boundary condition at our cadence, not a dynamical state. Eddy diffusion adds only $\sqrt{K \cdot t} \approx 50$ km/month at $K \approx 10^3 \text{ m}^2 \text{ s}^{-1}$ — negligible at $\frac{1}{4}^\circ$.

Worked example — the surface current at one pixel

Target: (u, v) at a $\frac{1}{4}^\circ$ pixel P , one month ahead. Which drivers, how each one moves the current, and what (v, τ) it carries. Live version: the Dependency-Cone Explorer artifact.

Driver	Mechanism → current	How it enters the stencil	v	τ	L_{corr}	Useful lags ($\Delta t = 1$ mo)	Reach: lag 0 → last useful lag
Wind stress	Ekman transport & surface drift — responds within ~1 day, locally	direct forcing; $\tau < \Delta t \Rightarrow$ lag 0 only, not itself predictable	10 m/s	10 d	1,500 km	lag 0 only	10,000 (cap) km
Air temperature / heat flux	net heat flux → mixed-layer T → density gradient → geostrophic current	fast driver integrated by a slow medium: inherits $\tau_{\text{ML}} \approx 6$ mo and, for old lags, ocean advection speed	10 m/s (atm.) 0.15 m/s (ocean)	6 mo	1,500 km	0 – 5 (6 of 24)	1,889 → 3,833 km
Ocean surface T (mixed layer)	horizontal density gradient → thermal wind	via the gradient $\Rightarrow$ neighbours needed ($\geq 3 \times 3$); advected at 0.15 m/s	0.15 m/s	8 mo	—	0 – 7 (8 of 24)	389 → 3,110 km
Interior T, S (below ML)	thermal wind integrated from depth; Rossby adjustment	via the gradient; 0.03 m/s $\Rightarrow$ 78 km per month — the 3×3 patch covers lag 0	0.03 m/s	5 yr	—	0 – 23 (24 of 24)	78 → 1,866 km
SSH	pressure gradient → barotropic geostrophic current	via the gradient; Rossby 0.03 m/s westward + Kelvin arm 2.5 m/s along coasts (anisotropic)	0.03 m/s	1 yr	—	0 – 11 (12 of 24)	78 → 933 km
Own history $(u, v \text{ at } P)$	momentum persistence, mesoscale eddies (westward drift ~0.03 m/s; eddies live 4+ months but pass through a fixed pixel in weeks–months)	direct; eddy memory ≈ 3 months	0.15 m/s	3 mo	—	0 – 2 (3 of 24)	389 → 1,166 km

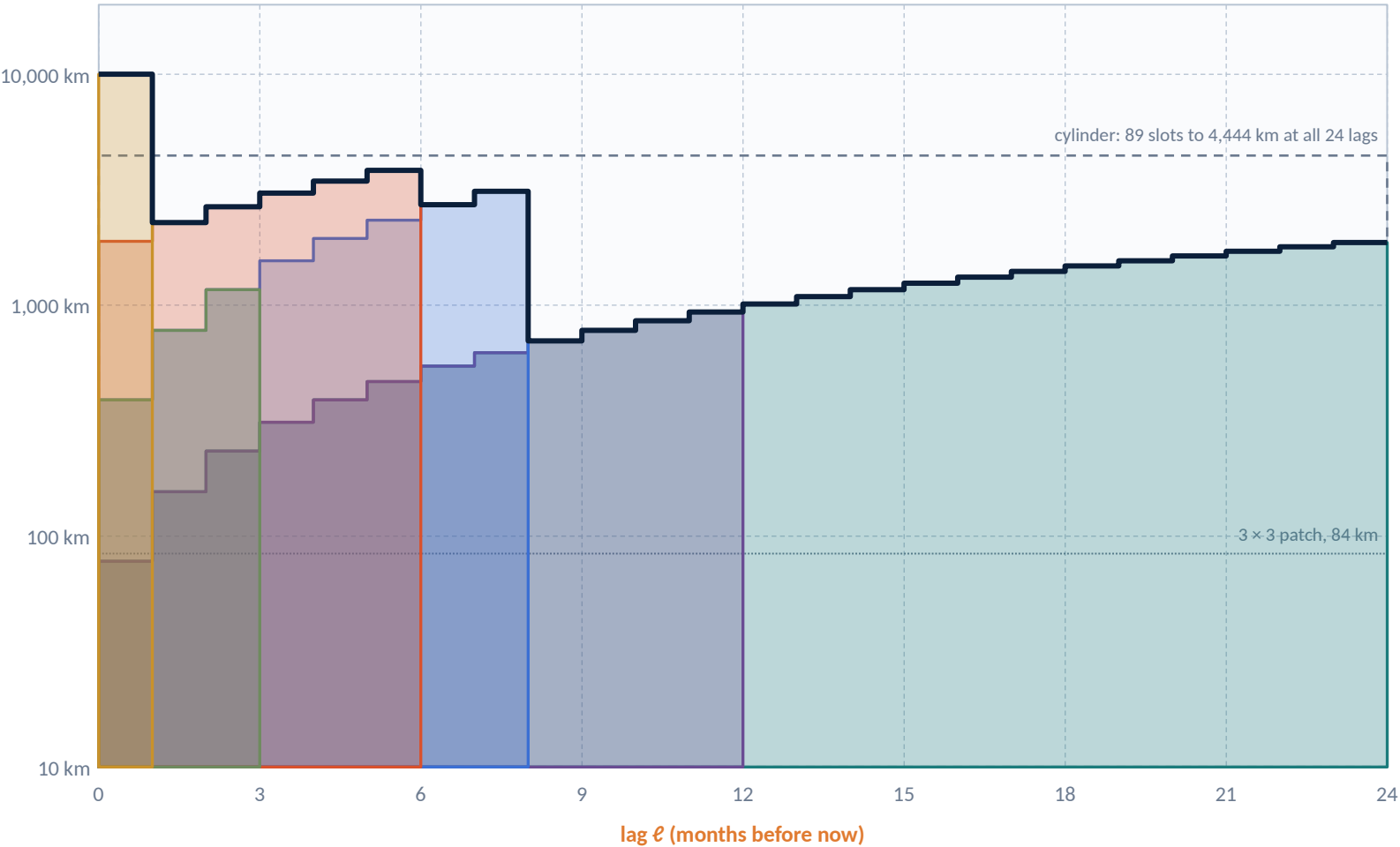
Two rules do all the work. $\text{reach}(\ell) = \min(\max(v \cdot (\Delta t + \ell), L_{\text{corr}}), 10,000 \text{ km})$ — a signal observed ℓ before now has $\Delta t + \ell$ to travel; a field is never less coherent than its own correlation length. A lag is useful iff $\Delta t + \ell \leq \tau$ (lag 0 is always kept). One exception: a fast driver acting through a slow medium (air temperature → heat flux → mixed layer) adds instead of maxing — $\text{reach} = L_{\text{corr}} + v_{\text{ocean}} \cdot (\Delta t + \ell)$ — because the flux pattern is laid down over L_{corr} and then carried away. Wind is the extreme case: at monthly cadence its memory is shorter than the step, so it enters as one wide sheet of forcing at lag 0 and nothing older carries information about next month's wind. Air temperature is the subtle case: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it delivered for months and carries it along at 0.15 m/s — so the channel inherits the ocean's τ and v .



Step 1 — each driver becomes a wedge in (lag, distance)

x: how long before now the input was observed · y: how far from P it may sit and still matter (log). Bold outline = the union the stencil should follow.

distance from P (log)



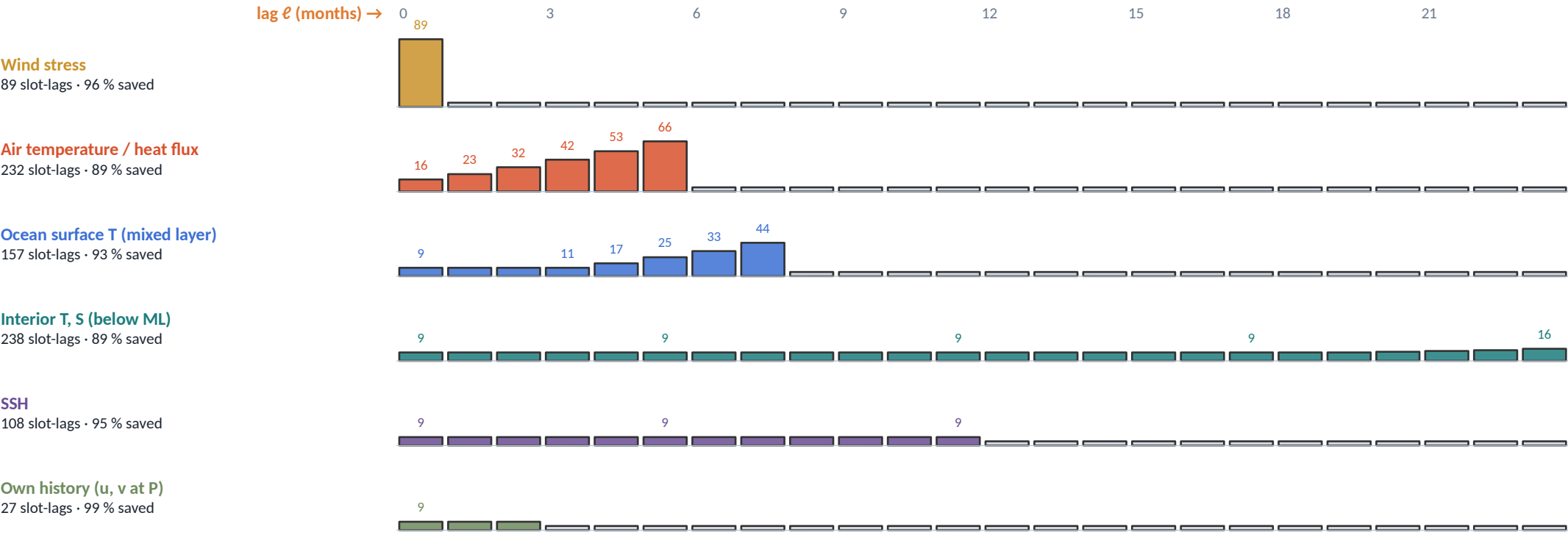
Drivers ($\Delta t = 1$ month)

- Wind stress**
lag 0 only · reach capped at 10,000 km
- Air temperature / heat flux**
lags 0–5 · reach 1,889 → 3,833 km
- Ocean surface T (mixed layer)**
lags 0–7 · reach 389 → 3,110 km
- Interior T, S (below ML)**
lags 0–23 · reach 78 → 1,866 km
- SSH**
lags 0–11 · reach 78 → 933 km
- Own history (u, v at P)**
lags 0–2 · reach 389 → 1,166 km

Reading. The union is an L: a tall thin spike at lag 0 (the atmosphere, wide but instant), a shoulder out to ~6–8 months (mixed-layer T and the integrated heat flux, 2–4,000 km), then a long low tail to 24 months (interior T/S at $\leq 1,900$ km). The cylinder over-covers the tail by 2–50 $\times$ and cannot cover the lag-0 spike at all.

Step 2 — the union becomes a per-channel, lag-dependent stencil

Slots needed at each lag = $89 \cdot (\text{reach} / 4,444 \text{ km})^2$, floored at the 9-slot 3×3 patch and capped at 89. Same 89-slot sunflower, different footprint per channel and lag.



851

cone-shaped slot-lags for the six channels

vs

12,816

for the cylinder ($89 \times 24 \times 6$)

→

93.4 % fewer

Bar height = slots needed at that lag (max 89). Grey stubs = lags outside the driver's memory: dropped. Numbers label the first lag, every sixth, and the first few changes.

What the shapes say. Wind is one full-width sheet at lag 0 and nothing else. Air temperature is a wide sheet for six lags (the mixed layer's memory). Surface T is a true cone: 9 slots at lag 0 growing to 44 at lag 7. Interior T/S is a thin 24-lag column at the 9-slot floor for a year and a half before it widens. SSH is a thin 12-lag column (plus the coastal arm the isotropic count ignores). Own history is three short lags. The cylinder spends 2,136 on each of them; the budget frees ~93 % of the slots for the ones that matter — or lets us afford 48-month history for the interior channels at no extra cost.

Step 3 — the physics behind each wedge

The stencil fixes the support; attention learns the weights. Four ablations on this target decide whether the support is right.

Ekman — wind → drift

Surface stress τ drives a transport $\approx \tau / (\rho f)$ at 90° to the wind within ~ 1 inertial period (≈ 1 day at 30°). Local, fast, shallow (tens of m). At monthly cadence this is same-step forcing: the wind of month $t+1$ is unknown, the wind of month t is already in the ocean state.

Thermal wind — T, S gradients → shear

$\partial u / \partial z = (g / (f \rho_0)) \partial \rho / \partial y$: horizontal density gradients set the vertical shear of the geostrophic current, integrated from depth. A gradient does not exist inside one pixel — this is why the 3×3 patch is the floor for T/S channels and why heat flux matters through its spatial pattern, not its local value.

Pressure gradient — SSH → surface geostrophy

$u_g = -(g/f) \partial \eta / \partial y$. SSH anomalies arrive as slow westward Rossby waves in the interior (0.03 m/s) and as fast Kelvin/coastal waves along boundaries (2.5 m/s): a thin column with one long arm — the one wedge the isotropic count under-states.

Advection & eddies — own history

Momentum persists and mesoscale eddies drift westward at ~ 0.03 m/s. Individual eddies live 4 months and more (Chelton et al. 2011), but at a fixed pixel the velocity decorrelates in weeks to a few months because they pass through — hence $\tau \approx 3$ months for own history. Upstream cells at 0.15 m/s $\times (\Delta t + \ell)$ carry what will arrive. Short cone, tilted upstream.

What the stencil fixes vs what attention learns

The cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only two physical facts per driver: how fast its signal travels and how long it is predictable. Everything else — which upstream cell matters this month, how strongly the SSH gradient projects onto u , whether the heat-flux pattern is the one that tilts the density field — is left to the attention weights inside the support. A support that is too small cannot be learned around; one that is too large costs slots and lets noise in. The four ablations to the right test whether the support is right, run with the roll battery under the corrected window-scope pool (paper v8, 2 Sep 2026) — never with the retired 'corridor AUC' of the contaminated heads.

The four ablations, specialised to this target

- 1 Interior T/S slots at 500 km reach instead of 4,444 km: predicted no loss (their wedge never exceeds 1,900 km).
- 2 Wind and heat-flux slots restricted to lags 0–1: predicted no loss ($\tau_{\text{atm}} < \Delta t$); the same cut on surface T: predicted loss (its wedge runs to lag 7).
- 3 Cone-shaped stencil (851 slot-lags) vs cylinder (12,816): predicted equal rolled skill at 7 % of the slots — and headroom for 48-month interior history.
- 4 Pentad codec ($\Delta t = 5$ d $\approx \tau_{\text{atm}}$): predicted one-step wind skill that the monthly model cannot show, because wind's own wedge now has lags > 0 .

Two codec seeds minimum; head numbers are never quoted from one. Explore the parameters live in the Dependency-Cone Explorer artifact (same formulas, editable v , τ , L_{corr} , Δt , anisotropy).

Related work I — the dependency cone vs. the literature

Who has used 'speed × time' to size what a model sees, and who has not (survey Aug 2026)

Work	What it does	Shares with the cone idea	Does not
Courant, Friedrichs & Lewy 1928 — domain of dependence	Numerical PDE stability: the numerical stencil must contain the physical domain of dependence $\Delta x \leq v \Delta t$.	The origin of the argument; our ' $\Delta x \leq v \Delta t$ ' is exactly this	No notion of memory τ ; about stability, not information
OceanNet — Chattopadhyay et al. 2024, Sci. Rep. (arXiv 2310.00813)	Regional FNO for Gulf Stream SSH, 4–5-day mean lead, 80 M params; justifies dropping lateral boundary conditions: 'Rossby waves at 40°N $\approx$ 1 cm/s ... eddies travel only 30–120 km from the open boundary' over the horizon.	Explicit wave-speed × time argument in an ocean ML model	Used to shrink a boundary, not to shape a per-channel stencil; single process
PARADIS 'Learning to Advect' — Pereira et al. 2026 (arXiv 2601.21151)	Neural semi-Lagrangian layer: gathers features at the departure point $x_d = x - \Delta t \cdot u(x)$; stability becomes a Lipschitz condition on u instead of CFL; a 20-parameter NSL layer moves a tracer a 3-layer U-Net's receptive field cannot.	Input support set by velocity × Δt , learned per point — the anisotropic, tilted cone made operational	One transport field for all variables; no τ truncation; atmosphere only
ClimODE — Verma et al., ICLR 2024	Neural ODE with the continuity/advection equation; learns a velocity field, but reads inputs with fixed 3×3 convs plus a global-attention branch.	Advection as inductive bias	Local/global mix fixed, not sized by $v \Delta t$ per variable
Limited-area ML weather: Neural-LAM (Oskarsson 2023/24), stretched grid (Nipen 2024), Diffusion-LAM 2025	Boundary forcing zone of a fixed 10 grid nodes (~100 km) at every step, or a smooth global→regional mesh.	A boundary halo is a cone cut by the domain edge — our 'window edge' problem	Halo width is static; not derived from wind speed × lead time
Global ML weather: Keisler 2022 (9 GNN hops / 6 h), GraphCast (multi-mesh, 16 layers), Pangu (1/3/6/24 h chain), FuXi (0–5/5–10/10–15 d cascade)	Per-step reach is set by hop count or window size; longer horizons handled by bigger steps or horizon-specific models.	Implicit speed bound per step; growing the cone with lead time by changing Δt	Same reach for every channel; no receptive-field-vs-lead-time ablation published
Per-variable tokenisation: ClimaX 2023, Aurora 2024/25, Prithvi WxC 2024, Stormer	Each variable gets its own embedding / patch tokens, aggregated by attention.	Variable-specific processing as a design pattern	Variable-specific embedding, not variable-specific spatial or temporal extent
Ocean emulators: Samudra 2024/25 (1°, 19 levels, 5-day step, ConvNeXt, atmosphere prescribed), ACE2-SOM (mirror: ML atmosphere + slab ocean), Xihe, GLONET	One medium is treated as prescribed forcing at a cadence set by the other's dynamics.	'Atmosphere as forcing' at slow cadence — our monthly- Δt inversion in practice	Not stated as a τ -vs- Δt rule; no channel-specific stencils
Earth 2 dependency-cone note (16 Aug) + this deck	Cone recursion shown bit-equal to a full roll with NaN poisoning (a property of the roll code, not a skill number); reach 222 vs 4444 km per month covers 29.5 % vs 100 % of the world ocean after 12 months (geometry); now: per-process (v , τ) cones, union rule, channel-group stencils.	—	Open: the τ -truncation + union → per-channel stencil principle was not found stated anywhere in our search; it is a hypothesis to test (four ablations), not a result

Reading. The 'speed × time' half of the idea has two clear precedents — OceanNet's Rossby-speed boundary argument and PARADIS's semi-Lagrangian gather — and the 'one medium as forcing' half is standard practice in ocean emulators. What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil shape. Absence in a one-day search is not absence in the literature; treat it as 'not yet found', and cite OceanNet and PARADIS as the nearest neighbours.

Related work II — the four-sphere plan vs. the literature

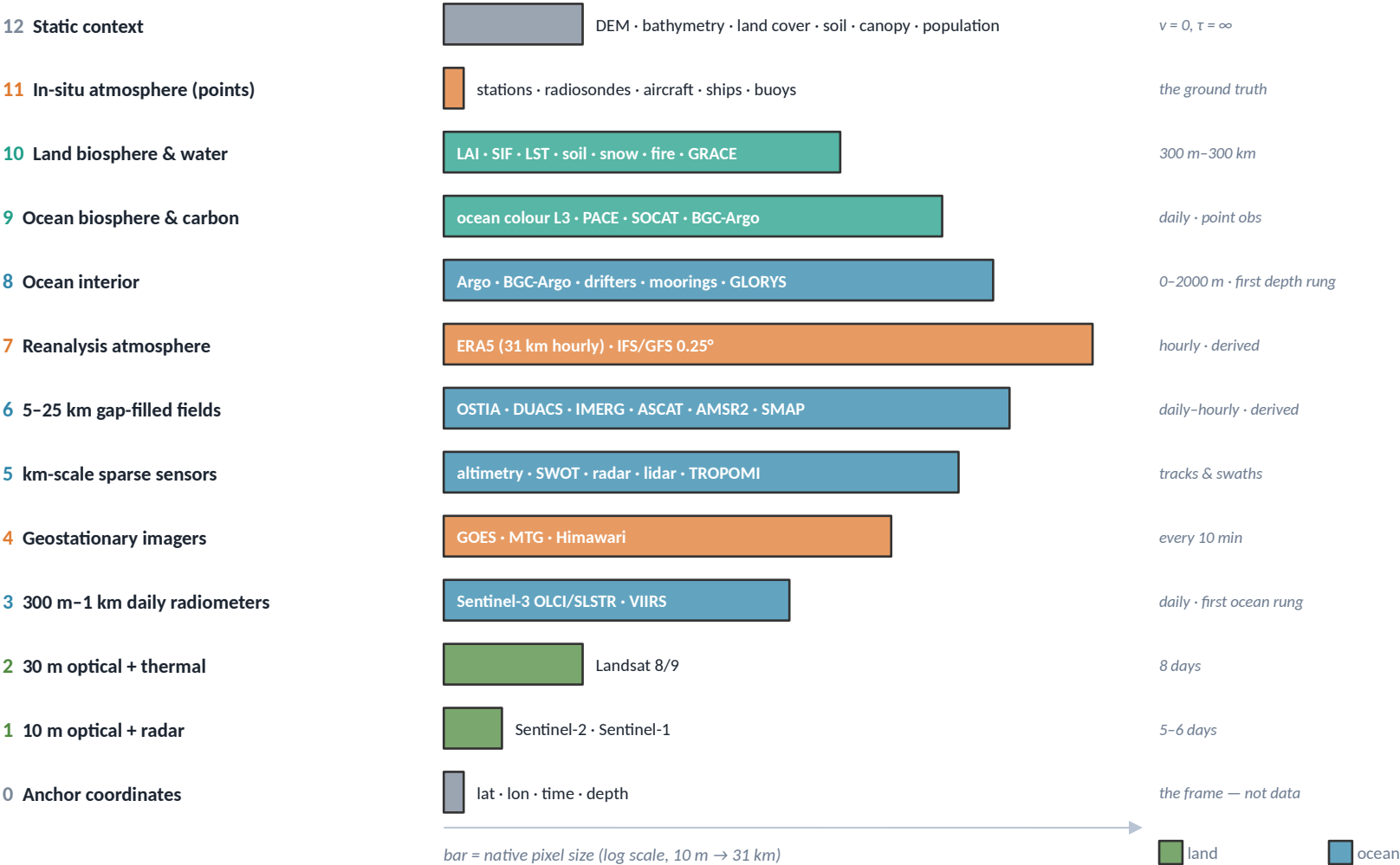
Coupled emulators, observation-native models, 'Earth-system' foundation models, biosphere ML, coupled data assimilation (survey Aug 2026)

Family • nearest works	What they do	Shares with our plan	Does not
Coupled emulators — Samudra → SamudrACE (Dheeshjith 2025; Duncan 2026), ACE2-SOM (2025), ACE2-NEMO (2026)	Samudra: 3-D global ocean emulator (T, S, currents at depth, 1°, 5-day step); SamudrACE couples it to Ai2's ML atmosphere ACE2. ACE2-SOM/NEMO couple an ML atmosphere to a slab / dynamical ocean.	SamudrACE is the closest structural analogue: an ocean-interior model first, atmosphere joined later	Emulators of a simulator, not embeddings of observations; no ocean colour, no land biosphere
AIFS surface ocean — Hahner, Zampieri et al., ECMWF 2026 (arXiv 2604.25559)	One encoder–processor–decoder forecasts atmosphere + SST, SSS, SSH, surface currents, waves, sea ice 'in a component-agnostic way'; slowly evolving ocean/ice fields get larger loss-scaling factors.	Joint multi-sphere state space; explicit handling of the timescale mismatch	Surface ocean only; forecast model, no reusable embedding; no biosphere
Aurora — Bodnar et al., Nature 2025 ('A foundation model for the Earth system'); Aurora 1.5	One pretrained 3-D Swin backbone, fine-tuned to air quality, ocean waves, cyclones.	One representation, many sphere-specific heads	Atmosphere-centric; no interior ocean, no two-way coupling, no biosphere. Note the title collision
Observation-native — Aardvark Weather (Allen et al., Nature 2025), GraphDOP (ECMWF 2024/25), 4DVarNet (Fablet et al. 2023/24)	Aardvark/GraphDOP learn forecasts or a latent state directly from raw observations, skipping reanalysis; 4DVarNet learns a variational solver that fuses sparse altimetry + SST into gap-free surface fields.	The recipe for taking Argo profiles, swaths and tracks natively — what 'combine all relevant data' needs	Atmosphere (Aardvark, GraphDOP) or 2-D surface ocean (4DVarNet); no Argo-depth encoder found
'Earth-system' FMs — ESFM (Ozdemir et al. 2026, arXiv 2605.00850), Copernicus-FM (2025), Prithvi family (EO, WxC, Surya), AlphaEarth (2025)	ESFM: Aurora-style backbone with per-variable tokens over ERA5, CMIP6, sparse MODIS and station data with missingness handling. Copernicus-FM unifies Sentinel sensors incl. S-5P. Prithvi: separate sphere-specific FMs. AlphaEarth: embedding field with ERA5-Land as an input.	Heterogeneous sources in one embedding; missingness as a first-class token (ESFM) — same primitive as our masked / never-measured tokens	All atmosphere- or land-centric; none has the ocean interior or the ocean biosphere as a target
Ocean biosphere / carbon ML — SOM-FFN (Landschützer), CSIR-ML6 (Gregor 2019), pCO ₂ -Residual (Bennington 2022), CANYON-B (Sauzède 2017), Granite-Geospatial-Ocean (2025)	Data-driven surface-pCO ₂ and carbon-sink maps from SST, SSS, chl, MLD; NN inference of nutrients/carbonate from T, S, O ₂ for BGC-Argo; one ocean-colour FM.	Exactly the physics → biology couplings step 2 of our sequence would learn end-to-end	Task models with hand-chosen predictors; no shared representation with the physical state
Coupled data assimilation — Penny & Hamill 2017 (BAMS); Penny et al. 2019 (JAMES); ECMWF CERA	The canonical statement that atmosphere (days) and ocean (months–years) memories differ by orders of magnitude; strongly vs weakly coupled DA; production systems still weakly coupled.	The multi-timescale problem is known and hard; our cone-per-sphere is a representation-side answer to it	Assimilation, not learned representation
Platforms & twins — NVIDIA Earth-2, Destination Earth (ECMWF/CMCC/ESA), EU Digital Twin of the Ocean (EDITO)	Generative km-scale weather tooling (NVIDIA); physics-based coupled km-scale twins (DestinE); ocean data/model platform (EDITO).	Same four-sphere integration goal at production scale	Not learned embeddings; NVIDIA 'Earth-2' is a name collision we must disambiguate

Reading. The nearest lineage on the physics side is Samudra → SamudrACE (ocean interior first, atmosphere second — but emulating a simulator); on the representation side it is ESFM and Copernicus-FM (heterogeneous sources, missingness tokens — but no ocean). Nobody starts from the ocean interior as an embedding, and nobody joins the ocean biosphere to a physical-state representation. Two names to disambiguate in any write-up: NVIDIA's 'Earth-2' platform and Aurora's 'A foundation model for the Earth system'.

What should a generic Earth embedding see?

The rule: every signal that is relevant to the four spheres at a point and cannot be derived from the other inputs. Thirteen rungs, finest first.



Three tests for a derived product

Observations (a reflectance, a radar echo, a float's thermometer) are irreducible: always in. A derived product (a gap-filled map, a reanalysis, an ML reconstruction) is admitted only if:

- 1 it carries information we do not ingest raw — ERA5 holds radiosondes, aircraft, radiances; GLORYS holds a full ocean model around the floats;
- 2 it is flagged derived and dropped from the input more often, so the model must also work without it;
- 3 its effective time is shifted by its assimilation window, so it is not a peek into the future (DUACS: ± 6 weeks; GLORYS: 7 days; ERA5: 12 h).

Dropped as derivable (targets only): NDVI and band ratios, solar geometry, slope/aspect, geostrophic currents from sea-level slope, per-scene class maps, Stokes drift.

Input ladder I — fine and fast (rungs 0-4)

Land at 10–30 m, the first ocean-covering rung at 300 m–1 km, and the atmosphere's 10-minute movie

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
0	Anchor coordinates — latitude, longitude, time of year, time of day, depth	Where and when we are looking. Not data, but the model needs it: 15 °C means something different in January at 60° N than in July at the equator.	—	The frame of reference; every other input is relative to it.
1	Sentinel-2 L2A (13 bands, 10/20/60 m, 5-day revisit with two satellites; three in orbit to end-2026) · Sentinel-1 GRD (VV/VH radar, 10 m pixels, 6-day repeat; S1C + S1D)	Sentinel-2: a colour-plus-infrared photograph of the sunlit surface, corrected for the atmosphere — crops, forests, soil, water, snow, buildings. Sentinel-1: a radar image, day or night, through cloud, of how rough and how wet the surface is — floods, sea-ice edges, ships, soil moisture.	○	Field-scale texture and the fine state of the land biosphere; radar sees through cloud and at night. Land and coast only.
2	Landsat 8/9 Collection 2 (surface reflectance + surface temperature, 30 m, 8-day combined; archive to 1982)	The same kind of photograph plus a thermal image — how warm the ground is — at field scale, with a forty-year archive.	○	Thermal at field scale; the long record for phenology and change.
3	Sentinel-3 OLCI (21 bands, 300 m, < 2 days) + SLSTR (500 m / 1 km, sea- and land-surface temperature, daily) · VIIRS NOAA-20/21 (375/750 m daily; MODIS retiring late 2026/27)	Daily whole-planet pictures in visible, infrared and thermal. Over the ocean the colour of the water says how much phytoplankton is in it; the thermal channels give the skin temperature of sea and land; hot spots are fires.	○	First rung that covers the ocean surface, and the first daily one: ocean colour (the biosphere), fires, snow, daily land temperature.
4	GOES-19/18 ABI · Meteosat-12 FCI · Himawari-9 AHI (16 channels, 0.5–2 km, every 10 minutes; full disks to about ± 60–70° latitude)	A continuous movie of clouds, water vapour and surface radiance from fixed points in the sky; clouds visibly form, move and dissolve between frames.	○	Atmospheric motion at minute scale — the only direct view of the wind field acting on clouds, and of the daily cycle. The fast end of the cone.

Input ladder II — fields and ocean memory (rungs 5-8)

Sparse kilometre-scale sensors, the gap-filled daily maps, the reanalysis atmosphere, and the first rung below the surface

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
5	SWOT (2 km swath sea level, 21-day) · nadir altimetry Sentinel-6/Jason-3/Sentinel-3 (~7 km along-track, ~10-day) · Sentinel-1 ocean winds (1 km on swaths) · ground weather radar (MRMS 1 km/2 min; OPERA 1 km/5 min) · GLM lightning (8 km) · GEDI & ICESat-2 lidar (25 m / 11 m footprints) · TROPOMI, OCO-2/3 (trace-gas columns)	Sea-surface height to a few centimetres along the satellite's track — the ocean's pressure map, whose slopes drive currents; rain seen from the ground every few minutes; where lightning strikes; the height of trees and of sea ice; the gases in the air column.	O	Sea level (hence currents), storm-scale rain, 3-D structure of vegetation and ice, greenhouse-gas columns. All sparse in space or time — the case where a set of dots beats a grid.
6	OSTIA SST (5 km daily; MUR 1 km) · DUACS sea level L4 (0.125° daily, with geostrophic u, v) · IMERG rain (0.1°, 30 min) · ASCAT winds (12.5/25 km, ≈ twice daily) · AMSR2 sea-ice concentration (10 / 6.25 km daily) · ice drift (62.5 km, 2-day) · SMAP soil moisture (36 km daily; 9 km 3-hourly model) · SSS L4 (weekly) · Copernicus waves (1/12°, 3-hourly, model)	The cleaned-up daily maps: sea temperature with the clouds filled in, sea level everywhere and not only under the track, rain everywhere, wind over the whole ocean, how much of the sea is ice and how fast it drifts, how wet the top centimetres of soil are.	D*	Completeness and a usable initial state for the slow media; they merge sources we do not ingest raw (AVHRR, microwave SST, every altimeter). Flag, mask heavily, time-shift by their windows. (*ASCAT, SMAP L3 and AMSR2 are swath observations on a grid: O.)
7	ERA5 (31 km native, 0.25° grid, hourly, 1940–, ~5-day latency): 2 m temperature, 10 m wind, sea-level pressure, precipitation, surface radiation, cloud cover, boundary-layer height, dewpoint, 37 pressure levels · real time: ECMWF IFS/AIFS open data or GFS (0.25°, 6-hourly)	The best physically consistent estimate of the whole atmosphere at every hour — temperature, wind, pressure, humidity, radiation, clouds, at the surface and aloft — made by running a weather model and continuously nudging it towards millions of observations.	D	The forcing of everything else: heat and momentum into ocean and land; the vertical structure of the air. ERA5's SST and sea ice are its inputs, not outputs — never use them as ocean observations.
8	Argo core (T, S profiles 0–2000 m every 10 days; 4,372 floats, ~150 k profiles/yr) · Deep Argo (to 4–6 km; 219) · BGC-Argo (O ₂ , NO ₃ , pH, chlorophyll, backscatter, light; 989) · moorings (TAO, OceanSITES) · drifters (1,317; velocity at 15 m + SST) · GLORYS12 reanalysis (1/12°, 50 levels, daily, 1993–) · Copernicus 1/12° analysis/forecast (real time)	What the ocean does below the surface: how warm and salty it is at each depth (heat content, density, the mixed layer), how fast and where it flows, where heat and fresh water are stored. Floats are sparse points; the reanalysis is a model's gridded estimate around them.	O + D	First rung below the surface: heat capacity, density gradients (thermal wind), the AMOC — the slow memory of the system. Only dots do this justice; gridding Argo throws away its depth resolution.

Input ladder III — biospheres, truth, stage (rungs 9–13)

Ocean carbon, land carbon and water, the point observations the reanalyses are fitted to, and the static boundary conditions

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
9	Ocean colour L3 (OC-CCI / GlobColour, 4 km daily, cloud gaps kept) · PACE OCI (hyperspectral 1.2 km, 2-day, 2024-) · SOCAT surface CO ₂ (44 M point obs) · BGC-Argo (rung 8) · Copernicus surface carbon (0.25° monthly, ML) · SeaFlux (1° monthly)	How much plant life is in the surface ocean and of what kind; how much CO ₂ the water holds relative to the air — whether the ocean is absorbing or releasing carbon; oxygen and nutrients at depth.	O + D	The ocean carbon sink and its drivers.
10	LAI / FAPAR (300 m 10-day) · SIF fluorescence (TROPOMI ~7 km daily) · LST (1 km daily) · evapotranspiration (500 m 8-day; GLEAM 0.1° daily — models) · soil moisture (ASCAT 12.5 km, ~2 h) · snow (500 m daily) · fire (VIIRS 375 m, < 3 h) · water storage (GRACE-FO, ~300 km monthly) · rivers (GRDC gauges; GloFAS 0.05° daily model; Hydroweb heights) · flux towers (half-hourly CO ₂ , water, heat)	How much leaf area there is and how actively it photosynthesises (fluorescence is the plants' own faint glow), how much water evaporates, how much snow and soil water is stored, where it burns, how much water sits in rivers, lakes and aquifers.	O + D	The land carbon and water cycles — the fourth sphere. Almost all of it is column-only: it does not move sideways.
11	Surface stations (ISD > 14,000 active, hourly; GHCN-Daily > 100,000) · radiosondes (~800 in real time, 00/12 UTC) · aircraft (AMDAR) · ships & buoys (ICOADS, NDBC) · tide gauges (GLOSS)	Thermometers, barometers and anemometers on the ground, balloons through the atmosphere, instruments on ships and buoys, sea-level gauges on coasts — the measurements rung 7 is fitted to.	O	Irreducible ground truth. Redundant while rung 7 is present; essential when it is masked — the model must rebuild the atmosphere from points.
12	Copernicus DEM (30 m) · GEBCO bathymetry (~450 m) · land cover (WorldCover 10 m; CCI 300 m annual) · SoilGrids (250 m) · canopy height (10 m / 1 m) · glacier outlines · population (100 m) · night lights (500 m) · roads	The stage: how high, how deep, what kind of surface and soil, how tall the trees, where the people are.	O / D	Boundary conditions: depth sets what currents can do, terrain sets rain and rivers, soil sets what water does. One dot each — $v = 0$, $\tau = \infty$.
13	Optional: aerosol optical depth (MAIAC 1 km daily) · MERRA-2 aerosols (0.5°, hourly, 3-week latency) · trace-gas columns (rung 5)	Dust, smoke and pollution in the air, and the greenhouse-gas columns above each point.	O + D	Radiative-forcing detail and the carbon-cycle link between land, ocean and air.

Observed, derived, or derivable — and the leakage trap

Why the source of each dot is a token the model sees, and why delayed-time products must be time-shifted

Always in — observations

A number a sensor produced. Reflectances, radar echoes, radar ranges to the sea surface, float thermometers, gauges, towers, balloons. Irreducible: nothing else contains them.

- Sentinel-1/2/3, Landsat, VIIRS, geostationary imagers
- altimeter tracks, SWOT swaths, lidar footprints
- Argo / BGC-Argo profiles, drifters, moorings
- ocean-colour L3 (gaps stay gaps), SOCAT
- stations, radiosondes, ships, buoys, tide gauges
- swath products on a grid (ASCAT, SMAP L3, AMSR2)

Admitted derived — flagged

A number a program produced from observations plus a model or a statistical prior. Admitted when it carries information we do not ingest raw. Every dot carries a source token; derived sources are dropped from the input at $\geq 50\%$ of anchors, so the model must also work from observations alone.

- ERA5 / IFS / GFS (radiosondes, aircraft, radiances inside)
- GLORYS12, Copernicus 1/12° analysis (full ocean model around the floats)
- OSTIA, MUR, DUACS L4, IMERG, SSS L4 (multi-sensor merges)
- SMAP L4, GloFAS, GLEAM, MOD16 (land models)
- Copernicus carbon, SeaFlux (ML reconstructions)

Dropped — derivable from the ladder

A function of inputs already present. Not an input; an optional target (a masked-channel loss on it costs nothing and gives free supervision).

- NDVI / EVI and other band ratios (rungs 1–3)
- solar zenith, day length, Coriolis f (position + time)
- slope, aspect, distance to coast (rung 12)
- geostrophic u, v and OSCAR (SSH, wind, SST)
- Dynamic World and per-scene classifiers (Sentinel-2)
- Stokes drift (wave spectrum)

The leakage trap: delayed-time products look backwards and forwards. A gap-filled map or a reanalysis for day t is built from observations on both sides of t . If the codec is trained to predict the present or the future from the past, a derived dot at lag ℓ secretly contains information from lag ℓ – (half its window) — sometimes from after the anchor. Windows to respect:

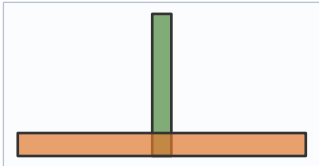
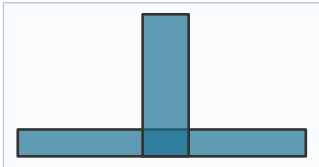
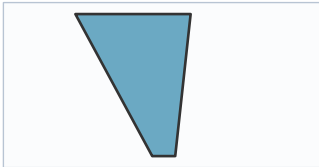
DUACS sea level L4 — delayed-time grids use a centred ± 6 -week window of tracks (the NRT stream only the past 7 weeks) · GLORYS12 — 7-day assimilation cycle · ERA5 — 12-hour assimilation window · OSTIA / MUR — a window of recent days · ocean-colour L4 — space-time interpolation across cloud gaps (use L3)

Rule: shift each derived source's effective time by its window, never use a delayed-time product at lag 0, and train the forecasting part of the objective on the near-real-time streams, which only see the past (ERA5T, OSTIA NRT, DUACS NRT, the Copernicus analysis). Like ERA5T vs ERA5, this yields two embeddings per anchor: an NRT one (honest for forecasting) and a final one.

Five cone families — one shape per rung

Apply $\Delta x \leq v \cdot (\Delta t + \ell)$, $\Delta t + \ell \leq \tau$ to every input: the shape of the context each channel needs differs by orders of magnitude

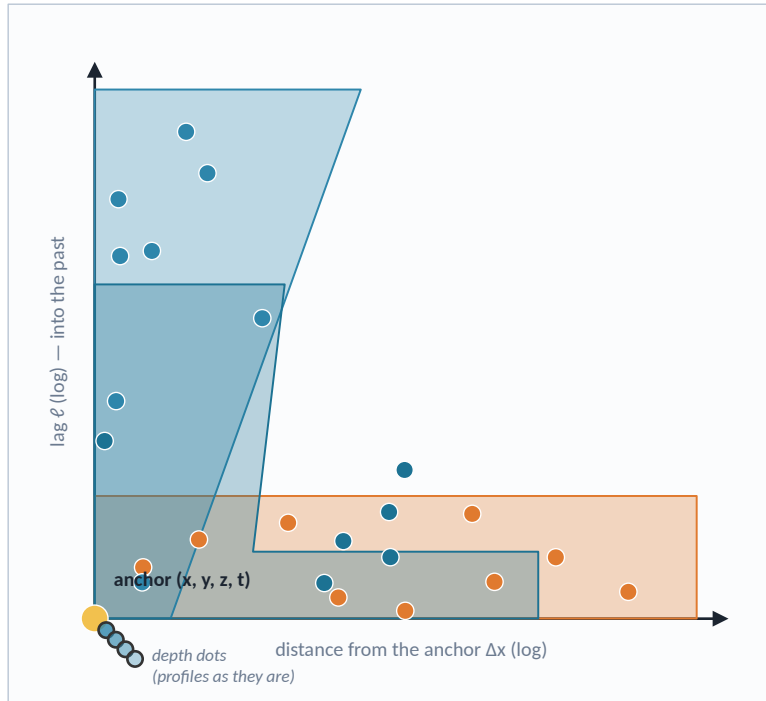
	Family	Carrier — what moves the information	v (order of magnitude)	τ — memory at a fixed point	Reach after 1 day · 1 week · 1 month	Ladder rungs
A	Fast, wide, short	wind, synoptic weather	10 m/s $\approx$ 860 km/day	3–10 days (rain: hours to 1 day)	860 km · 6,000 km · capped at 10,000 km	4 · 5 (radar, lightning) · 6 (IMERG, ASCAT) · 7 (wind, pressure, cloud)
B	Slow, narrow, long	ocean currents and eddies; Rossby waves for sea level; Kelvin / coastal waves along boundaries	currents 0.1–0.2 m/s $\approx$ 13 km/day · Rossby 0.03 m/s $\approx$ 2.6 km/day westward · coastal 2.5 m/s	eddies, sea level 2–6 months · interior T, S 1–10 yr (thermocline), decades (deep)	13 · 90 · 400 km (currents) · 3 · 18 · 80 km (Rossby)	5 (altimetry, SWOT) · 6 (DUACS, SSS) · 8 (Argo, GLORYS, drifters)
C	L-shaped union	fast atmospheric forcing on a slow medium: the forcing arm is A, the memory arm is B	A at short lags; B at long lags	SST, SSS 3–6 months (mixed-layer inertia) · sea ice seasonal + multi-year thickness · chlorophyll 1–4 weeks	forcing arm 860 km at lag 0; memory arm 100 km + 13 km/day	3 (SST) · 6 (OSTIA, MUR, ice) · 9 (chlorophyll)
D	Column-only (land)	nothing horizontal — soil, plants, snow stay put; forced from above by A; rivers carry water downstream ($\sim$ 1 m/s along the network); sea ice drifts 5–10 km/day	0 horizontally (river network $\approx$ 90 km/day along channels)	surface soil moisture 1–3 weeks, root zone months · vegetation weeks to seasons · snow seasonal · water storage months to years · fire days	L_corr $\approx$ 1–10 km at every lag, plus the atmosphere disc at lags $\leq \tau_{\text{atm}}$	1 · 2 · 10 · part of 12
E	Static	none	0	∞	one dot	12



Glyphs: Δx sideways, lag upwards, present at the bottom. Depth is a sixth axis, not a family: the mixed layer (top 10–100 m, deeper in winter) is stirred within hours and belongs to the surface cone; below it memory grows with depth — months at 100–300 m, years at 500–1,500 m, decades below 2,000 m — while vertical speeds are $\sim$ 1 m/day. Depth dots are log-spaced (0, 10, 30, 60, 100, 200, 400, 700, 1000, 1500, 2000 m), dense near the surface, each band with its own τ ; the atmosphere mirrors this at 1000, 850, 500, 250 hPa. Consequence: fine rungs want a small dense patch and a long local history; coarse physical rungs a large sparse far field and a short (A) or thin-and-long (B) history; the interior depth and years. No fixed grid $\times$ window serves all three — a per-channel dot sampler does.

Proposal — a cone-native codec: dots in, embedding out

The codec reads (Δx , Δy , depth, lag, channel, value) samples from each channel's cone and predicts the dots it does not see — including future ones



wind (A) • currents, SSH (B) • SST (C) — each channel samples its own cone

1 Dot sampler (fixed by physics)

Per channel group: stratified log-polar $\times$ log-lag draws inside the cone, slot budget from the family. Point, profile and track channels contribute their actual observations. Fine imagery arrives as tokens from the local codec. Missing dots are simply absent.

2 Dot $\rightarrow$ token

value embedding + Fourier features of (Δx , Δy in km, z in m, ℓ in days — all log) + channel embedding + source flag (observed swath / point / L4 / reanalysis / model / local token).

3 Encoder (Perceiver-style)

$K = 128-256$ latents cross-attend to $N \approx 2,000-4,000$ dots, then 4-8 self-attention blocks over the latents. Cost $O(N \cdot K)$, a few GFLOPs per anchor — below one ViT-Base image forward (≈ 17 GFLOPs).

4 Pool $\rightarrow$ e_{dyn} (256-D, int8)

the dynamic embedding: the state at the anchor, with its tendency and its context.

5 Objective: masked dots, including the future

A held-out dot is a query token (coordinates + channel, no value); a light decoder cross-attends from queries to the latents and predicts a distribution (Gaussian or bins; NLL / CRPS). Masking schemes mixed per batch:

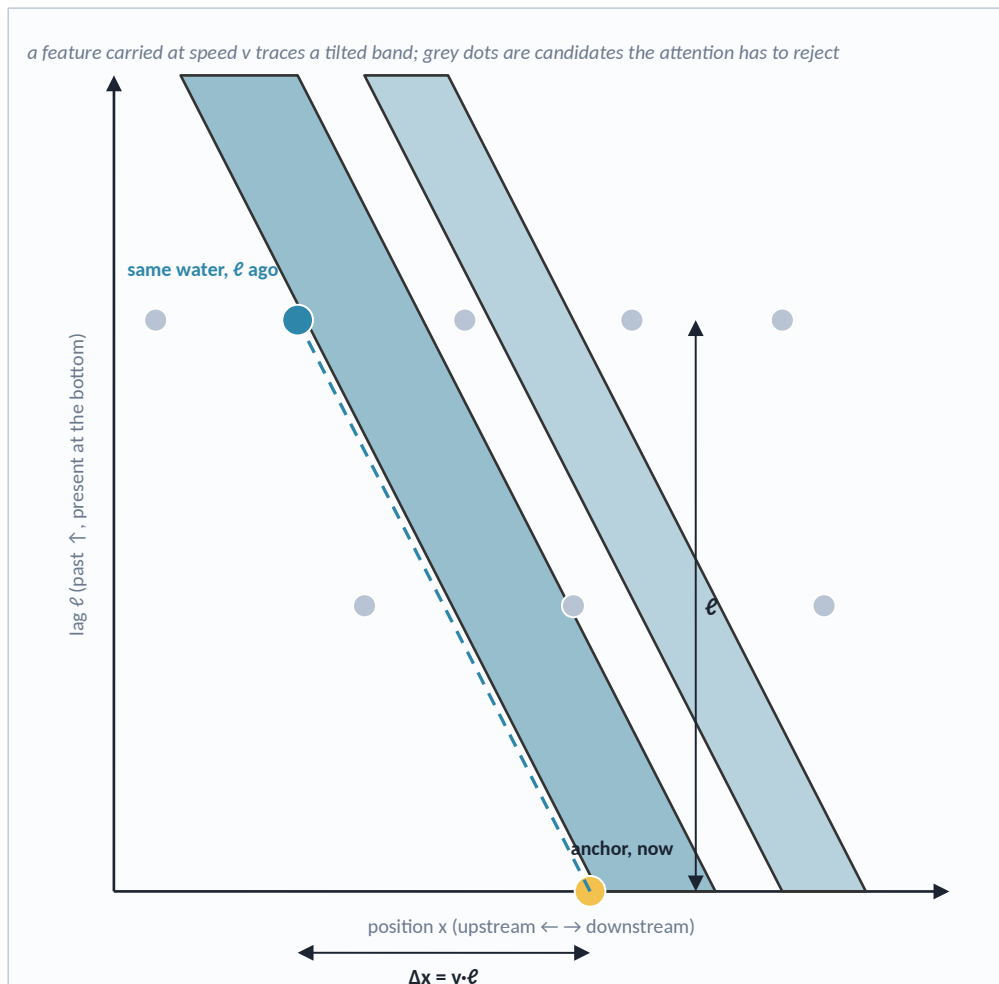
- channel drop — remove a whole channel group (derived ones more often)
- lag-band drop — hide the recent past, predict it from the older past = forecasting inside pretraining
- future dots — queries at negative lags; the embedding becomes state + tendency
- sector drop — hide upstream; depth-band drop — predict interior from surface and back
- anchor reconstruction — always predict the anchor's own channels

Never-measured vs masked survives on the target side: no loss where nothing was ever observed. Slow fields up-weighted (AIFS-ocean practice).

Why a set encoder and not a 4-D transformer over (x, y, z, t) grids: the input is ragged — a profile here, a swath there, a 10 m tile in the corner — and the cone is not a box. A set of dots handles both without padding, and retires the 'never-measured vs masked' distinction on the input side. Two products per anchor: an NRT embedding (observations + near-real-time streams) and a final one, exactly as ERA5T and ERA5; because the cone only looks backwards, an embedding is final once its sources have arrived and never needs recomputing.

Where velocity comes from — two dots of the same field at two lags

A photograph does not show speed; two photographs a known time apart do. Attention over dots with relative coordinates can implement feature tracking.



The argument

If a field is carried along, $f(x, t + l) \approx f(x - v \cdot l, t)$: the best predictor of the anchor's value is the dot at displacement $-v \cdot l$ at lag l . The cross-correlation between the anchor's neighbourhood now and the field l ago peaks at $\Delta x = v \cdot l$. That is the maximum-cross-correlation method oceanographers use to track features in SST images (Emery et al. 1986). A query attending over dots with relative $(\Delta x, l)$ encodings can implement exactly this search — and the winning ratio $\Delta x / l$ is a velocity readout. Tendency (two lags at one place) and convergence (dots around the anchor) follow the same way. A per-pixel snapshot codec cannot do any of this; it has one lag.

Making sure the capacity is used, not merely available

- Flow as maskable channels — drifter velocities, sea-ice drift, ASCAT / ERA5 wind, river discharge, and (flagged) OSCAR / GLORYS u, v are inputs, so the masked objective asks the embedding to produce velocity whenever the velocity channel is hidden.
- Multiple lags of the same field by construction — the B and C budgets always contain lags > 0 , and the cone guarantees the upstream dots are in the set.
- Optional advected-sampling prior — draw a fraction of the dots along $x - \hat{v} \cdot l$ using a first-guess velocity (geostrophic u, v ; ERA5 wind): PARADIS's semi-Lagrangian gather used as a sampler, not a layer.

The test: a velocity probe

A ridge head from the frozen embedding to drifter velocity or GLORYS u, v at the anchor, with derived channels masked at test time. Prediction: the snapshot embedding scores near climatology ($R^2 \approx 0$); the cone codec does not. This is the cheapest decisive experiment on the list.

Is it zero-sum? Thin codec + thick stage 2 vs cone codec + thin stage 2

A: embed one pixel-month, let a stage-2 stencil connect embeddings (today's Earth 2 and every GeoFM in the survey). B: embed the cone, keep stage 2 thin.

	A — thin codec, thick stage 2	B — thick (cone) codec, thin stage 2
Cost per embedding	Minimal: one pixel-month (27 tokens for $3 \times 3 \times 3$ months)	30–100 × more input per anchor ($\approx 3,000$ dots) — still $\approx$ one ViT-B image forward
Reusability, caching	Embedding = this place, this time; computed once, never changes	Embedding = this place, this time and its past context; final once its sources have arrived (NRT and final versions)
Interpretability	Clear provenance: this pixel	Provenance is a set of dots; attention maps recover it, with more work
Dynamics: velocity, tendency, transport	Not representable from a snapshot; stage 2 must rebuild it from compressed codes	Representable in the codec (previous slide); supervised by flow channels
Information bottleneck	Codec compresses before it can know what stage 2 needs; sub-pixel phase and small gradients that carry velocity may be dropped as noise — what the code drops, stage 2 cannot recover	Compression happens after the dynamics are visible; the objective (predict future and upstream dots) tells the codec what to keep
Sparse channels: Argo, tracks, stations	A lone profile with no context yields a near-empty code; gridding loses depth resolution	Points enter natively as dots and are contextualised at encode time
Cone geometry	Re-learned per task and per channel by stage 2 ($24 \times 89 \times$ channels tokens, each time)	Fixed once by physics, shared by all tasks; 7 % of the cylinder's slots
Sample efficiency of probes	Attention head must learn the geometry from labels (AMOC: ~ 20 years of monthly labels)	Geometry already inside; ridge / linear probes should suffice more often
Static and mapping tasks	Ideal: snapshot semantics (land cover, LCZ, canopy)	Neutral to slightly worse — context can blur a static class; expose the local code alongside
Forecasting in pretraining	Not available: no time axis in the codec	Native: future dots as targets
Training complexity, shortcut risk	Simple per-pixel pipeline; nothing to copy	Heterogeneous dot sampler, bucketing, leakage control; derived channels can be copied — needs source flags and drop rates
Comparability with published GeoFMs	Direct	Indirect: a different unit of embedding

Green = advantage, red = cost. A wins on cost, caching, provenance, simplicity; B on everything that involves time. Next slide: why this is not a symmetric trade.

Not zero-sum — the asymmetry, and the split that follows from it

The trade is about where the bottleneck sits relative to the dynamics, not about how much work there is in total

Two facts break the symmetry

1 B contains A. A cone codec that ignores every dot but the anchor's own is a per-pixel codec. Whatever A can represent, B can; the reverse fails because of the bottleneck: information the per-pixel code discards before the dynamics are visible is gone for stage 2 (data-processing inequality). In capability the trade is one-sided; the price is compute and the cleanliness of the embedding's meaning.

2 Compute is paid in different places. A pays little per embedding but re-pays the context assembly in every stage-2 model, for every task, every time. B pays once per anchor and amortises across tasks — provided the cone is generic, which is exactly what a physics-set support promises: it depends on the channel, not on the task.

The principled split

Into the embedding: everything whose relevance is decided by physics — the cone (v, τ per channel), depth coupling, the local past. Identical for every task.

Into stage 2: everything whose relevance is decided by the task — which section to integrate over (AMOC), which basin to sum (carbon), which teleconnection matters (ENSO's Pacific state, 10,000 km and six months away, outside any single cone), which label.

Where A stays right: purely static targets (land cover, soil, canopy height, LCZ), pixel-level anomaly detection, and very-high-volume or on-board inference (Prithvi-EO's in-orbit deployment). Keep the local code for these.

What existing evidence says: nothing yet. The attribution matrix's 3×3 codec-vs-raw contrast (0.672 vs 0.659) is a single-seed probe number inside the probe noise band, so it is consistent with parity; and every rolled-skill number from before 2 Sep 2026 was withdrawn with the paper reset (heads trained under the endpoint pool had seen the held-out years). The v8 paper's null ladder — a Linear Inverse Model in pixel space against one in embedding space — is the first measurement of the bottleneck row, and Phase 0 is built to run after it.

The hybrid: split by scale, not by fiat

Raw observations and admitted derived products

the thirteen rungs; point observations stay points; every dot carries a source flag

ladder
rungs 0-13

Local codec — per tile-date, cached, snapshot semantics

the existing per-pixel / 3×3 MAE codec generalised to a tile (64×64 px at 10 m, STP-style pyramid or FlexiViT tokens); masked-pixel objective; a few 64-D tokens per tile-date reused by every anchor whose cone contains the tile

e_loc 64-D
mapping tasks

Cone codec — per anchor, dynamic semantics

reads local tokens + coarse dots + native point observations inside each channel's cone; masked-dot objective with future dots; the momentum features live here

e_dyn 256-D
prediction tasks

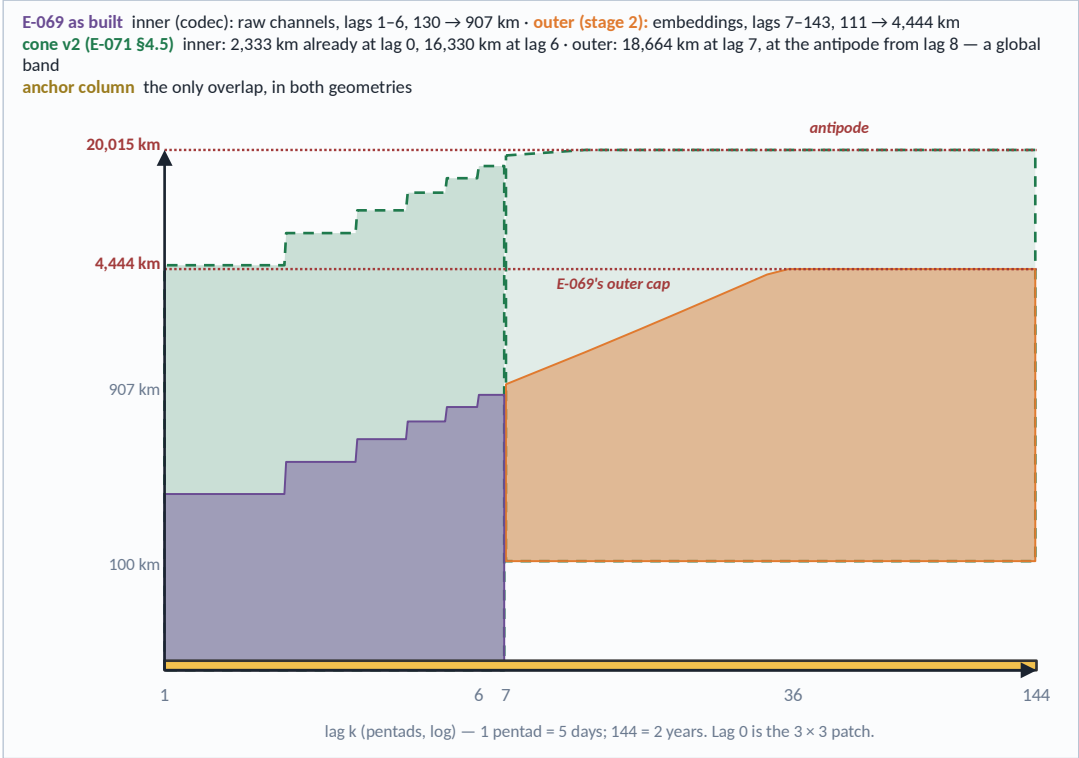
Stage 2 — thin

teleconnections beyond the cone (attention over distant anchors' e_dyn), aggregation along sections and over basins, task heads; the probe ladder stays (ridge $\rightarrow$ MLP $\rightarrow$ attention) with the expectation that more tasks stop at ridge

AMOC, carbon,
warming, biosphere

Two stencils, one cone — built, run, and re-cut for v2

E-069 ran five seeds (3 + 2 objectives) on the 0.3 m/s geometry; H1 is refuted, history buys persistence, not velocity. Under cone v2 the split between the stencils moves from distance to time — E-071 §4.5.



E-069: the codec owns the near field, stage 2 the far field. **v2:** the codec owns the last 30 days planet-wide; stage 2 owns 35 days – 2 years planet-wide. The overlap is still exactly the anchor column, and the empty annulus for $k \leq 6$ still holds by construction.

What five seeds said (EXPERIMENTS.md § E-069, 3–4 Sep)

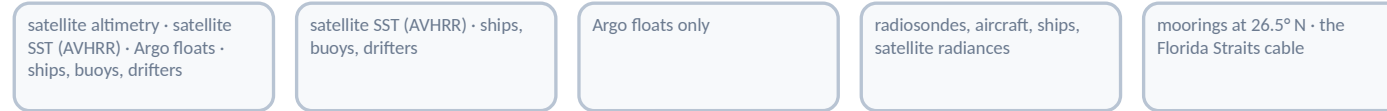
- 1 • H1 refuted.** The velocity probe — a ridge from the 32-number embedding to the anchor's own current, with the current channels hidden from the encoder — reads AT OR BELOW the raw-patch bar for the cone on both components at every seed, while the snapshot twin sits at the bar. History buys no velocity beyond geostrophy-from-the-sea-surface-height-patch at 7.05M parameters and 20k steps, under either objective. E-069 masking: cone cur_u 0.60 / 0.10 / 0.62 against a twin at ≈ 0.70 ; $\text{cur}_v \approx 0$ at every seed. E-069b lag0-drop objective: cone 0.63 / 0.66 and 0.58 / 0.59 against bars 0.64 / 0.71 and 0.67 / 0.70. A CPU synthetic-advection test finds no displacement primitive at the scales tried — a reading, not a closure: the field's certified ceiling was 0.16.
- 2 • What history DOES buy.** A hidden sea-surface-height channel is reconstructed at ≈ 0.36 of its bar against the twin's 0.75–0.90; hidden sea surface temperature 0.53–0.66 against 0.99–1.27; forecasts at +1 and +2 pentads 7–8 % better. Persistence and one-step forecast, not velocity.
- 3 • Consequence for v2.** Widening the reach 18× does not touch the mechanism H1 tested. v2's case rests on rolled skill (H2) from far-field context, and on the persistence signal being worth more when it is global, harmonic-anomaly and profile-aware. The velocity lever is capacity, the objective, or an explicit displacement primitive — not geometry.
- 4 • Two open decisions, deferred.** The outer ring's dot count per lag, on the same per-octave rule as the inner cone; and $K = 144$ pentads (two years) against a log-spaced eight-year outer window for the deep western boundary current's advective path.

Reading. The original split gave the codec the near field and stage 2 the far field. With the reach sized from the fastest mechanism the codec sees the planet within a week, and the honest description of the two stencils becomes thirty days of raw channels against two years of embeddings — both global. The identities the coverage test asserts (union = the whole cone, overlap = the anchor column) survive unchanged; what changed is what they mean.

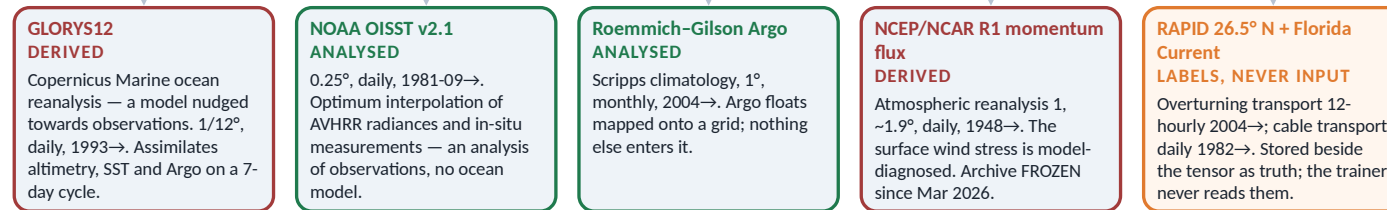
What feeds the cone codec today — four products, 42 channels, one axis

family 4 recipe r3 — the tensor the codec eats: North Atlantic 100° W–20° E, 0–70° N · 0.25° · 5-day bins · 1982-01-01 → 2024-12-31 · 84,405 ocean cells · built once and sha-pinned (run #535)

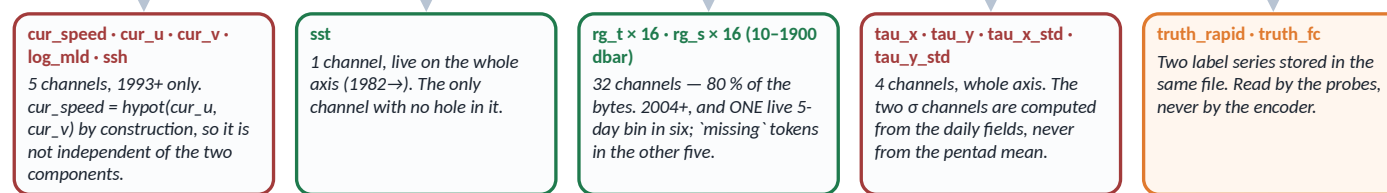
0 • OBSERVATIONS *what an instrument actually measured*



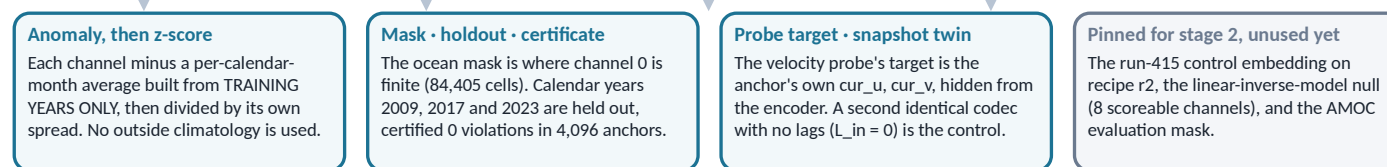
1 • PRODUCTS WE DOWNLOAD *four inputs + two labels; green = observation-led, red = model-derived*



2 • CHANNELS IN THE TENSOR *42 of them, plus the two labels; live-bin caveats in italics*



3 • DERIVED INSIDE TRAINING *nothing here is downloaded — it is computed from the tensor itself*



▲ The app's 1991–2020 sea-surface-temperature normal is deliberately NOT the anomaly baseline: it averages over the held-out years, so it would leak the test period into training.

▲ Derived products peek forward by half their assimilation window — the observations a reanalysis may look at around each day (GLORYS: 7 days). 5 of the 42 channels; they need a time shift.

Reading. Today's codec sees the ocean's momentum forcing (wind stress) and its density structure in one column, and nothing else: no heat or freshwater forcing, no observed velocity, no boundaries — no ice, no bathymetry — and nothing living. Two thirds of the channel count is a single monthly Argo product that is live in one 5-day bin out of six. The additions on the right are ordered by information per byte and by what the model cannot derive from what it already has; the first three are also the three the E-069 plan (the cone codec being trained now) would use unchanged.

Missing from “the ocean and its biosphere” — additions, ranked

1 Buoyancy forcing — ERA5 net short- and long-wave radiation, sensible and latent heat, evaporation – precipitation, wind-stress curl. The only forcing entirely absent today, and it also replaces the frozen NCEP momentum flux.

2 Ocean interior at 0.25°, in 3-D — GREP temperature, salinity and both velocity components at 8 depths, 1993→ (+20 GB). Velocity at depth, and six times as many live subsurface bins. The data ladder's #1.

3 Observed sea level — DUACS altimetry sea-level anomaly, absolute dynamic topography and geostrophic velocities, 0.125°, daily, 1993→: the observation GLORYS is fitted to, as its own independent channel.

4 Statics bathymetry and its gradient, the Coriolis parameter f and its gradient β , distance to coast and to the 1000 m isobath, mean dynamic topography. ~3 MB; the ladder's #2.

5 Sea-ice concentration — OSTIA / OSI SAF the northern boundary condition of the window. Today the model sees ice-covered water as ordinary water.

6 Drifters — Global Drifter Program 15 m velocity and SST as native dots: the observed velocity the H1 probe (does the embedding know the current?) should really be scored on.

7 Biosphere ocean-colour chlorophyll (OC-CCI 4 km 1997→, PACE 2024→), BGC-Argo oxygen / nitrate / chlorophyll profiles as dots, SOCAT surface CO₂. The ladder rejected chlorophyll as an INPUT for the overturning head — “no mechanistic path at these timescales” — so the biosphere enters first as what the embedding must PREDICT, and as a held-out sphere.

8 Sea-surface salinity — SMOS / SMAP 2010→, short, but it is the freshwater lid. And RAPID's depth-resolved moc_vertical and boundary T/S files are already downloaded and unused.

Ranked by information per byte and by what the model cannot derive from what it already holds.

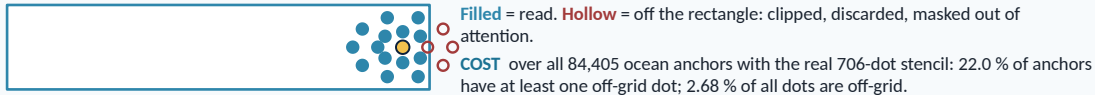
Boundaries — four edges, four rules, and two of them now change

Findings of 3 Sep; decisions of 4 Sep (Chris): the dataset is global, and the anomaly baseline is a day-of-year harmonic fit — E-071.

A · SPACE — the window edge (100° W – 20° E, 0–70° N)

FINDING, 3 SEP a dot outside the rectangle is INVALID, never wrapped: the tensor is a basin, so a wrap “would put the Iberian shelf one cell west of Florida” (ml/cone_sampler.py:19–23). The index is clipped, the value discarded, and valid = False becomes the ATTENTION MASK (ConeSampler.sample :293; the lag-0 3×3 patch likewise, _patch :263). Anchors are drawn uniformly over ocean cells with NO edge margin (ml/train_cone.py:358), so an edge anchor sees a truncated, one-sided cone.

The asymmetry: nothing tells the model that the missing dots are missing because of OUR rectangle rather than because of the ocean.



DECISION → GLOBAL (E-071 §1) family 7 — the first tensor covering the whole globe rather than the North Atlantic window — runs to the POLES: $721 \times 1,440$ cells, $-90...90^\circ$ N; longitude WRAPS ($xx = (x + dx) \bmod W$), and every dot is placed on the sphere as a destination point, so a pole crossing falls out of the formula and nothing is ever clipped. The 2.68 % off-grid dots become real dots and the 22 % truncated anchors see a full cone. ASPECT = 0.71 was measured on the North Atlantic and is re-measured per latitude band.

B · SPACE — land

UNCHANGED — these rules were right

RULE a dot on land is a REAL token whose value was never observed: obs = False → the codec's miss token — the same miss_tok PixelMAE (the pixel-level masked auto-encoder this codec inherits) uses — NOT an invalid/masked token.

The distinction is the one the whole architecture rests on: “the data never observed this” is information; “this dot does not exist” is not. One is a token the model can learn from, the other is a hole in the attention matrix.

COST 8.17 % of all dots land on land — on-grid, unobserved. Combined with the edge, **10.85 % of the 706 dots at a typical anchor carry no value.**

Land dots are still spent from the budget: they occupy their slot and are attended to, so a coastal anchor pays for its geography twice.



the 706 dots at one anchor

■ 89.15 % carry a value ■ 8.17 % land → miss token ■ 2.68 % off-grid → invalid

C · TIME — the archive ends, and the holdout blocks

UNCHANGED — these rules were right

RULE 1 — the ends of the axis an anchor whose cone runs off either end of the tensor — it needs bins $t-6 \dots t+2$ — is INADMISSIBLE, not silently short (ConeSampler.admissible, :314). Nothing is padded and nothing is trained on half a cone.

RULE 2 — the holdout —holdout-scope window (the only scope implemented, ml/train_cone.py:116) says every bin the cone touches — six pentads back and both future targets — must be a training bin. A held-out year therefore casts a **shadow of eight pentads** around itself — six back and two forward, the exact span of the cone — and no training anchor can peek into it by any path. A brute-force certify() (:337), written deliberately as a separate loop rather than as a reuse of the admissibility test, counts violations; run #537 (the E-069 cone-codec build) measured 0 violations in 4,096 anchors.

COST OF THE SHADOW development split (calendar years 2009, 2017, 2023 held out): 2,923 training bins → 2,891 admissible anchor bins, 32 bins lost (1.1 %). Frozen protocol (2008–09, 2016–17, 2021–24 — train ≤ 2020, test the terminal years): 2,557 → 2,533, 24 bins lost (0.9 %). Cheap, and it is what makes the holdout real.

Held-out EVALUATION anchors are different on purpose: their dots MAY come from held-out bins — otherwise the held-out set would be a second training pool, not a measurement.

D · TIME — the calendar month: the edge we introduced ourselves

(i) **The anomaly baseline.** Every channel is a departure from a per-calendar-month climatology built on TRAINING YEARS ONLY (ml/trainprobe.py::anomaly_transform :140), and a pentad's month label is the month of its FIRST day (ml/build_family4.py:960). So **411 of the 3,142 bins (13.1 %) straddle a month boundary**, and 106 of those have only ONE of their five days inside the month whose climatology is subtracted — a sawtooth at every boundary, of order the month-to-month climatology difference, which in the seasonal thermocline is not negligible.

(ii) **Argo's monthly cadence.** Roemmich–Gilson is written into the single pentad holding the 15th of each month and is missing in the other five; forward-filling was REJECTED (E034_pentad_tensor.md:88–100). **252 live pentads, 2004–2024 — one in six.** A 35-day inner window holds ONE live Argo bin, occasionally two, and which one depends on where the anchor sits inside the month.

The seasonal context token itself is the bin's opening day-of-year (cone_sampler.pentad_doy), so it crosses month and year boundaries smoothly — that one is fine.

DECISION → DAY-OF-YEAR HARMONIC CLIMATOLOGY (E-071 §2) $\text{clim}(\tau) = a_0 + \sum_{k=1..3} [a_k \cos k\tau + b_k \sin k\tau]$, with τ taken from the pentad's MID-day over a 365.2422-day year, fitted per cell and per channel on training bins only: 7 coefficients instead of 12 box means, one pass with 35 accumulators instead of 24. K per channel is chosen by held-out variance explained (the mixed-layer depth's spring step may need $K = 4-6$). No bin is ever charged for a month it barely touches, and Argo's 252 live pentads are fitted together instead of ~21 per box. Verification: the last-pentad-of-month minus first-of-next residual, a sawtooth today, should be zero.

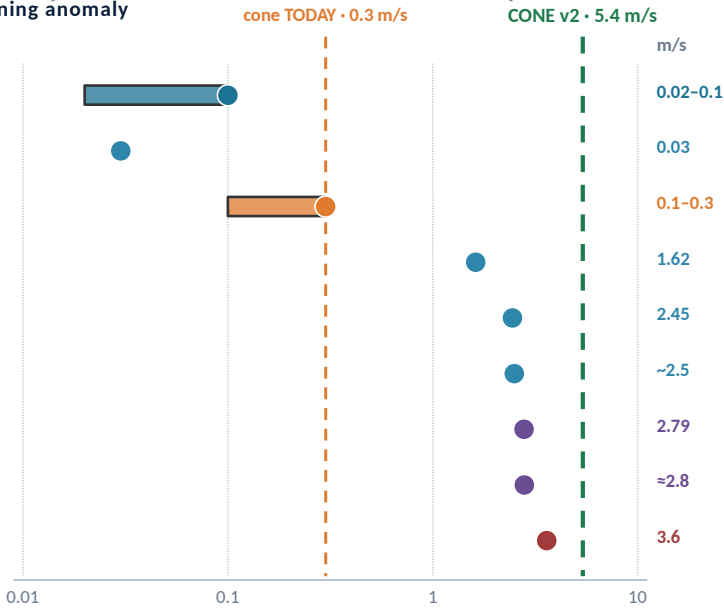
Reading. Three of the four edges were handled by making the missing thing explicitly missing — an invalid token is dropped from attention, a miss token says the world was not observed, an inadmissible anchor is not trained on — and those three rules stay. The other two edges do not get handled better; they disappear. The window edge goes with the global tensor (E-071 §1); the month edge goes with the harmonic baseline (E-071 §2). Both are decided and specified, and both are UNIMPLEMENTED — nothing here is measured yet.

The speeds the cone did not have — and the speeds it will

Decision (Chris, 4 Sep): size every family from the MAXIMUM measured speed of its fastest mechanism, $\times 1.5$ — not a typical speed, not an estimate (E-071 §4).

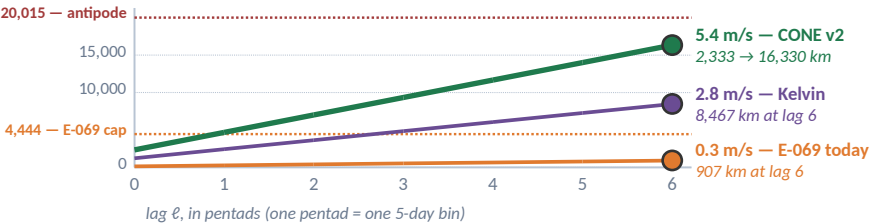
MEASURED MAXIMA — what moves an overturning anomaly between latitudes. Log axis, 0.01 $\rightarrow$ 10 m/s

- Deep Western Boundary Current
standard oceanography — the slow return limb
- Long baroclinic Rossby wave
Chelton et al. 2011 — westward, the interior
- Eddies & boundary currents
family B TODAY — `ml/cone.py:97, v_ms = 0.3`
- Kuroshio off Taiwan
our GLORYS bake, monthly 1° · `data/currents_y`
- Agulhas Current core
Wikipedia, Agulhas Current
- Gulf Stream core
NOAA Ocean Service — about 9 km/h
- Somali Current, our own bake
July 2020, 2.5° N 47.5° E · `data/currents_y`
- First-baroclinic Kelvin wave
Wikipedia; Chelton 1998 · Pacific in ~2 months
- Somali Current, peak
Wikipedia — 7 knots, south-west monsoon



OFF THIS RULER, TO THE RIGHT. Barotropic (surface) gravity wave ≈ 200 m/s and the jet stream ≈ 100 m/s (family A, wind stress). Both cross the planet inside one 5-day bin, so neither is a cone at all: E-071 §4.3 gives those families a RING of 24 dots at log-spaced radii out to the antipode — bearings, not density.

HOW FAR A SIGNAL GETS IN SIX PENTADS — reach = $v \cdot \Delta t \cdot (1 + \ell)$, capped at the antipode



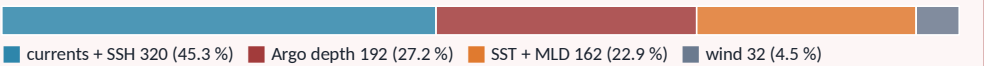
Why the buffer is 1.5, and why it is spent on the WIDE side. The tensor is a 5-day mean on a 0.25° grid and under-reads peaks; the maxima above are literature and reanalysis values, not the true extreme; and a warming ocean may run faster. A cone that is too WIDE only costs tokens. A cone that is too NARROW asserts that the driver cannot have arrived, and the model has no way to discover that the assertion was wrong.

Reading. The cone was a one-speed model of a multi-speed ocean. Cone v2 sizes each family from the fastest thing that actually carries it — the Somali Current's 3.6 m/s for the ocean, the jet stream for the wind — with the 1.5 buffer spent on the wide side, because a cone that is too narrow excludes true causes silently while one that is too wide only costs tokens. Nothing here is measured. And after E-069's H1 (slide 44), the case for v2 is rolled skill and persistence context, not velocity: a codec that could not read a displacement at 130 km will not read one at 2,333 km.

1 • Argo had no spatial reach at all — now it has 26 profile tokens

`ml/cone.py::channel_family (:122)` assigns `rg_t*` / `rg_s*` — the 32 Roemmich–Gilson temperature and salinity channels — to family B (0.3 m/s), but `channel_dots (:261)` OVERRIDES that and returns the ANCHOR COLUMN ONLY. Their effective reach is 0 km at every lag, so no propagation of a subsurface anomaly toward the anchor is visible at any speed — and those channels are still 192 of the 706 dots (27.2 %), a quarter of the budget on one vertical column. The stated reason was token economy: Argo is live one pentad in six.

DECISION $\rightarrow$ E-071 §3 the two most recent LIVE pentads (lags 0...11, known from the calendar, not from the data), 13 locations each — the anchor plus a 12-dot sunflower at the v2 reach — and each location is ONE profile token carrying 16 T + 16 S values with the pressure ladder as an inner axis. That is 26 tokens instead of 192, every one of them observed. Later, with native floats: the k nearest live profiles inside the cone, no sunflower at all.



2 • The design speeds — one per family, E-071 §4.2

family	fastest mechanism that carries it	max	$\times 1.5 = v_{\text{design}}$	reach / pentad
B — ocean	Somali Current (advection) $\geq$ first-baroclinic Kelvin (adjustment)	3.6 m/s	5.4 m/s	2,333 km
C — surface	atmosphere at lags 0-1, then the ocean	— / 3.6	global / 5.4 m/s	— / 2,333 km
A — wind	the jet stream (storms translate at 10-20 m/s)	~100 m/s	global	within one pentad

- (i) Dot placement becomes LOG-RADIAL, 36 dots per lag for the ocean families. Uniform-area placement at 24 dots gives 4.8 dots per octave of distance over 907 km but only 2.5 over 16,330 km — the near field, where the eddies live, would be starved by a wider cone at the same budget.
- (ii) Token budget 748 $\rightarrow$ 1,600 per anchor (B 888 · C 444 · A 200 · Argo 26 · lag-0 patch 42): about $2.1\times$ the encoder cost, and linear in a Perceiver (64 latents cross-attend to the tokens). The fallback that brings it back to ~500 is one token per (lag, location) carrying all channels as a value vector — kept OUT of v2 so that v2 measures the geometry and nothing else. E-069 keeps its geometry — its five seeds are done and its H1 verdict stands (slide 44). Cone v2 lands on family 7 — the first tensor covering the whole globe rather than the North Atlantic window.

Land, ice and air — nothing is dark by design

Chris, 4 Sep: “why not add some data on land, too? ... sparse for now ... land surface temperature, satellite imagery, radar reflection, air temperature across heights” and “some of our channels should be available for both the oceans and the land”. E-071 §6: a channel is a quantity and an instrument, never a sphere.

THE SHARED SET — one quantity, one instrument, two surfaces. A per-cell SPHERE CODE (ocean · land · ice sheet · inland water) says which retrieval filled the cell, so one channel means sea-surface temperature at one anchor and land-surface temperature at the next.

quantity	over ocean	over land / ice	source, at 0.25° pentads
surface (skin) temperature	ERA5 skin T (shared); SST stays its own observed channel	ERA5 skin T (shared); LST stays its own observed channel	ERA5 skin temperature is the ONE shared channel (same model field everywhere); OSTIA/OISST SST (a foundation temperature, tiny diurnal range) and MODIS LST (clear-sky radiometric skin at overpass, 10–20 K diurnal swing) measure different things and stay separate — shared only when measurand AND instrument match (Chris, 4 Sep)
near-surface air	2 m T, 2 m dew point, 10 m wind	the same	ERA5 — this row retires the FROZEN NCEP R1 wind source
air aloft — “across heights”	T, wind, humidity at 850 / 500 / 250 hPa	the same	ERA5 pressure levels; IGRA radiosondes (~800 stations, 00/12 UTC) as native dots later
pressure, precipitation, radiation, fluxes	pressure, precip, net SW + LW, latent + sensible heat	the same	ERA5; IMERG 0.1° as the observed precipitation channel, 2000→
frozen fraction	sea-ice concentration	snow-cover fraction; ice sheet ≡ 1	OSI SAF / NSIDC sea ice 25 km 1978→ · MOD10A1 snow 500 m daily 2000→ — ONE channel
albedo	≈ constant, plus the ice	vegetation, snow, soil, ice	MCD43A3 500 m daily 2000→ — one channel; the ocean value is the constant plus the ice
radar cross-section σ ⁰	wind and waves roughen it	soil moisture and vegetation set it	ASCAT 25 km twice daily 2007→ — the SAME instrument over both. The “radar reflection” channel, raw
optical reflectance	ocean colour	vegetation, snow, soil	MODIS / VIIRS 500 m — red, near-IR, one short-wave IR, raw, 2000→. Chlorophyll and NDVI become TARGETS
L-band brightness temperature	sea-surface salinity	soil moisture	SMAP 36 km 2015→ — one instrument; the raw T_B is the channel, both retrievals are targets
mass / water storage	ocean bottom pressure	terrestrial water storage; ice-sheet mass	GRACE / GRACE-FO mascons, monthly 2002→ (2017–18 gap) — already one global field
surface height anomaly	sea-level anomaly (DUACS 0.125°, 1993→)	ice-sheet elevation change; lake and river height	The same radar altimeters: DUACS over sea, CryoSat-2 (2010→) / ICESat-2 (2018→) over ice, Hydroweb over lakes
surface velocity	currents (GLORYS u, v)	ice velocity (ITS_LIVE); land ≡ 0	One (u, v) pair. On land it is exactly zero — which is information: nothing advects
the column below	T, S profiles (Argo, 0–2,000 m)	soil layers (ERA5-Land, SMAP L4); firn on the ice sheets	ONE profile-token type (E-071 §3) with a sphere-specific depth ladder

Sphere-specific channels stay sphere-specific, and are **miss** elsewhere with a clear conscience: mixed-layer depth and sea-surface height (ocean); ESA CCI soil moisture 0.25° daily 1978→ — it covers the whole axis — snow water equivalent, leaf area, GLEAM evapotranspiration (land); ice velocity and elevation change (ice).

Reading. Today a land cell is told “the world was not observed here” 42 times per pentad, which is false. Cone v2 defines channels by what is measured and by which instrument, and lets a sphere code say what surface lies under the cell — so most of the input ladder already crosses the coastline: the same radar, the same reflectance, the same altimeter, the same column token. Chris’s correction of 4 Sep sets the bar for “shared”: the measurand AND the instrument must match. ERA5 skin temperature is therefore the one shared temperature channel, and the observed SST and land-surface temperature stay separate.

1 • Antarctica was never unobserved — the GRID was

E-070 chose ~80...80° because that is GLORYS12’s extent — an ocean product stops where the ocean stops — and §1 of the plan wrongly let it read as **no data**. Cone v2’s grid is **~90 ... 90°, 721 × 1,440**. The ice-sheet rows are filled by ERA5 (skin and 2 m temperature, winds, pressure, snowfall, radiation — the surface mass balance’s inputs), MODIS LST and albedo in the clear-sky months, GRACE mass, CryoSat-2 (2010→) and ICESat-2 (2018→) elevation change, ITS_LIVE velocity, and around the continent the sea-ice concentration and drift.

Why it matters for the overturning: Antarctic meltwater and sea-ice formation set Antarctic Bottom Water, the lower limb of the global overturning — and the ice sheet’s freshwater flux is its largest uncertainty.

2 • Cone families for the new spheres (E-071 §6.3)

atmosphere A⁺ — global at once, lags 0–2, the 24-dot log-spaced ring to the antipode.

land surface state L — v = 0: the land does not advect, so the reach is the correlation length alone (~300–500 km) and the memory is months. A column plus a 12-dot disc at every inner lag and into the outer window. The SAME channel over an OCEAN anchor follows family C at 5.4 m/s — the one place a family depends on the anchor’s sphere.

ice sheet I — column only, monthly cadence, the firn profile token.

The sphere code (ocean · land · ice · inland water, from the rung-12 static mask) is a coordinate the codec reads, exactly as depth is.

3 • Cost, and the token change this forces (§6.4–6.5)

One global channel at 0.25° × 5 d × 1982–2024 ≈ 6.5 GB float16; the shared set of ~20 channels ≈ 130 GB, plus ~5 land-only ≈ 33 GB, on top of the 62 GB ocean tensor. Two sources — ERA5 and MODIS — supply three quarters; ERA5 needs the free CDS account.

With ~65 channels, per-channel tokens would be ~14,000 per anchor. So v2 makes a TOKEN = one (lag, location) carrying the full value vector of EVERY channel there, with a per-channel observed mask inside and the sphere code as a coordinate — the §3 profile token generalised. **≈ 300 tokens per anchor whatever the channel count, below E-069’s 748.**

PHASE L0 — “sparse for now”: six channels, two sources, all gap-free, all 1982→, all global to the poles, ~40 GB — ERA5 skin T, 2 m T, 10 m u/v (retiring the frozen NCEP R1), surface pressure, precipitation; plus the frozen fraction and ESA CCI soil moisture. MODIS / GRACE / SMAP / ASCAT (2000→) follow as L1, once the harmonic climatology has shown it handles a channel that starts eighteen years into the axis.

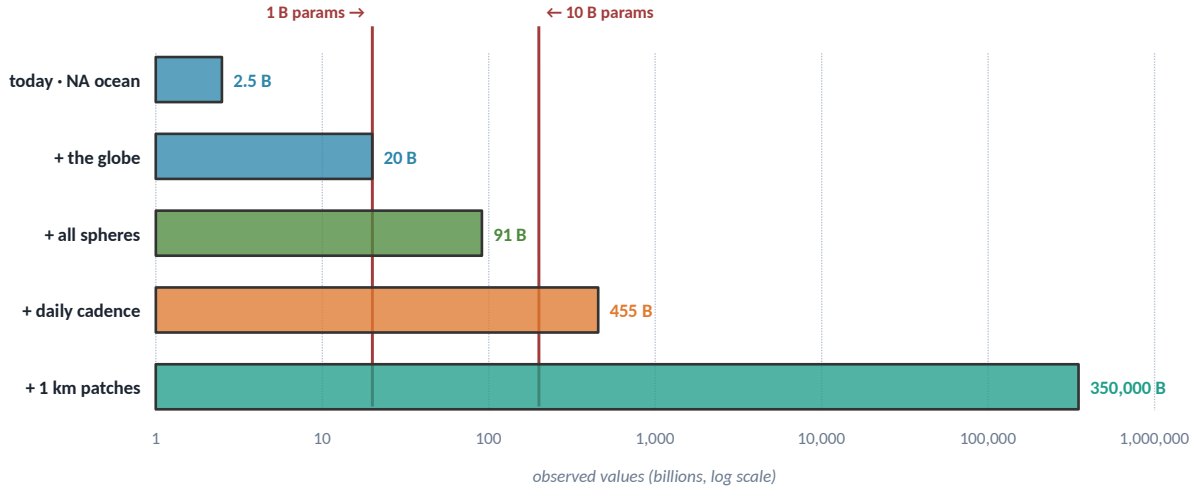
The Earth foundation model — the data decides the size

Chris, 4 Sep: a 1–10 B model, maybe MoE, a Chinchilla ladder, and a study of how to design the training data. E-072: at today’s data the size is small and measured; the ladder is a ladder of DATA.

ASSUMPTIONS (E-072 §2.2). 0.25° pentads 1982-01 → 2024-12 (3,142 bins); the global grid 721 × 1,440; ~686 k ocean and ~352 k land + ice cells; channel sets as in E-071 §6; a token carries 65 channels. “Ceiling” is values ÷ 20, the programme’s stated Chinchilla anchor — and it IS a ceiling, because 0.25° cells are correlated over hundreds of kilometres.

rung of data	observed values	tokens (÷ 65)	ceiling, values / 20	what it takes
North Atlantic ocean pentads, r3 — today	2.5 B	0.04 B	0.12 B params	exists
global ocean pentads, family 7	20 B	0.31 B	1.0 B	E-070 Phases B–F (the pull is done; the tensor is not built)
+ the shared 20 and land-only 5 channels of E-071 §6	91 B	1.4 B	4.5 B	ERA5 (needs the CDS account), MODIS, ASCAT, SMAP, GRACE, OSI SAF, CCI soil moisture: ~170 GB
daily cadence, all spheres	455 B	7.0 B	23 B	the daily global tensor is 1.3 TB (E-070 §8); the streaming loader E-033 Phase 3 deferred
1 km daily, all spheres (rung 3 radiometers)	350,000 B	5,400 B	17,000 B	not a tensor: a patch-encoder feed (§2.1); compute-bound for the first time

The same ladder on a log axis — observed values, in billions. A ceiling of N parameters needs 20·N values, so the 1 B and 10 B targets are the two guides: 20 B and 200 B.



Reading. At one basin, a quarter degree, five-day bins and 42 ocean channels, the model size is set by the data and it is small — the width ladder is flat from 7.6 M to 206.66 M, the linear model wins from 15 days out, and every held-out minimum arrives inside 2,000 steps. A 1–10 B model is therefore not a training decision but a DATA-DESIGN decision: the globe, the other spheres, daily cadence and the kilometre rungs are what raise the value-count ceiling from ~0.12 B parameters to 1 B, 4.5 B, 23 B and beyond — and the compute stays cheap until the kilometre rung.

What the programme has MEASURED (E-072 §1)

The width ladder is flat. 7.6 M → 40.4 M → 206.66 M stage-2 heads roll at corridor accuracy 0.103 / 0.104 / 0.105 (E-062-R0). At today’s data, capacity is not the axis — more parameters buy nothing until the data grows.

The LIM beats every head from 15 days. A 200-mode linear inverse model — a plain linear forecast fitted to the same field — out-forecasts the learned head at every lead ≥ 15 d and beats damped persistence from 30 d (E-066). The null ladder is mandatory at every rung.

Held-out minima arrive inside 2,000 steps and the next 198,000 make it worse (E-059, E-060) — the regime is data-limited: early stopping, ≤ 4 epochs, never a long schedule.

Argo is 80 % of the bytes for ~1 % of the information. The 32 upsampled Argo channels carry ~0.28 GB of information and are null at every lead (E-062). Information per byte decides the data design — never bytes.

The independence correction (E-072 §2.3)

A 0.25° cell is correlated with its neighbours over ~300 km and, at the surface, over 2–6 months. In the North Atlantic that is ~150 cells × ~15 pentads per independent sample, so the 2.5 B raw values are of order 1–10 M INDEPENDENT ones — which is exactly the regime in which a 7.6 M head equals a 206 M one.

So the multipliers buy less than they say. The globe adds new REGIMES and the spheres add new PHYSICS (land does not advect; the atmosphere decorrelates in days) — both worth their face value. Daily cadence inside a pentad is mostly the same sample five times and is worth perhaps ×1.5. The kilometre rung adds genuinely new fine-scale variance.

The honest reading: globe and spheres first, then resolution, cadence last.

The scaling ladder and the recipe

E-072 §3–4. Six rungs, each bought only on the previous rung’s read-out under the frozen protocol; and the training recipe this programme has already paid to learn, plus the two things the new data forces.

COMPUTE PRICING. 6-N-T-epochs FLOPs; an RTX 4090 at 165 TFLOP/s bf16 (E-033’s measurement) at 40 % utilisation; a v5e-4 TPU node at 4 × 197 TFLOP/s at the same utilisation. Four epochs — the data-constrained limit (Muennighoff et al. 2023) — except at the top rung.

rung	N	data	tokens × epochs	4090-h	v5e-4 h	the question it must answer before the next rung is bought
L0	7 M	NA pentads (E-069’s ConeMAE)	0.15 B	< 1	< 1	done: persistence yes, velocity no
L1	30–125 M	NA pentads, cone v2 geometry + harmonic anomaly + profile tokens	0.15 B	< 1	< 1	does the geometry change (E-071) move the per-family held-out loss and the rolled skill at leads 1–3 beyond the LIM?
L2	0.3–1 B	global ocean pentads (family 7)	1.2 B	~30	~7	G1–G3 of E-070: does the globe transfer (Kuroshio, ACC), and does the North Atlantic number hold at matched steps?
L3	1–4.5 B, MoE	+ all spheres (E-071 §6)	5.6 B	~570	~120	does a sphere-routed mixture of experts beat a dense model of the same ACTIVE parameters; do land and atmosphere improve the ocean’s rolled skill, and vice versa?
L4	4–23 B, MoE	daily, all spheres	14 B (2 epochs)	~3,500	~740	the first compute-bound rung; is the daily cadence worth ×5 bytes on the terminal holdout?
L5	10 B+	+ kilometre patches	—	compute-bound	—	only after L4 has a positive answer

Rules of the ladder

- 1 Each rung is bought only on the PREVIOUS rung’s read-out, under the frozen protocol (train ≤ 2020, test 2021–2024, longitude holes), with the null ladder beside it — climatology, persistence, damped persistence, the LIM in pixel space and in the rung’s own embedding space, nearest-analogue retrieval, and the rung below.
- 2 A rung whose held-out minimum arrives inside 2,000 steps at FULL data is over-parameterised — and the next rung is not bought.
- 3 The width ladder (flat 7.6 M → 206 M today) is RE-MEASURED at L2 before L3 exists: it is the one measurement that separates a data-limited rung from a compute-limited one.

Reading. Every rung reports the same null ladder — climatology, persistence, damped persistence, the LIM at K = 200 in pixel space and in the rung’s embedding space, nearest-analogue retrieval and the rung below — as MSSS per lead, per channel, per scope, with block-bootstrap intervals and amplitude calibration reported separately. And the arithmetic on the left says the rest: compute is cheap until the kilometre rung; the data is the expensive part.

Architecture (§4.1)

Encoder: a Perceiver-style transformer over LOCATION TOKENS — cross-attention into a fixed latent set, 64 × 256 latents at 7 M, widening with the ladder. At L3 the feed-forward block becomes a **mixture of experts routed by sphere code and channel group** (ocean / land / ice / atmosphere), top-2, with a small load-balancing loss — the router already sees the sphere, so routing is partly deterministic.

Decoder: the queryable Gaussian decoder reading z ALONE (E-069 closed the degeneracy where a [z + latents] memory lets the bottleneck carry nothing). Stage 2 stays a separate sequence model over embeddings — global rings, K = 144 pentads.

Objective (§4.2)

Masked reconstruction under the E-069b masking plan (a dropped channel hidden at the present-day patch; hidden dots from the recent-lag band and a bearing wedge) **plus future targets at +1 and +2 pentads, scored by Gaussian negative log-likelihood**, so the model is graded on its stated uncertainty. The per-family predict-the-mean bar is logged at every eval. The fair-CRPS FGN head (E-052 / E-057) is the probabilistic option once a deterministic level exists.

Optimiser & schedule (§4.3)

AdamW, β₁ 0.9, β₂ 0.95, weight decay 0.1, gradient clip 1.0 on standardised inputs; bf16 with fp32 master weights; **2,000 warm-up steps, cosine to 10 % of peak**. Peak learning rate 3 × 10^{−4} at 7 M (E-069) and 4 × 10^{−4} at the 1280 × 20 heads (E-054); at L2 a three-point sweep at the smallest width, carried up by μP-style scaling — the per-layer parametrisation that keeps the optimal learning rate fixed across width, and the one thing that makes a 4.5 B run a one-shot rather than a search. **Early stopping at the held-out minimum, its step recorded; ≤ 4 epochs.**

Noise (§4.4)

z-noise 0.7 on the forecaster’s input embeddings stays — measured +0.045 / +0.050 on the rolled field (E-036 / E-037). Its mechanism is DENOISING: the forecaster learns the manifold of plausible embeddings rather than the identity. Re-measure the dose at L3+; the optimum moved with the lattice in E-065 and will move with width.

What scale does NOT fix (§4.5)

H1 — recovering the anchor’s own velocity from the displacement of tracers across the cone — FAILED at 7 M under two objectives, five seeds, and on a synthetic advection field. A displacement head or a relative-position attention bias is a MODELLING arm, tested at L1 on the synthetic field first; never promised from scale.

Read-outs: El Niño, AMOC, colour — and what each needs first

E-072 §5–6. Six read-outs, one frozen representation. Each row names the label it is scored against, the data that must land before the head is worth training, and the null it must beat.

Every read-out is a head on the frozen representation plus the rolled forecast — never a separate model — so adding one costs a probe, not a run.

read-out	the label	the data prerequisite	the null it must beat
El Niño	Niño-3.4 / ONI, and the 2026 event's peak and exit — our own OISST bake (–0.57 Jan → +2.09 Jul 2026) reconciled with CPC's ONI	the tropical Pacific is already in family 7 (E-070); the SUBSURFACE (GODAS / ORAS5, or the warm-water volume) and daily winds are the additions ranked in CONE_DATA_AND_ENSO §4.5 — without them the head is an SST-persistence forecast	the IRI/CPC plume; the LIM; the spring-barrier skill curve
AMOC	RAPID at 26.5° N, then MOVE, OSNAP, SAMBA and the Florida cable — the truth series is already in the tensor (truth_rapid, truth_fc)	E-071's buoyancy forcing (ERA5 heat and freshwater fluxes) and the 3-D interior (GREP) — the two additions the data ladder ranks first; and the terminal holdout, because the number is low-frequency (E-062's 3-year roll finding)	damped persistence; the LIM; the wind-stress ridge bar
surface colour	chlorophyll — OC-CCI 4 km from 1997, PACE from 2024 — as a TARGET channel: reflectance is the input, colour the target (E-071 §6.1)	the reflectance bands in the shared set, and BGC-Argo profile tokens	persistence; the seasonal harmonic; a ridge on reflectance
sea ice	concentration, extent, the September minimum — OSI SAF / NSIDC, already the frozen-fraction channel	nothing new: it is in Phase L0	damped persistence; the LIM
land water	GRACE storage, soil moisture, snow	GRACE, ESA CCI soil moisture, MOD10 snow — Phase L0 / L1 of E-071 §6.5	persistence; the harmonic
the atmosphere	2 m temperature and precipitation at 5–30 days — ERA5, IMERG	Phase L0	ECMWF's own open-data forecast — a strong, free null

The order of work is the order of the prerequisites

Sea ice and land water are FREE the day Phase L0 exists — six gap-free ERA5 / ice / soil channels, ~40 GB. **El Niño** needs the Pacific subsurface. **AMOC** needs the buoyancy forcing and the interior. **Colour** needs the reflectance bands and the biosphere sphere.

- What to do first, in one list (E-072 §6)
- 1 E-071 on the North Atlantic (rung L1): cone v2 geometry + harmonic anomaly + profile tokens, per-family loss and rolled skill against the LIM — the cheapest measurement that says whether the data half of this plan moves anything.
 - 2 Family 7's Phases B–F (the global tensor is pulled, not built) and the L2 rung with E-070's gates G1–G3 — with the width ladder re-measured there.
 - 3 Phase L0 of E-071 §6 (six ERA5 / ice / soil channels, ~40 GB) and the first sea-ice and land-water heads — the cheapest new read-outs.
 - 4 The Pacific subsurface and daily winds → the El Niño head, scored on the 2026 event's peak and exit against the IRI plume — the read-out with a public benchmark.
 - 5 L3: the sphere-routed mixture of experts at 1–4.5 B on the all-spheres tensor, with the μ P learning-rate transfer measured at L2.
- Nothing above rung L1 is dispatched by this plan; each rung is bought on the one below it.

Reading. Every read-out is a head on ONE frozen representation plus the rolled forecast, never a separate model — which is why the list can grow without a new training run each time. The order of work is therefore the order of the prerequisites, not the order of how interesting the read-out is: sea ice and land water the day Phase L0 lands, El Niño once the Pacific subsurface is in, AMOC once the buoyancy forcing and the interior are, colour once the reflectance bands are.

Phase 0 first — the decisive experiment fits the data we already have

Then one sphere per phase, in the agreed order: ocean surface + atmosphere as forcing, interior, ocean biosphere, land, fast atmosphere

Phase 0 — the A/B on today's tokens (weeks). Same tokens the stage-2 pool reads today: Earth 2 ocean pixel-months, 3×3 codec, sunflower-89 $\times$ 24 months. Train a cone codec over those tokens with channel drop + lag-band drop + anchor reconstruction. Compare at equal total context under the corrected window-scope pool (paper v8, 2 Sep 2026), after its null ladder exists, with $n \geq 3$ seeds per arm:

A current codec + stage-2 attention head • **B** cone codec + ridge probe • **B'** cone codec + attention head

Verdict metric: rolled skill — MSSS vs climatology and vs damped persistence per lead, trained and held-out longitudes separately, block-bootstrap intervals, the LIM null beside every number. Diagnostics (not verdicts): a velocity probe (GLORYS u, v at the anchor; drifters where available; derived channels masked at test) and the RAPID transport probe.

Predictions: $B \geq A$ on rolled skill with a linear head, and B beats the embedding-space LIM where A does not; velocity probe $B \gg A$; static probes equal. If B needs B' to match A, the 'work in the embedding' argument loses on sample efficiency and the hybrid keeps a thick stage 2. Effects inside the tier's replicate band are written as consistencies, never levels.

Phase	Sphere	What enters	New probes
1	Ocean surface + atmosphere as forcing	rungs 6 + 7 as gridded dots (OSTIA, DUACS, ASCAT, IMERG, AMSR2 ice; ERA5 / IFS open data); L-shaped and A-family cones; pentad anchors at 25 km	SST and SSH forecast skill at 1–4 weeks vs persistence and vs the Copernicus 1/12° forecast; sea-ice edge
2	Interior	Argo core / Deep / BGC and moorings as native dots; GLORYS flagged derived, dropped 50 %; depth-band drop objective	AMOC again; mixed-layer depth; heat content 0–700 m
3	Ocean biosphere	ocean-colour L3 (gaps stay gaps), PACE, SOCAT, BGC-Argo variables	carbon-flux reconstruction vs SeaFlux / Copernicus carbon (held out, not input)
4	Land	local codec on Sentinel-2/-1 and Landsat tiles (rungs 1–2); rung-10 fields as dots; column cones with the atmosphere hat	LAI / SIF and soil-moisture forecasts, fire risk; standard GeoFM benchmarks on e_{loc} for comparability
5	Fast atmosphere	geostationary imagers and ground radar at 10-min cadence through a second local codec (spatio-temporal tubelets); family-A disc at hourly Δt	nowcasting skill; wind-driven SST and ice response

Ablations that run in every phase

- 1 cone-in vs cone-on-top at equal context
- 2 drop lag bands
- 3 drop the far field
- 4 drop depth
- 5 derived channels masked at test
- 6 cone $\times$ 2 vs cone $\times$ $\frac{1}{2}$ on v and τ

Engineering first: a pre-materialised cone index per channel family (which tiles, dates and profiles fall in which anchors' cones). The sampler, not the network, is the bottleneck — the network is under an hour per year of 25 km pentad ocean embeddings at 100 TFLOP/s, about a day for 1 km monthly land.

Data continuity 2026-27 — build the archive sensor-agnostic

What the verification pass found switching off or changing (agency notices, 2 Sep 2026). Channel-drop training turns a missing sensor into a masked channel, not a failure.

Source	What changes	Consequence for the ladder
Sentinel-1	S1B failed Dec 2021; S1C launched 5 Dec 2024, S1D 4 Nov 2025 (data open 17 Apr 2026); S1A terminated 29 Jun 2026	Constellation is S1C + S1D at 6-day repeat, timeline shifted one day; radar record has a 2022–24 single-satellite gap
Sentinel-2	Three satellites (S2A/B/C) only until end-2026; S2C is prime	Revisit drops back to 5 days in 2027 — do not tune the local codec to the 2025–26 cadence
MODIS → VIIRS	Terra/Aqua shut down late 2026 / early 2027; Suomi-NPP data delivery ends 1 Nov 2026	Rung 3 continues on NOAA-21 (primary) and NOAA-20; the 2000–2026 MODIS archive is training data with a hard end
Passive microwave	DMSP SSMIS retiring ~Sep 2026; NSIDC NRT SSMIS sea ice retired 18 Jun 2026; NOAA AMSR2 NRT ends Sep 2026 for AMSR3 (GOSAT-GW, launched 28 Jun 2025)	NRT sea-ice concentration and drift now depend on one 2012 satellite (AMSR2) until AMSR3 products are operational
Altimetry	Sentinel-6B launched 17 Nov 2025, reference hand-over date not yet published; DUACS L4 now 0.125°; SWOT L4 only experimental	Sea-level rung stable; treat the 0.25° legacy grid and the new 0.125° grid as one channel with a resolution flag
Climate records that stop	OC-CCI v6 ends 31 Dec 2024; ESA CCI soil moisture and snow are annual retrospective releases; OSCAR final ends Aug 2022; FLUXNET2015 frozen (Feb 2020)	Use the NRT twins for 2025+ (GlobColour L3, ASCAT / SMAP, OSCAR NRT, ICOS / AmeriFlux); expect version seams
Licences to watch	EN4 non-commercial; GRDC research-only, no redistribution; OPERA radar for EUMETNET members; GLEAM commercial use by approval; ECMWF 9 km IFS licensed (0.25° open data free)	Keep licensed sources out of any embedding meant for release; they can still supervise probes

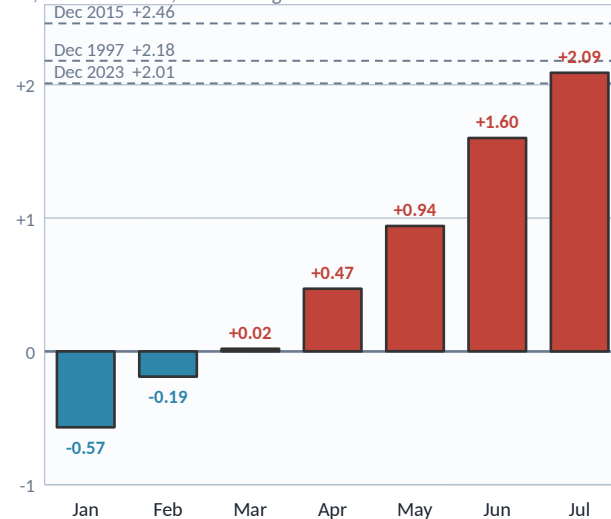
Also verified: ERA5's SST and sea ice are prescribed inputs (HadISST2, OSI SAF, OSTIA), so they must never be scored as ocean observations; ERA5-Land has no data assimilation at all — a land model replayed on ERA5 forcing; Copernicus waves are MFWAM, not WAVEWATCH; Copernicus surface carbon is 0.25°, not 1°; the OPERA composite is now 1 km / 5 min (CIRRUS), not the retired 2 km / 15 min ODYSSEY. Float counts are the live OceanOPS figures of 2 Sep 2026 and move daily.

El Niño 2026 — what the catalog predicts, and what to add

Our own OISST bake says Niño-3.4 (the central-Pacific sea-temperature anomaly that defines the event) went -0.6°C in January to $+2.1^{\circ}\text{C}$ in July 2026; NOAA's Climate Prediction Center, 13 Aug: El Niño Advisory, > 90 % chance of a very strong event, 69 % chance of exceeding $+2.5^{\circ}\text{C}$ in Oct-Dec — stronger than any event since 1950

Niño-3.4 monthly anomaly, 2026

5° S–5° N, 170° W–120° W, 500 one-degree cells



Computed here from data/oisst_y (NOAA's daily satellite-and-buoy sea-temperature analysis, block-meant to 1° monthly) against data/oisst_clim, the app's 1991–2020 normal. The Climate Prediction Center's July value is $+1.4^{\circ}\text{C}$ — a different sea-temperature reconstruction (ERSSTv5), a different averaging and possibly the relative index CPC now favours; the 0.7°C gap is unexplained here, and is itself the reason to carry the official index beside our own bake.

Already in the catalog

274 records, compiled for the Atlantic overturning circulation — its 'amoc' flag mis-sorts for El Niño. what → signal · limit

OISST v2.1 → the label itself, all four Niño boxes. Baked global 1° monthly 1981-09 → 2026-07.

GLORYS12 surface currents + mixed-layer depth → equatorial advection. Baked 1993 → 2026-05 · the depth levels are not baked.

GPCP 1979→ / IMERG 2000→ → the rainfall shift towards the dateline that is El Niño's atmospheric expression · only a climatology is baked from GPCP; IMERG is live.

MERRA-2 10 m wind → trade winds · monthly tiles only, so a westerly wind burst (a 5–15 day reversal) is averaged away.

NASA sea-surface height → Kelvin and Rossby waves · the app's tiles END 2019-01.

OSCAR surface currents → zonal advection · tiles 2014-10 → 2024-07 only.

CERES → convection · tiles end 2018-10, and it is monthly NET radiation, not the daily outgoing-longwave field the index needs.

TAO/TRITON + PIRATA moorings ('gtmba') → the single most El-Niño-specific record in the catalog · catalogued only for its Atlantic side, and flagged not-renderable.

Catalogued but unused ORAS5 1958→ · SODA3 · ERSSTv5 (the substrate of the official index) · HadISST · EN4 1900→ · ERA5 · CFSR · the SEAS5 and NMME seasonal hindcasts 1981/82→.

The machine-learning tensors are North-Atlantic-only family 7 — the first tensor covering the whole globe at the same 0.25° , 5-day grid (experiment E-070) — is mid-fetch, and its own plan names the equatorial waveguide as a target regime.

Add, ranked

1 Warm Water Volume and 20 °C-isotherm depth NOAA PMEL, monthly 1980→. How much warm water is stacked up in the equatorial Pacific — the ~6-month-lead predictor; recharge-discharge theory IS this number.

2 GODAS or ORAS5 NOAA's operational ocean analysis ($1/3^{\circ} \times 1^{\circ}$, 40 levels, 5-day and monthly, 1980→, keyless OPeNDAP), or ECMWF's ORAS5 (1958→, 5 members): the subsurface temperature and thermocline the tensor has nowhere in the Pacific.

3 NOAA interpolated outgoing longwave radiation 2.5° , daily, 1974→. Cold cloud tops mark deep convection — the Walker-circulation and Madden-Julian-Oscillation diagnostic CERES does not replace.

4 Index series, as labels and as nulls the Oceanic Niño Index (the official definition, on ERSSTv5, 1950→), weekly Niño-1+2/3/4, the Southern Oscillation Index and its equatorial twin, the Multivariate ENSO Index, the RMM Madden-Julian index (the wind-burst trigger), the Pacific Decadal Oscillation and Pacific Meridional Mode (spring precursors).

5 Daily winds ERA5 10 m u and v (0.25° , hourly, 1940→) or CCMP 6-hourly 1987→ — westerly wind bursts and wind-stress curl at the cadence at which they actually happen.

6 TAO/TRITON moorings as native dots on the Pacific side temperature and salinity with depth, currents, winds along 137°E – 95°W , 1980→. The array is already catalogued, as 'gtmba'.

7 DUACS sea level directly daily 0.125° , 1993→ — equatorial Kelvin and Rossby waves, and a proxy for warm-water volume.

8 Benchmarks the IRI/CPC forecast plume (2002→) and the NMME and SEAS5 hindcasts already in the catalog and unused — what any El Niño head must beat, and the spring-barrier skill curve to beat it on.

9 In the app a 'nino.json' bake beside eei.json and gistemp.json (~40 lines next to 'oisst_monthly()' in scripts/refresh_data.py), and the four Niño boxes drawn on the globe.

10 The cone codec on family 7, with an El Niño head sea-surface height, mixed-layer depth, the temperature/salinity column and wind stress are already the classic feature set. What is missing is the rainfall/longwave channel and the label.

Reading. Onset is behind us — the event began in March 2026 — so “predict this year's El Niño” now means three different things, and each needs different data. Hindcast the ONSET from the November 2025 state (a -0.6°C , mildly La-Niña-ish surface with a recharged subsurface): that is a test of whether the subsurface memory was readable. Forecast the PEAK amplitude and month (October–December) and the 2027 exit. And forecast the TELECONNECTIONS — the rainfall, drought and cyclone shifts that are what anyone outside the Pacific actually feels. All three need the subsurface first, the daily winds second, and an honest null beside them: persistence, a linear inverse model, and the operational forecast plume.

Family 8: the measurement is never missing, only far away

A proposal, 7 Sep 2026. For a sparse channel, carry the k nearest observations in the cone — with their distance — instead of a grid cell that is usually empty. Dense channels keep the sunflower.

A • family 7 — a grid cell is present or absent

the sunflower — the dense-channel stencil: a 3×3 patch plus dots, dense near the anchor, sparse far from it



the Argo group is written into 252 of 3,142 pentads (8 %); at every other bin the cone reads a miss token.

The token

(value, Δx , Δy , Δt , n_R) — the observation, its offset east and north in km, how long ago in days, and how many observations lay within the search radius. **Distance is a feature, never a mask.**

Three rules that make it safe

One-sided in time: only observations at or before the anchor's pentad (no leakage into a forecast).

Bounded: a search radius $R_{\max}$ and window $T_{\max}$ per channel; only past that bound does a “none within R ” token appear — the same miss token as today, made rare.

k from the data, not by hand: choose k so that the k -th neighbour typically sits at one correlation length (~ 300 km at the surface).

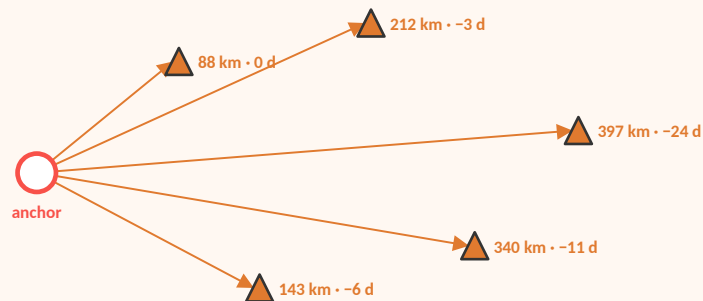
The numbers for Argo

$\approx 4,000$ active floats on a 10-day cycle $\Rightarrow \approx 2,000$ profiles per pentad worldwide, one per $\approx 180,000$ km² of ocean $\Rightarrow$ nearest profile of order 200 km in a single pentad; with $T_{\max} = 30$ days, six pentads of profiles $\Rightarrow \approx 90$ km, and the fifth neighbour at ≈ 200 km. **So $k = 5$ within 30 days is one correlation length.** Before 2004 there is nothing to find — and “none within R ” is then the honest, measured answer, not a gap.

What it is not

Not a replacement for the cone: dense gridded channels (currents, SST, the reanalysis) keep the sunflower, which encodes structure at several scales that k nearest neighbours would collapse to the 3×3 . Family 8 is a **HYBRID** — and it is no longer a tensor: a tensor for the dense groups plus an observation store (t , lat, lon, channel, value) and a precomputed neighbour index. That is the door to raw observations — profiles, moorings, altimeter tracks — without gridding them first.

B • family 8 — the k nearest observations, however far



▲ = an Argo profile, $k = 5$

always k tokens per sparse channel; each carries (value, Δx , Δy , Δt , local density).

Reading. Family 7 spends 80 % of its bytes on an Argo group that is absent in 92 % of pentads and gridded before the model sees it; family 8 spends a few tokens per anchor to make the interior always present and always honest about how far away the nearest measurement is. It is E-071's profile tokens taken to their conclusion, and the same primitive Aardvark and GraphDOP use to forecast from raw observations.

Family 7 vs family 8, side by side — and the one experiment that decides

Same grid, same pentads, same dense channels. What changes is how a SPARSE channel enters the cone.

aspect	family 7 (as built)	family 8 (proposed)
Representation	one array per group, [T,H,W,C] float16	dense groups unchanged + an observation store + a neighbour index per anchor
A sparse channel at an anchor	the nearest 1° cell, present in 8 % of pentads, else NaN → miss token	always its k nearest observations, each with (Δx, Δy, Δt, n_R)
What “missing” means	absent at this cell and bin	no observation within (R_max, T_max) — rare, and true
Time direction	the bin's own value	one-sided: at or before the anchor's pentad, never after
Tokens per sparse channel	1 (mostly a miss token)	k, fixed (3 / 5 / 7), regardless of density
Dense channels	sunflower + 3×3 patch	unchanged
Anomaly	per-channel harmonic climatology at the cell	the same climatology evaluated at the observation's own (lat, lon, day)
Argo source	Roemmich–Gilson gridded, monthly, 1°	raw profiles (the ~2,000 per pentad), no mapping step
Storage	52.9 GB	52.9 GB + a point store of order 1–3 GB + the index
Sampler cost	a strided gather	a k-NN lookup, precomputed once per anchor set; trivial with a per-bin KD-tree
What the model can learn from it	value	value AND how much to trust it — distance and density are the uncertainty proxy, and the model can learn that a 3-day-old profile 80 km away beats a 3-week-old one 400 km away
Risks	miss tokens dominate the Argo channels (83–97 % missing in the North Atlantic mapping)	unbounded search returns junk (hence R_max, T_max); density bias between basins (hence n_R); dense channels do not benefit (hence hybrid); it stops being a single tensor

The experiment that decides it, before any tensor is built

One arm inside the next cone-codec wave on family 7: replace the Argo group's 32 grid channels by k = 5 profile tokens from the raw Argo archive (T_max 30 d, R_max 1,000 km), everything else identical to the continuous twin. **Read-out:** held-out per-family loss against the predict-the-mean bar (the E-069b decomposition), the subsurface probe, and — the falsifier — whether the anchor family's loss on hidden interior channels drops below the twin's. If it does not, distance-as-a-feature bought nothing and family 8 is not built. Cost ≈ one seed, ~1.7 h on a 4090. **What must not be forgotten:** E-044c arm A3 measured that removing the Argo targets made the forecast WORSE (0.570 vs ~0.50) — the interior carries signal, and this is the cheapest way to carry more of it.

Reading. Family 8 is not a bigger tensor; it is a different contract between the sampler and the observing system. The decision is one ablation away and costs about a dollar.

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Ocean-including models OceanSAR-1: huggingface.co/galeio-research/OceanSAR-1 · arxiv.org/abs/2504.06962 · OceanSAR-2: arxiv.org/abs/2601.07392 · Copernicus-FM: arxiv.org/abs/2503.11849 · huggingface.co/datasets/wangyi111/Copernicus-Pretrain · github.com/zhu-xlab/Copernicus-FM · Aurora: microsoft.github.io/aurora · wekeo.copernicus.eu (ocean FM workspace)

Benchmarks & head-to-head tables OlmoEarth Tables 2, 3, 7 (arxiv.org/abs/2511.13655) · TerraMind Table 6 / PANGAEA (arxiv.org/abs/2504.11171; arxiv.org/abs/2412.04204) · Prithvi-EO 2.0 Tables A1–A4 / GEO-Bench (arxiv.org/abs/2412.02732; arxiv.org/abs/2306.03831) · TESSERA Table 1 (arxiv.org/abs/2506.20380) · GEO-Bench-2 ranks (arxiv.org/abs/2511.15658) · AlphaEarth & TESSERA for LCZ mapping (arxiv.org/abs/2606.20034) · Granite-Ocean Table 1 (arxiv.org/abs/2509.21273) · OceanSAR-1 Tables 1–3 (arxiv.org/abs/2504.06962)

Related work OceanNet arxiv.org/abs/2310.00813 · PARADIS arxiv.org/abs/2601.21151 · ClimODE arxiv.org/abs/2404.10024 · Neural-LAM arxiv.org/abs/2309.17370 · stretched grid arxiv.org/abs/2409.02891 · Keisler arxiv.org/abs/2202.07575 · GraphCast arxiv.org/abs/2212.12794 · Pangu arxiv.org/abs/2211.02556 · FuXi arxiv.org/abs/2306.12873 · Samudra arxiv.org/abs/2412.03795 · SamudrACE [doi 10.1029/2025GL119340](https://doi.org/10.1029/2025GL119340) · ACE2-SOM arxiv.org/abs/2412.04418 · ACE2-NEMO arxiv.org/abs/2603.28704 · AIFS surface ocean arxiv.org/abs/2604.25559 · Aurora [doi 10.1038/s41586-025-09005-y](https://doi.org/10.1038/s41586-025-09005-y) · Aardvark [doi 10.1038/s41586-025-08897-0](https://doi.org/10.1038/s41586-025-08897-0) · GraphDOP arxiv.org/abs/2412.15687 · 4DVarNet [doi 10.1029/2023MS003609](https://doi.org/10.1029/2023MS003609) · ESFM arxiv.org/abs/2605.00850 · Penny & Hamill 2017 BAMS · CSIR-ML6 [doi 10.5194/gmd-12-5113-2019](https://doi.org/10.5194/gmd-12-5113-2019) · CANYON-B [doi 10.3389/fmars.2017.00128](https://doi.org/10.3389/fmars.2017.00128)

Also checked Prithvi WxC (arxiv.org/abs/2409.13598) — atmosphere-only, no ocean downstream tasks found; NASA marine-debris work — not Prithvi-based. Earth 2 numbers quoted from paper/paper.tex (attribution matrix, probe ladder).